Problem 8-1

The horizontal force is **P**. Determine the normal and frictional forces acting on the crate of weight *W*. The friction coefficients are μ_k and μ_s.

Given:

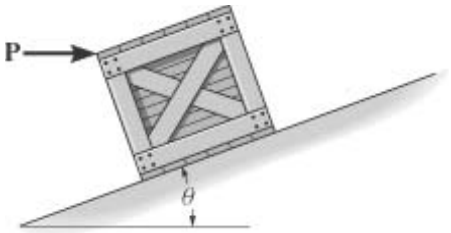
$W = 300$ lb

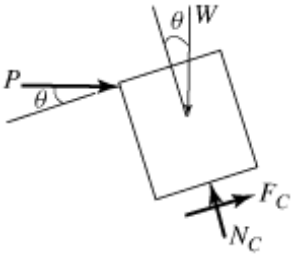
$P = 80$ lb

$\mu_s = 0.3$

$\mu_k = 0.2$

$\theta = 20$ deg





Solution:

Assume no slipping:

$\Sigma F_x = 0;$ $\quad P\cos(\theta) - W\sin(\theta) + F_C = 0$

$F_C = -P\cos(\theta) + W\sin(\theta)$ $\quad F_C = 27.4\,\text{lb}$

$\Sigma F_y = 0;$ $\quad N_C - W\cos(\theta) - P\sin(\theta) = 0$

$N_C = W\cos(\theta) + P\cdot\sin(\theta)$ $\quad N_C = 309\,\text{lb}$

Check $\quad F_{Cmax} = \mu_s N_C$ $\quad F_{Cmax} = 92.8\,\text{lb}$

$F_{Cmax} > F_C$

Problem 8-2

Determine the magnitude of force **P** needed to start towing the crate of mass *M*. Also determine the location of the resultant normal force acting on the crate, measured from point *A*.

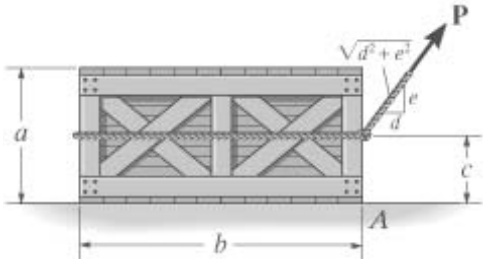
Given:

$M = 40$ kg $\quad c = 200$ mm

$\mu_s = 0.3$ $\quad d = 3$

$a = 400$ mm $\quad e = 4$

$b = 800$ mm



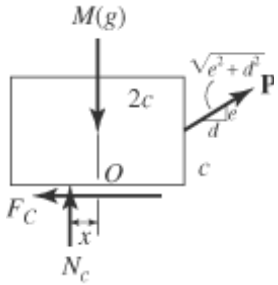
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Solution:

Initial guesses: $N_C = 200\ \text{N}$ $P = 50\ \text{N}$

Given

$\Sigma F_x = 0;$ $\left(\frac{d}{\sqrt{d^2+e^2}}\right)P - \mu_s N_C = 0$



$\Sigma F_y = 0;$ $N_C - M g + \frac{e P}{\sqrt{d^2+e^2}} = 0$

$\begin{pmatrix} N_C \\ P \end{pmatrix} = \text{Find}(N_C, P)$

$N_C = 280.2\ \text{N}$ $P = 140\ \text{N}$

$\Sigma M_O = 0;$ $-\mu_s N_C\left(\frac{a}{2}\right) - N_1 x + \left(\frac{e P}{\sqrt{d^2+e^2}}\right)\left(\frac{b}{2}\right) = 0$

$$x = \frac{-1}{2}\,\frac{\mu_s N_C\, a\sqrt{d^2+e^2} - e\,P\,b}{N_C\sqrt{d^2+e^2}} \qquad x = 123.51\ \text{mm}$$

Thus, the distance from A is $A = x + \frac{b}{2}$ $A = 523.51\ \text{mm}$

Problem 8-3

Determine the friction force on the crate of mass M, and the resultant normal force and its position x, measured from point A, if the force is **P**.

Given:

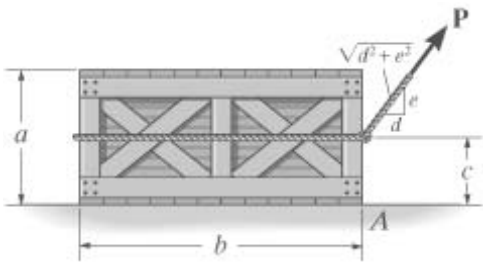
$M = 40\ \text{kg}$ $\mu_s = 0.5$

$a = 400\ \text{mm}$ $\mu_k = 0.2$

$b = 800\ \text{mm}$ $d = 3$

$c = 200\ \text{mm}$ $e = 4$

$P = 300\ \text{N}$



Solution:

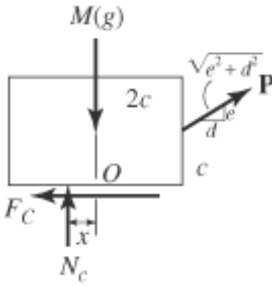
Initial guesses: $F_C = 25\ \text{N}$ $N_C = 100\ \text{N}$

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Given

$\Sigma F_x = 0; \qquad P\left(\frac{d}{\sqrt{d^2 + e^2}}\right) - F_C = 0$

$\Sigma F_y = 0; \qquad N_C - Mg + P\left(\frac{e}{\sqrt{d^2 + e^2}}\right) = 0$



$\begin{pmatrix} F_C \\ N_C \end{pmatrix} = \text{Find}(F_C, N_C) \qquad F_{Cmax} = \mu_s N_C$

Since $F_C = 180.00$ N > $F_{Cmax} = 76.13$ N then the crate slips $\qquad F_C = \mu_k N_C \qquad \begin{pmatrix} F_C \\ N_C \end{pmatrix} = \begin{pmatrix} 30.5 \\ 152.3 \end{pmatrix}$ N

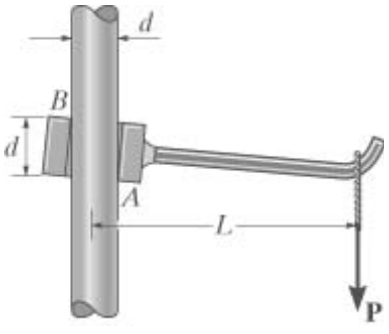
$\Sigma M_O = 0; \qquad -N_C\, x + P\left(\frac{e}{\sqrt{d^2 + e^2}}\right)a - P\left(\frac{d}{\sqrt{d^2 + e^2}}\right)c = 0$

$x = -P\left(\frac{-e\,a + d\,c}{N_C\sqrt{d^2 + e^2}}\right)$

Since $x = 0.39$ m $< \frac{b}{2} = 0.40$ m Then the block does not tip. $\qquad x_1 = a + x \qquad x_1 = 0.79$ m

---

Problem 8-4



The loose-fitting collar is supported by the pipe for which the coefficient of static friction at the points of contact A and B is μ_s. Determine the smallest dimension d so the rod will not slip when the load **P** is applied.

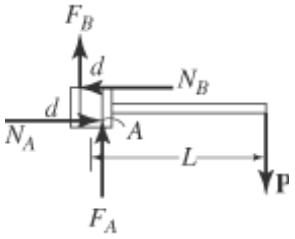
Given:

$\mu_s = 0.2$

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Solution:

$\Sigma M_A = 0; \quad N_B\, d - P\left(L - \frac{d}{2}\right) - \mu_s\, N_B\, d = 0 \qquad N_B = \frac{P\left(L - \frac{d}{2}\right)}{\left(1 - \mu_s\right) d}$



$\Sigma M_B = 0; \quad N_A\, d + \mu_s\, N_A\, d - P\left(L + \frac{d}{2}\right) = 0 \qquad N_A = \frac{P\left(L + \frac{d}{2}\right)}{\left(1 + \mu_s\right) d}$

$\Sigma F_y = 0; \quad \mu_s\left(N_A + N_B\right) - P = 0 \qquad \mu_s \frac{P}{d}\left(\frac{L + \frac{d}{2}}{1 + \mu_s} + \frac{L - \frac{d}{2}}{1 - \mu_s}\right) = P$

Thus, $d = 2\mu_s L$ $k = 2\mu_s$ $d = kL$

Problem 8-5

The spool of wire having a mass M rests on the ground at A and against the wall at B. Determine the force **P** required to begin pulling the wire horizontally off the spool. The coefficient of static friction between the spool and its points of contact is μ_s.

Units Used:

kN $= 10^3$ N

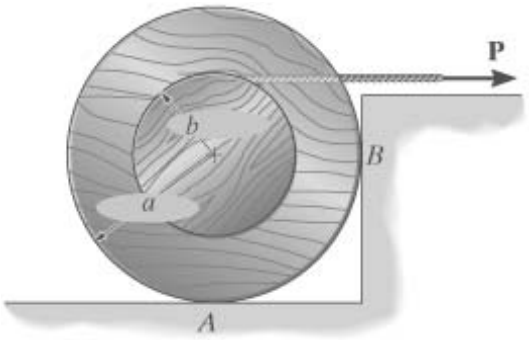
Given:

$M = 150$ kg

$\mu_s = 0.25$

$a = 0.45$ m

$b = 0.25$ m



Solution:

Initial guesses: $P = 100$ N $F_A = 10$ N $N_A = 20$ N $N_B = 30$ N $F_B = 10$ N

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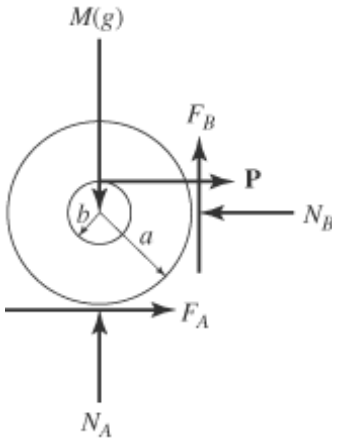
Given

$\Sigma F_y = 0; \quad N_A + F_B - M\,g = 0$

$\Sigma F_x = 0; \quad F_A - N_B + P = 0$

$\Sigma M_B = 0; \quad -P\,b + M\,g\,a - N_A\,a + F_A\,a = 0$

$F_A = \mu_s\,N_A \qquad F_B = \mu_s\,N_B$



$$\begin{pmatrix} P \\ F_A \\ F_B \\ N_A \\ N_B \end{pmatrix} = \text{Find}\left(P, F_A, F_B, N_A, N_B\right) \qquad \begin{pmatrix} F_A \\ N_A \\ F_B \\ N_B \end{pmatrix} = \begin{pmatrix} 0.28 \\ 1.12 \\ 0.36 \\ 1.42 \end{pmatrix} \text{kN} \qquad P = 1.14\,\text{kN}$$

Problem 8-6

The spool of wire having a mass M rests on the ground at A and against the wall at B. Determine the forces acting on the spool at A and B for the given force P. The coefficient of static friction between the spool and the ground at point A is μ_s. The wall at B is smooth.

Units Used:

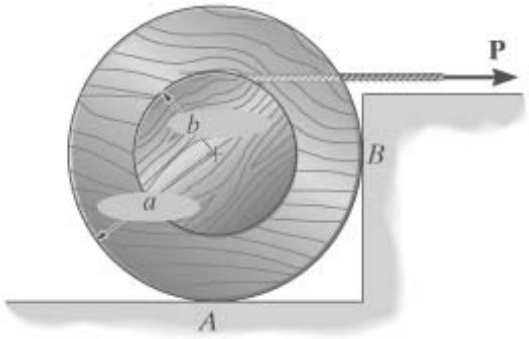
$\text{kN} = 10^3\ \text{N}$

Given:

$P = 800\ \text{N} \quad a = 0.45\ \text{m}$

$M = 150\ \text{kg} \quad b = 0.25\ \text{m}$

$\mu_s = 0.35$



Solution: Assume no slipping

Initial guesses : $F_A = 10\text{N} \qquad N_A = 10\text{N} \qquad N_B = 10\text{N} \qquad F_{Amax} = 10\text{N}$

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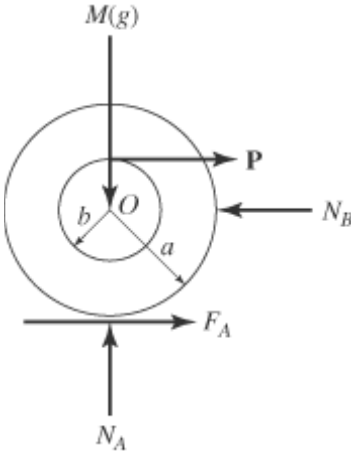
Given

$\Sigma F_x = 0;$ $F_A - N_B + P = 0$

$\Sigma F_y = 0;$ $N_A - M g = 0$

$\Sigma M_0 = 0;$ $-P\,b + F_A\,a = 0$

$F_{Amax} = \mu_s\, N_A$



$$\begin{pmatrix} F_A \\ F_{Amax} \\ N_A \\ N_B \end{pmatrix} = \text{Find}\left(F_A, F_{Amax}, N_A, N_B\right) \qquad \begin{pmatrix} F_A \\ F_{Amax} \end{pmatrix} = \begin{pmatrix} 444 \\ 515 \end{pmatrix} \text{N}$$

If $F_A = 444$ N $< F_{Amax} = 515$ N
then our no-slip assumption is good.

$$\begin{pmatrix} N_A \\ F_A \end{pmatrix} = \begin{pmatrix} 1.47 \\ 0.44 \end{pmatrix} \text{kN} \qquad N_B = 1.24\,\text{kN}$$

Problem 8-7

The crate has a mass *M* and is subjected to a towing force **P** acting at an angle θ_1 with the horizontal. If the coefficient of static friction is μ_s, determine the magnitude of **P** to just start the crate moving down the plane.

Given:

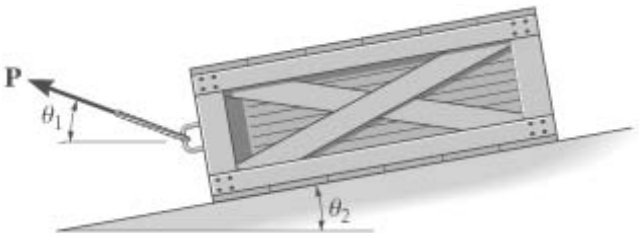
$M = 350$ kg

$\theta_1 = 20$ deg

$\theta_2 = 10$ deg

$\mu_s = 0.5$

$g = 9.81 \dfrac{\text{m}}{\text{s}^2}$



Solution:

Initial guesses: $N_C = 10$N $P = 20$N

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Given

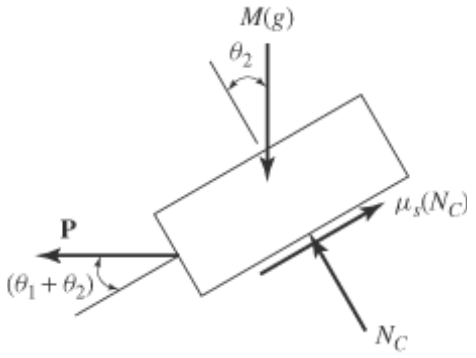
$$\Sigma F_x = 0; \qquad P\cos(\theta_1 + \theta_2) - \mu_s N_C + M g \sin(\theta_2) = 0$$

$$\Sigma F_y = 0; \qquad N_C - M g \cos(\theta_2) + P \sin(\theta_1 + \theta_2) = 0$$

$$\begin{pmatrix} N_C \\ P \end{pmatrix} = \text{Find}(N_C, P)$$

$$N_C = 2891 \text{ N}$$

$$P = 981 \text{ N}$$



Problem 8-8

The winch on the truck is used to hoist the garbage bin onto the bed of the truck. If the loaded bin has weight W and center of gravity at G, determine the force in the cable needed to begin the lift. The coefficients of static friction at A and B are μ_A and μ_B respectively. Neglect the height of the support at A.

Units Used:

$$\text{kip} = 10^3 \text{ lb}$$

Given:

$$W = 8.5 \text{ kip}$$

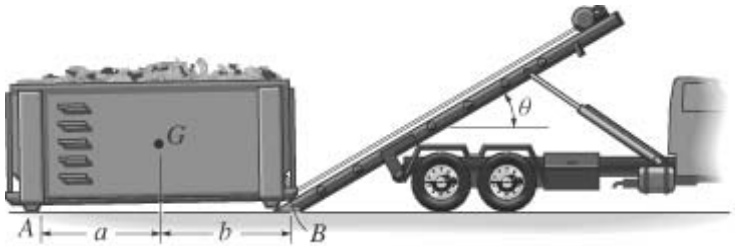
$$\mu_A = 0.3$$

$$\mu_B = 0.2$$

$$a = 10 \text{ ft}$$

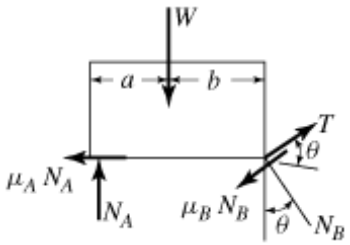
$$b = 12 \text{ ft}$$

$$\theta = 30 \text{ deg}$$



Solution:

The initial guesses are $T = 1 \text{ lb}$ $N_B = 1 \text{ lb}$ $N_A = 1 \text{ lb}$



Given

$$\circlearrowleft + \Sigma M_B = 0; \qquad W\, b - N_A\,(a + b) = 0$$

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$\overset{+}{\rightarrow} \Sigma F_x = 0; \qquad T\cos(\theta) - \mu_B N_B \cos(\theta) - N_B \sin(\theta) - \mu_A N_A = 0$

$+\uparrow \Sigma F_y = 0; \qquad N_A - W + T\sin(\theta) + N_B \cos(\theta) - \mu_B N_B \sin(\theta) = 0$

$\begin{pmatrix} T \\ N_A \\ N_B \end{pmatrix} = \text{Find}(T, N_A, N_B) \qquad \begin{pmatrix} N_A \\ N_B \end{pmatrix} = \begin{pmatrix} 4.64 \\ 2.65 \end{pmatrix} \text{kip} \qquad T = 3.67\,\text{kip}$

Problem 8-9

The motorcyclist travels with constant velocity along a straight, horizontal, banked road. If he aligns his bike so that the tires are perpendicular to the road at A, determine the frictional force at A. The man has a mass M_C and a mass center at G_C, and the motorcycle has a mass M_m and a mass center at G_m. If the coefficient of static friction at A is μ_A, will the bike slip?

Given:

$M_C = 60\ \text{kg}$

$M_m = 120\ \text{kg}$

$\mu_A = 0.4$

$\theta = 20\ \text{deg}$

$g = 9.81 \dfrac{\text{m}}{\text{s}^2}$



Solution: Assume no slipping

$\Sigma F_y = 0; \quad N_A - (M_m + M_C) g \cos(\theta) = 0$

$N_A = (M_m + M_C)\, g \cos(\theta)$

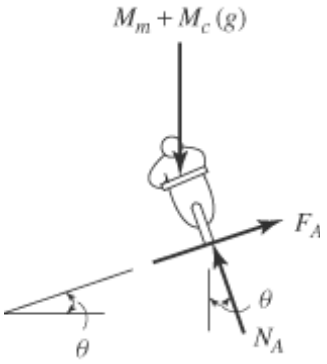
$N_A = 1659\ \text{N}$

$\Sigma F_x = 0; \quad F_A - (M_m + M_C)\, g \sin(\theta)$

$F_A = (M_m + M_C)\, g \sin(\theta)$

$F_A = 604\ \text{N}$

$F_{Amax} = \mu_A N_A$

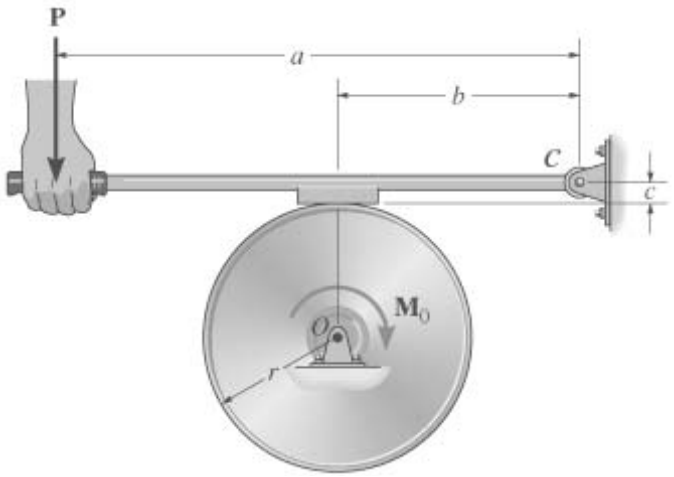


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Check: If $F_A = 604$ N $< F_{Amax} = 664$ N
then our no-slip assumption is good.

Problem 8-10

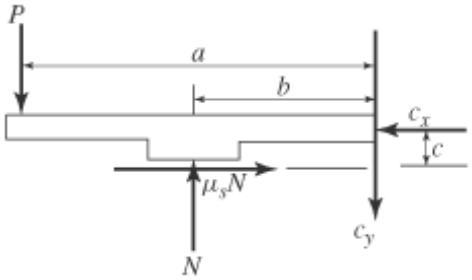
The block brake is used to stop the wheel from rotating when the wheel is subjected to a couple moment M_0 If the coefficient of static friction between the wheel and the block is μ_s, determine the smallest force **P** that should be applied.



Solution:

$$\curvearrowleft + \Sigma M_C = 0; \qquad P\,a - N\,b + \mu_S\,Nc = 0$$

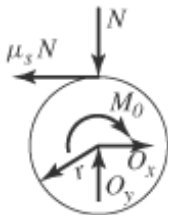
$$\mathrm{N} = \frac{P\,a}{b - \mu_S\,c}$$



$$\curvearrowleft + \Sigma M_O = 0; \qquad \mu_S\,Nr - M_O = 0$$

$$\frac{\mu_S\,P\,a\,r}{b - \mu_S\,c} = M_O$$

$$P = \frac{M_O\left(b - \mu_S c\right)}{\mu_S\,r\,a}$$

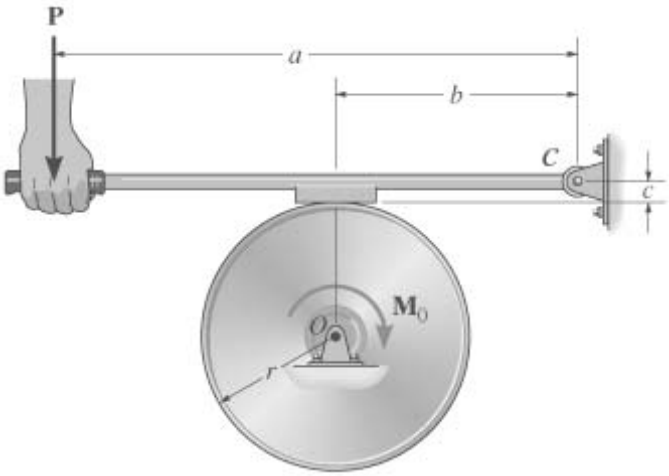


Problem 8-11

The block brake is used to stop the wheel from rotating when the wheel is subjected to a couple

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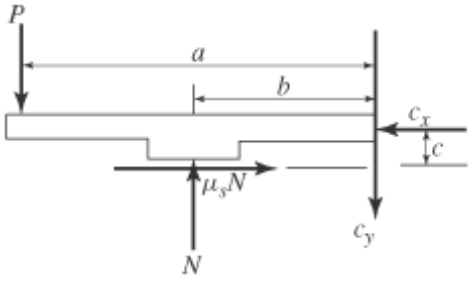
moment M_0 If the coefficient of static friction between the wheel and the block is μ_s, show that the brake is self locking, i. e., $P \le 0$, provided $\frac{b}{c} \le \mu_s$



Solution:

$\circlearrowleft + \Sigma M_C = 0; \quad P\,a - N\,b + \mu_S\,Nc = 0$

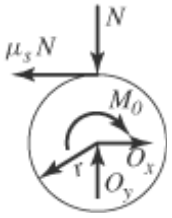
$$N = \frac{P\,a}{b - \mu_S\,c}$$



$\circlearrowleft + \Sigma M_O = 0; \quad \mu_S\,Nr - M_O = 0$

$$\frac{\mu_S\,P\,a\,r}{b - \mu_S\,c} = M_O$$

$$P = \frac{M_O\left(b - \mu_S\,c\right)}{\mu_S\,r\,a}$$

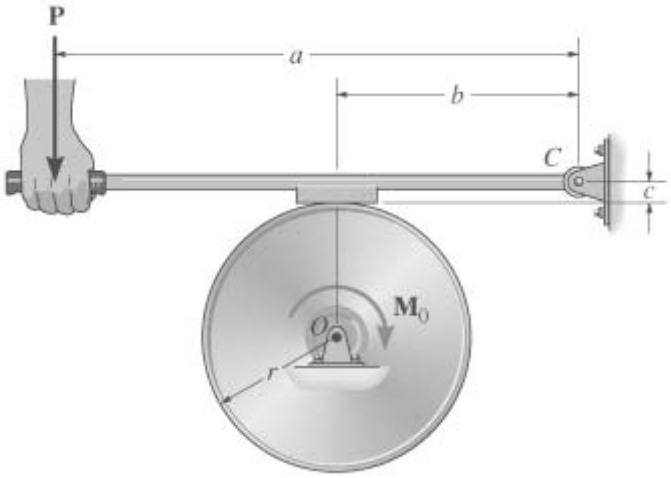


$P < 0$ if $\left(b - \mu_S c\right) < 0$ i.e. if $\frac{b}{c} < \mu_S$

Problem 8-12

The block brake is used to stop the wheel from rotating when the wheel is subjected to a couple moment M_0 If the coefficient of static friction between the wheel and the block is μ_s, determine the smallest force **P** that should be applied if the couple moment $\mathbf{M_O}$ is applied **counterclockwise**.

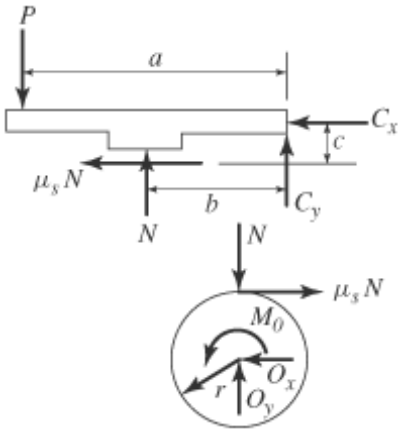
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Solution:

$$\circlearrowleft + \Sigma M_C = 0; \qquad P\ a - N\ b - \mu_S\ N\ c = 0$$

$$N = \frac{P\ a}{b + \mu_S\ c}$$



$$\circlearrowleft + \Sigma M_O = 0; \qquad -\mu_S\ N\ r + M_O = 0$$

$$\frac{\mu_S\ P\ a\ r}{b + \mu_S\ c} = M_O$$

$$P = \frac{M_O\left(b + \mu_S\ c\right)}{\mu_S\ ra}$$

Problem 8-13

The block brake consists of a pin-connected lever and friction block at B. The coefficient of static friction between the wheel and the lever is μ_s and a torque $\mathbf{M}$ is applied to the wheel. Determine if the brake can hold the wheel stationary when the force applied to the lever is (a) P_1 (b) P_2.

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Given:

$\mu_s = 0.3$

$M = 5$ N·m

$a = 50$ mm

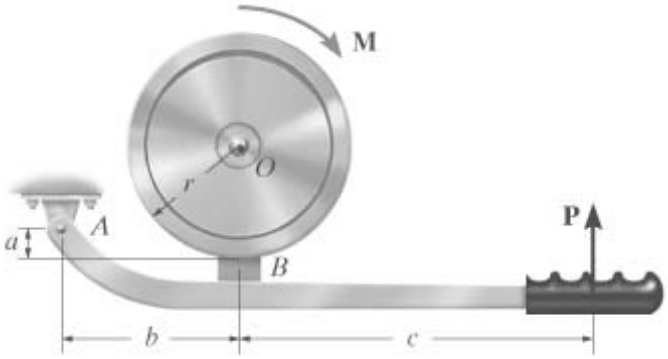
$b = 200$ mm

$c = 400$ mm

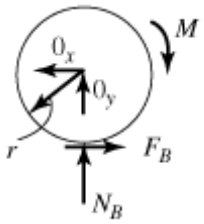
$r = 150$ mm

$P_1 = 30$ N

$P_2 = 70$ N



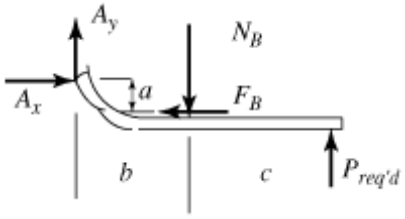
Solution: To hold lever:



$\circlearrowleft + \Sigma M_O = 0; \quad F_B\, r - M = 0$

$$F_B = \frac{M}{r} \qquad F_B = 33.333 \text{ N}$$

Require $$N_B = \frac{F_B}{\mu_s} \qquad N_B = 111.1 \text{ N}$$



Lever,

$\circlearrowleft + \Sigma M_A = 0; \quad P_{Reqd}\,(b + c) - N_B\, b - F_B\, a = 0$

$$P_{Reqd} = \frac{N_B\, b + F_B\, a}{b + c} \qquad P_{Reqd} = 39.8 \text{ N}$$

(a) If $P_1 = 30.00$ N $> P_{Reqd} = 39.81$ N then the break will hold the wheel

(b) If $P_2 = 70.00$ N $> P_{Reqd} = 39.81$ N then the break will hold the wheel

---

Problem 8-14

The block brake consists of a pin-connected lever and friction block at B. The coefficient of static friction between the wheel and the lever is μ_s and a torque **M** is applied to the wheel. Determine if the brake can hold the wheel stationary when the force applied to the lever is (a) P_1 (b) P_2. Assume that the torque **M** is applied **counter-clockwise.**

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Given:

$\mu_s = 0.3$

$M = 5\ \text{N}\cdot\text{m}$

$a = 50\ \text{mm}$

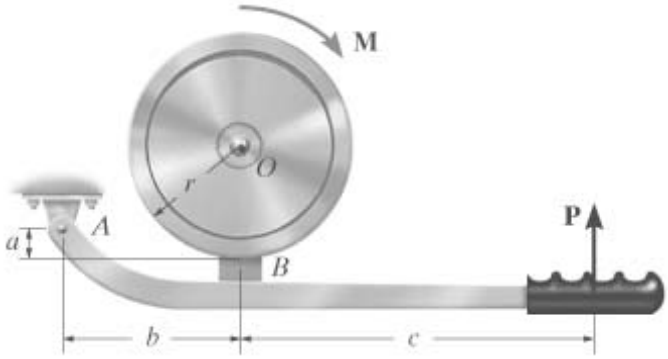
$b = 200\ \text{mm}$

$c = 400\ \text{mm}$

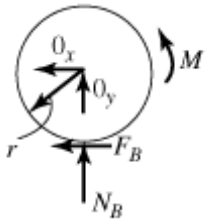
$r = 150\ \text{mm}$

$P_1 = 30\ \text{N}$

$P_2 = 70\ \text{N}$



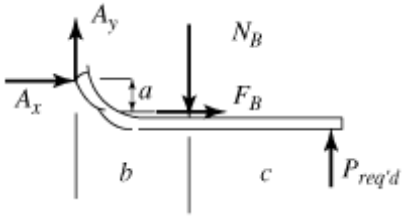
Solution: To hold lever:



$\circlearrowleft + \Sigma M_O = 0; \qquad F_B\, r - M = 0$

$$F_B = \frac{M}{r} \qquad F_B = 33.333\ \text{N}$$

Require $$N_B = \frac{F_B}{\mu_s} \qquad N_B = 111.1\ \text{N}$$



Lever,

$\circlearrowleft + \Sigma M_A = 0; \qquad P_{Reqd}\,(b + c) - N_B\, b + F_B\, a = 0$

$$P_{Reqd} = \frac{N_B\, b - F_B\, a}{b + c} \qquad P_{Reqd} = 34.3\ \text{N}$$

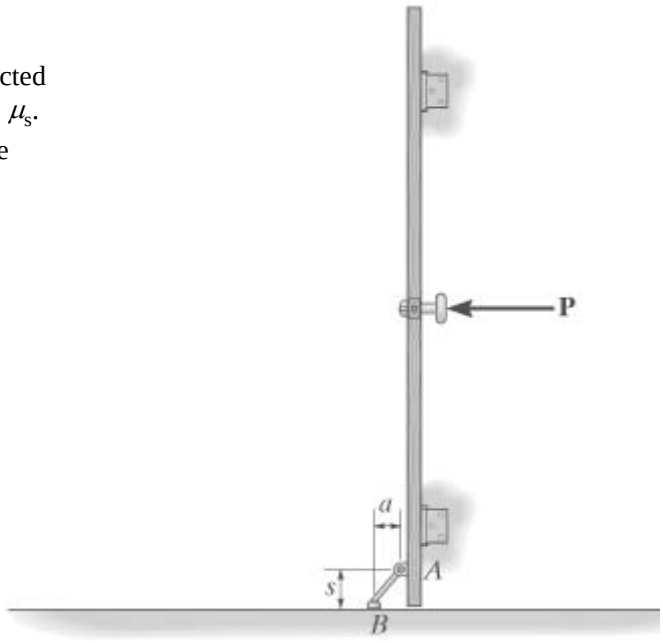
(a) If $P_1 = 30.00\ \text{N} > P_{Reqd} = 34.26\ \text{N}$ then the break will hold the wheel

(b) If $P_2 = 70.00\ \text{N} > P_{Reqd} = 34.26\ \text{N}$ then the break will hold the wheel

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Problem 8-15

The doorstop of negligible weight is pin connected at A and the coefficient of static friction at B is μ_s. Determine the required distance s from A to the floor so that the stop will resist opening of the door for any force **P** applied to the handle.



Given:

$\mu_s = 0.3$

$a = 1.5$ in

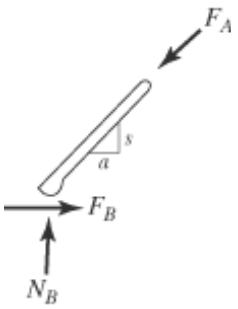
Solution:

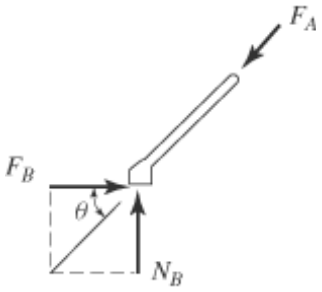
$$\Sigma F_y = 0; \quad N_B - \left(\frac{s}{\sqrt{s^2 + a^2}}\right) F_A = 0$$

$$\Sigma F_x = 0; \quad \mu_s N_B - \left(\frac{a}{\sqrt{s^2 + a^2}}\right) F_A = 0$$

$$\left(\frac{\mu_s s}{\sqrt{s^2 + a^2}}\right) F_A - \left(\frac{a}{\sqrt{s^2 + a^2}}\right) F_A = 0$$

$$\mu_s s = a \qquad s = \frac{a}{\mu_s} \qquad s = 5.00 \text{ in}$$





Problem 8-16

The chair has a weight W and center of gravity at G. It is propped against the door as shown. If the coefficient of static friction at A is μ_A, determine the smallest force **P** that must be applied to the handle to open the door.

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Given:

$\mu_A = 0.3$

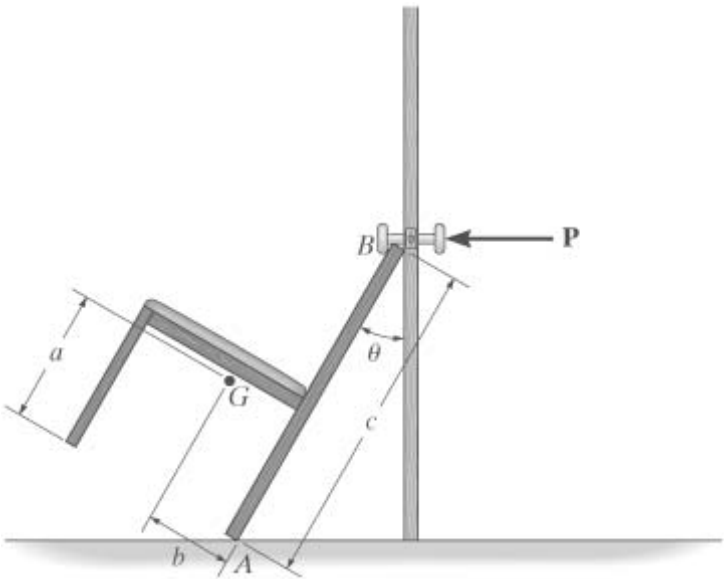
$a = 1.20$ ft

$b = 0.75$ ft

$c = 3$ft

$\theta = 30$ deg

$W = 10$ lb



Solution:

Guesses $B_y = 1$ lb $N_A = 1$ lb $P = 1$ lb

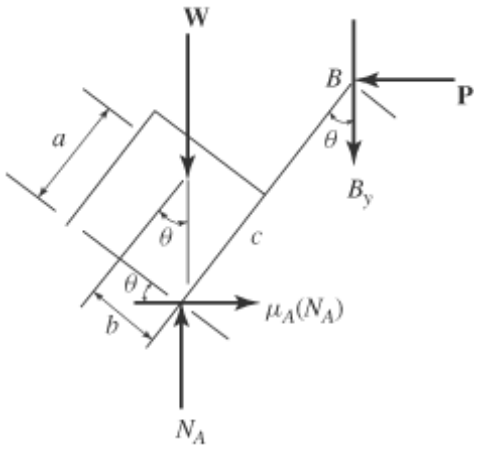
Given

$\Sigma F_x = 0; \quad -P + \mu_A N_A = 0$

$\Sigma F_y = 0; \quad N_A - W - B_y = 0$

$\Sigma M_B = 0; \quad \mu_A N_A c \cos(\theta) - N_A c \sin(\theta) + W\left[(c-a)\sin(\theta) + b\cos(\theta)\right] = 0$

$$\begin{pmatrix} B_y \\ N_A \\ P \end{pmatrix} = \text{Find}(B_y, N_A, P)$$



$B_y = 11.5$ lb

$N_A = 21.5$ lb

$P = 6.45$ lb

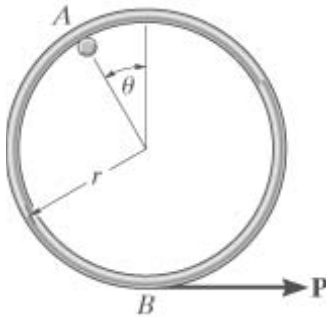
Problem 8-17

The uniform hoop of weight W is suspended from the peg at A and a horizontal force **P** is slowly applied at B. If the hoop begins to slip at A when the angle is θ, determine the coefficient of static friction between the hoop and the peg.

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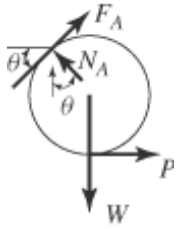
Given:

$\theta = 30$ deg



Solution:

$\Sigma F_x = 0; \quad \mu N_A \cos(\theta) + P - N_A \sin(\theta) = 0$



$P = (\mu \cos(\theta) - \sin(\theta))N_A$

$\Sigma F_y = 0; \quad \mu N_A \sin(\theta) - W + N_A \cos(\theta) = 0$

$W = (\mu \sin(\theta) + \cos(\theta))N_A$

$\Sigma M_A = 0; \quad -W\, r \sin(\theta) + P\,(r + r\cos(\theta)) = 0$

$W \sin(\theta) = P(1 + \cos(\theta))$

$(\mu \sin(\theta) + \cos(\theta))\sin(\theta) = (\sin(\theta) - \mu \cos(\theta))(1 + \cos(\theta))$

$\mu = \dfrac{\sin(\theta)}{1 + \cos(\theta)} \qquad \mu = 0.27$

Problem 8-18

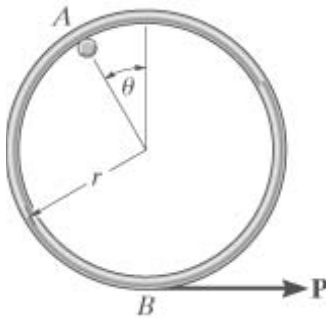
The uniform hoop of weight W is suspended from the peg at A and a horizontal force **P** is slowly applied at B. If the coefficient of static friction between the hoop and peg is μ_s, determine if it is possible for the hoop to reach an angle θ before the hoop begins to slip.

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Given:

$\mu_S = 0.2$

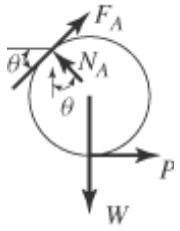
$\theta = 30 \text{ deg}$



Solution:

$\Sigma F_x = 0; \quad \mu N_A \cos(\theta) + P - N_A \sin(\theta) = 0$

$P = (\mu \cos(\theta) - \sin(\theta)) N_A$



$\Sigma F_y = 0; \quad \mu N_A \sin(\theta) - W + N_A \cos(\theta) = 0$

$W = (\mu \sin(\theta) + \cos(\theta)) N_A$

$\Sigma M_A = 0; \quad -W r \sin(\theta) + P (r + r \cos(\theta)) = 0$

$W \sin(\theta) = P (1 + \cos(\theta))$

$(\mu \sin(\theta) + \cos(\theta))\sin(\theta) = (\sin(\theta) - \mu \cos(\theta))(1 + \cos(\theta))$

$$\mu = \frac{\sin(\theta)}{1 + \cos(\theta)} \qquad \mu = 0.27$$

If $\mu_S = 0.20 < \mu = 0.27$ then it is not possible to reach $\theta = 30.00 \text{ deg}$.

Problem 8-19

The coefficient of static friction between the shoes at A and B of the tongs and the pallet is μ_{s1} and between the pallet and the floor μ_{s2}. If a horizontal towing force **P** is applied to the tongs, determine the largest mass that can be towed.

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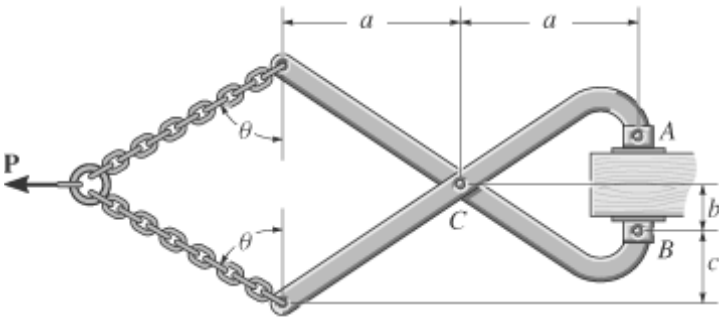
Given:

$\mu_{s1} = 0.5$ $a = 75$ mm

$\mu_{s2} = 0.4$ $b = 20$ mm

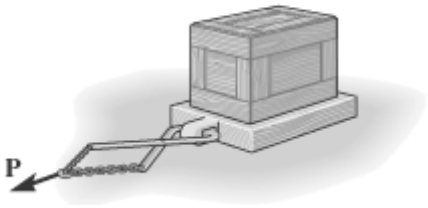
$P = 300$ N $c = 30$ mm

$g = 9.81 \frac{\text{m}}{\text{s}^2}$ $\theta = 60$ deg



Solution:

Assume that we are on the verge of slipping at every surface.

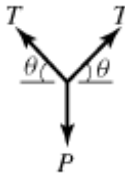


Guesses

$T = 1\text{N}$ $N_A = 1\text{N}$

$F = 1\text{N}$ $N_{ground} = 1\text{N}$

$F_A = 1\text{N}$ $mass = 1\text{kg}$



Given

$2\ T \sin(\theta) - P = 0$

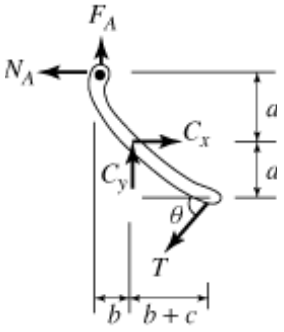
$-T \sin(\theta)\ (b + c) - T \cos(\theta)\ a - F_A\ b + N_A\ a = 0$

$F_A = \mu_{s1}\ N_A$

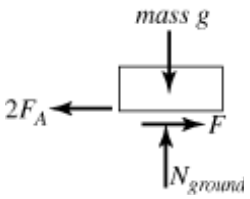
$2\ F_A - F = 0$

$N_{ground} - mass\, g = 0$

$F = \mu_{s2}\ N_{ground}$



$$\begin{pmatrix} T \\ N_A \\ F_A \\ F \\ N_{ground} \\ mass \end{pmatrix} = \text{Find}\left(T, N_A, F_A, F, N_{ground}, mass\right)$$



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$$\begin{pmatrix} T \\ N_A \\ F_A \\ F \\ N_{ground} \end{pmatrix} = \begin{pmatrix} 173.21 \\ 215.31 \\ 107.66 \\ 215.31 \\ 538.28 \end{pmatrix} \text{ N} \qquad mass = 54.9 \text{ kg}$$

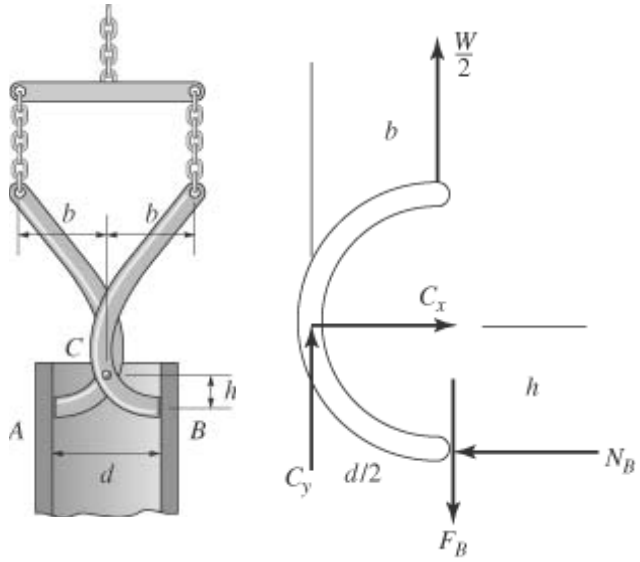
Problem *8-20

The pipe is hoisted using the tongs. If the coefficient of static friction at A and B is μ_s, determine the smallest dimension b so that any pipe of inner diameter d can be lifted.

Solution:

$$W - 2\,F_B = 0$$

$$\left(\frac{W}{2}\right) b - N_B\, h - F_B \left(\frac{d}{2}\right) = 0$$



Thus

$$F_B = \frac{W}{2}$$

$$N_B = \frac{W\,(2\,b - d)}{4h}$$

Require

$$F_B \le \mu_s N_B \qquad \frac{W}{2} \le \frac{\mu_s\, W\,(2b - d)}{4\,h} \qquad 2\,h \le \mu_s\,(2b - d) \qquad b > \frac{h}{\mu_s} + \frac{d}{2}$$

Problem 8-21

A very thin bookmark having a width a. is in the middle of a dictionary of weight W. If the pages are b by c, determine the force **P** needed to start to pull the bookmark out.The coefficient of static friction between the bookmark and the paper is μ_s. Assume the pressure on each page and the bookmark is uniform.

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Given:

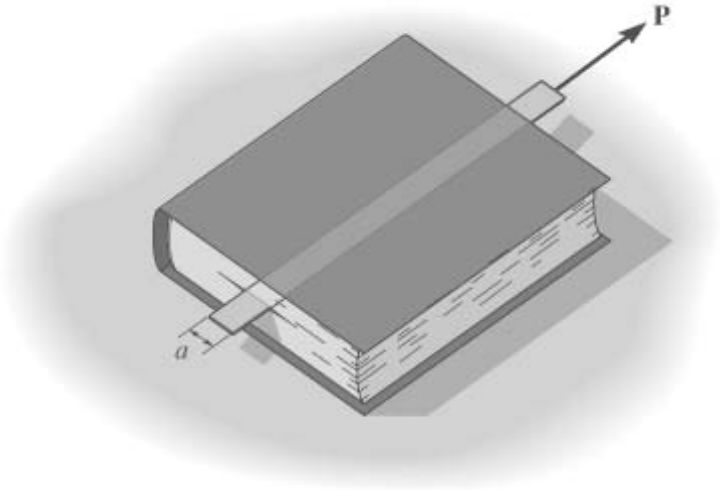
$a = 1$ in

$W = 10$ lb

$b = 8$ in

$c = 10$ in

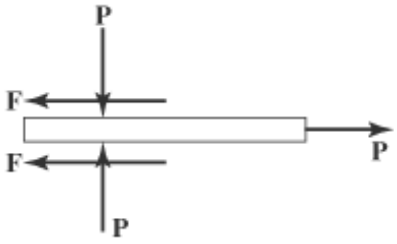
$\mu_S = 0.7$



Solution:

Pressure on book mark :

$$P = \frac{1}{2}\,\frac{W}{b\,c} \qquad P = 0.06\,\frac{\text{lb}}{\text{in}^2}$$



Normal force on bookmark: $N = P\,c\,a$

$$F = \mu_S\,N \qquad F = 0.44\,\text{lb}$$

$$\Sigma F_x = 0; \qquad P - 2F = 0 \qquad P = 2F \qquad P = 0.88\,\text{lb}$$

Problem 8-22

The uniform dresser has weight W and rests on a tile floor for which the coefficient of friction is μ_s. If the man pushes on it in the direction θ, determine the smallest magnitude of force **F** needed to move the dresser. Also, if the man has a weight W_{man}, determine the smallest coefficient of static friction between his shoes and the floor so that he does not slip.

Given:

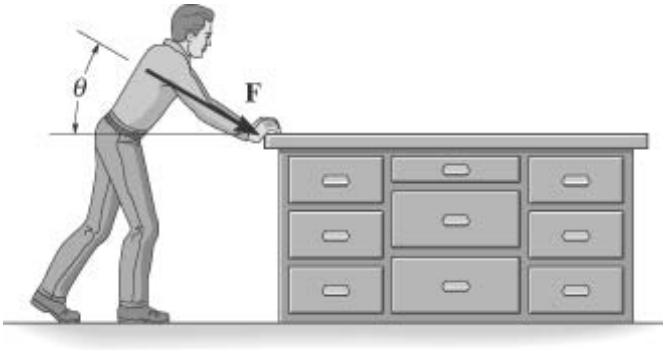
$W = 90$ lb

$\mu_S = 0.25$

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$W_{man} = 150$ lb

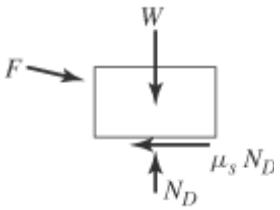
$\theta = 0$ deg



Solution:

Dresser: Guesses $N_D = 1\text{lb}$ $F = 1\text{lb}$

Given

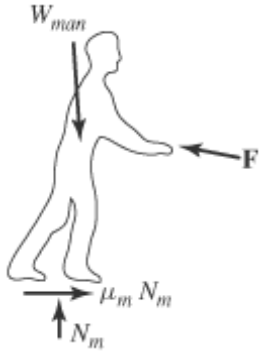


$+\uparrow \Sigma F_y = 0; \qquad N_D - W - F\sin(\theta) = 0$

$\xrightarrow{+} \Sigma F_x = 0; \qquad F\cos(\theta) - \mu_s N_D = 0$

$\begin{pmatrix} N_D \\ F \end{pmatrix} = \text{Find}(N_D, F) \qquad F = 22.50\,\text{lb}$

Man: Guesses $N_m = 1\text{lb}$ $\mu_m = 0.2$



Given

$+\uparrow \Sigma F_y = 0; \qquad N_m - W_{man} + F\sin(\theta) = 0$

$\xrightarrow{+} \Sigma F_x = 0; \qquad -F\cos(\theta) + \mu_m N_m = 0$

$\begin{pmatrix} N_m \\ \mu_m \end{pmatrix} = \text{Find}(N_m, \mu_m) \qquad \mu_m = 0.15$

Problem 8-23

The uniform dresser has weight W and rests on a tile floor for which the coefficient of friction is μ_s. If the man pushes on it in the direction θ, determine the smallest magnitude of force **F** needed to move the dresser. Also, if the man has a weight W_{man}, determine the smallest coefficient of static friction between his shoes and the floor so that he does not slip.

Given:

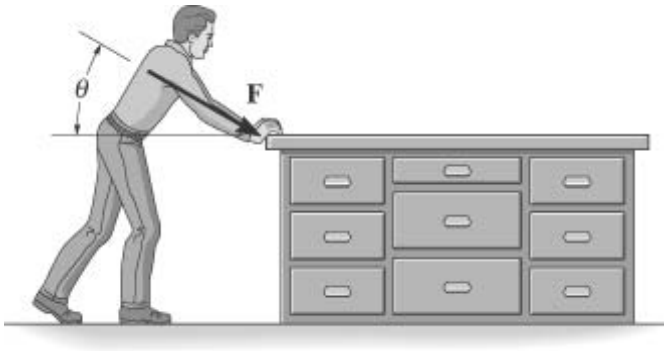
$W = 90$ lb

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$\mu_s = 0.25$

$W_{man} = 150$ lb

$\theta = 30$ deg



Solution:

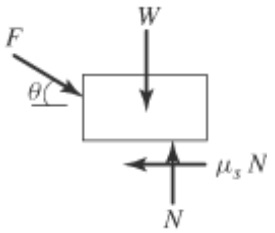
Dresser: Guesses $N_D = 1$lb $F = 1$lb

Given

$+\uparrow \Sigma F_y = 0; \quad N_D - W - F\sin(\theta) = 0$

$\xrightarrow{+} \Sigma F_x = 0; \quad F\cos(\theta) - \mu_s N_D = 0$

$\begin{pmatrix} N_D \\ F \end{pmatrix} = \text{Find}(N_D, F)$ $F = 30.36\,\text{lb}$



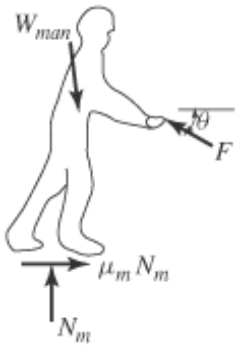
Man: Guesses $N_m = 1$lb $\mu_m = 0.2$

Given

$+\uparrow \Sigma F_y = 0; \quad N_m - W_{man} + F\sin(\theta) = 0$

$\xrightarrow{+} \Sigma F_x = 0; \quad -F\cos(\theta) + \mu_m N_m = 0$

$\begin{pmatrix} N_m \\ \mu_m \end{pmatrix} = \text{Find}(N_m, \mu_m)$ $\mu_m = 0.195$



Problem 8-24

The cam is subjected to a couple moment of **M**. Determine the minimum force **P** that should be applied to the follower in order to hold the cam in the position shown.The coefficient of static friction between the cam and the follower is μ_s. The guide at *A* is smooth.

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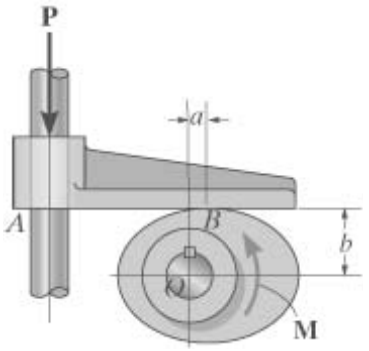
Given:

$a = 10$ mm

$b = 60$ mm

$M = 5$ N·m

$\mu_S = 0.4$

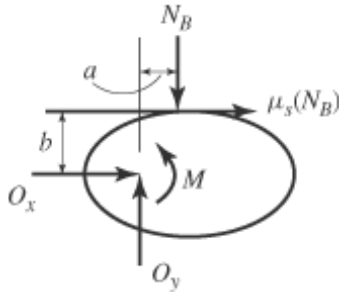


Solution:

$\Sigma M_0 = 0; \quad M - \mu_S N_B b - a N_B = 0$

$$N_B = \frac{M}{\mu_S b + a}$$

$N_B = 147.06$ N

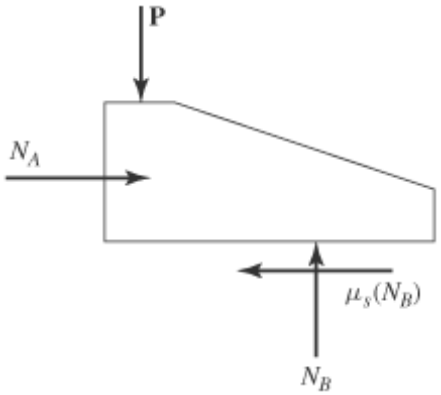


Follower:

$\Sigma F_y = 0; \quad N_B - P = 0$

$P = N_B$

$P = 147$ N



Problem 8-25

The board can be adjusted vertically by tilting it up and sliding the smooth pin *A* along the vertical guide *G*. When placed horizontally, the bottom *C* then bears along the edge of the guide, where the coefficient of friction is μ_s. Determine the largest dimension *d* which will support any applied force **F** without causing the board to slip downward.

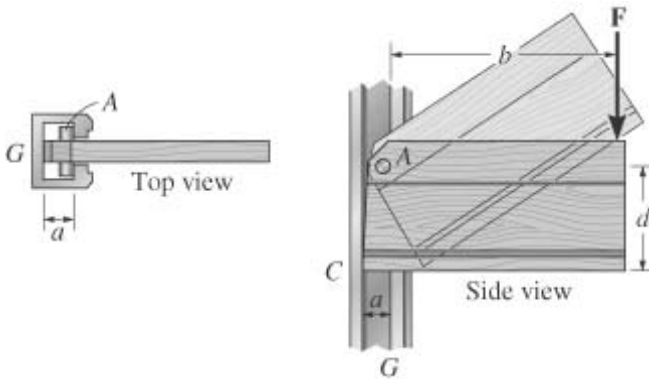
Given:

$\mu_S = 0.4$

$a = 0.75$ in

$b = 6$ in

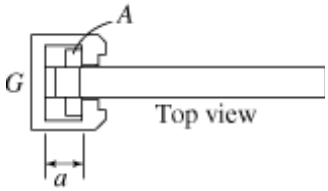
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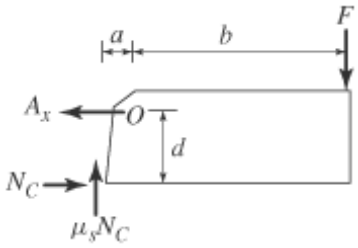
Solution:

$+\uparrow \Sigma F_y = 0;$

$\mu_s N_C - F = 0$



$+\circlearrowleft \Sigma M_A = 0;$



$-F\ b + d\ N_C - \mu_s\ N_C\ a = 0$

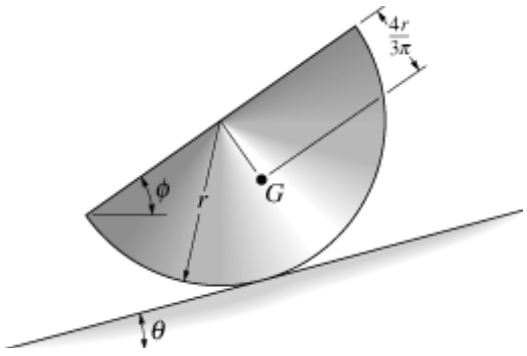
Solving we find $-\mu_s\ b + d - \mu_s\ a = 0$ $d = \mu_s\ (a + b)$ $d = 2.70\ \text{in}$

Problem 8-26

The homogeneous semicylinder has a mass *m* and mass center at *G*. Determine the largest angle θ of the inclined plane upon which it rests so that it does not slip down the plane. The coefficient of static friction between the plane and the cylinder is μ_s. Also, what is the angle ϕ for this case?

Given:

$\mu_s = 0.3$



Solution:

The semicylinder is a two-force member:

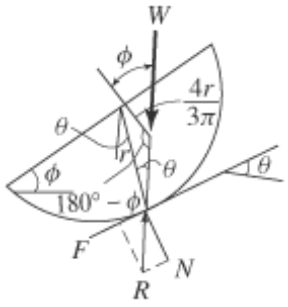
Since $F = \mu\ N$

$$\tan(\theta) = \frac{\mu_s\ N}{N} = \mu_S$$

$$\theta = \operatorname{atan}(\mu_s)$$

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$\theta = 16.7\,\text{deg}$



Law of sines

$$\frac{r}{\sin(180\ \text{deg} - \phi)} = \frac{\frac{4r}{3\pi}}{\sin(\theta)}$$

$$\phi = \operatorname{asin}\left(\frac{3\pi}{4}\sin(\theta)\right)$$

$\phi = 42.6\,\text{deg}$

Problem 8-27

A chain having a length L and weight W rests on a street for which the coefficient of static friction is μ_s. If a crane is used to hoist the chain, determine the force **P** it applies to the chain if the length of chain remaining on the ground begins to slip when the horizontal component is P_x. What length of chain remains on the ground?

Given:

$L = 20\ \text{ft}$

$W = 8\ \frac{\text{lb}}{\text{ft}}$

$\mu_S = 0.2$

$P_X = 10\ \text{lb}$



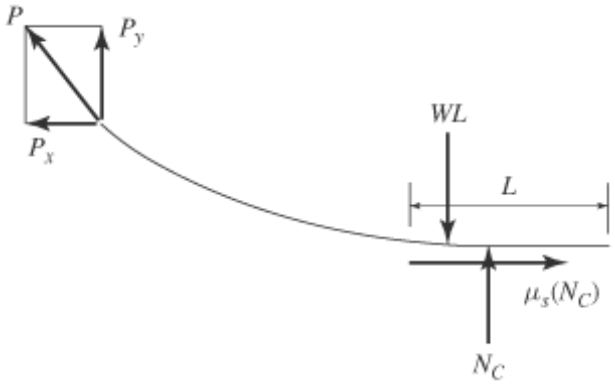
Solution:

$\Sigma F_x = 0; \quad -P_X + \mu_S\, N_C = 0$

$$N_C = \frac{P_X}{\mu_S}$$

$N_C = 50.00\,\text{lb}$

$\Sigma F_y = 0; \quad P_y - W\,L + N_C = 0$



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$$P_y = W\,L - N_C$$

$$P_y = 110.00\,\text{lb}$$

$$P = \sqrt{P_x^{\,2} + P_y^{\,2}}$$

$$P = 110\,\text{lb}$$

The length on the ground is supported by $N_C = 50.00\,\text{lb}$thus

$$L = \frac{N_C}{W}$$

$$L = 6.25\,\text{ft}$$

Problem 8-28

The fork lift has a weight W_1 and center of gravity at G. If the rear wheels are powered, whereas the front wheels are free to roll, determine the maximum number of crates, each of weight W_2 that the fork lift can push forward. The coefficient of static friction between the wheels and the ground is μ_s and between each crate and the ground is μ'_s.

Given:

$W_1 = 2400\ \text{lb}$

$W_2 = 300\ \text{lb}$

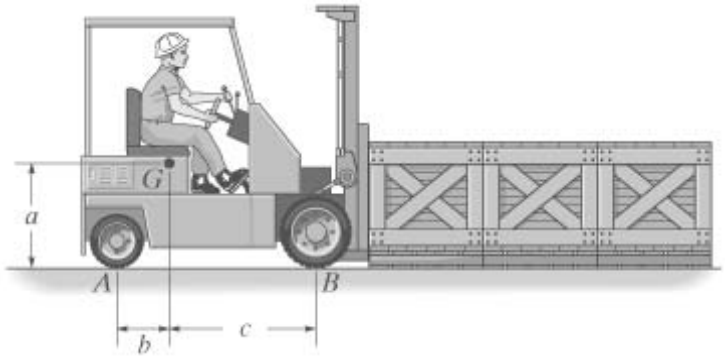
$\mu_s = 0.4$

$\mu'_s = 0.35$

$a = 2.5\ \text{ft}$

$b = 1.25\ \text{ft}$

$c = 3.50\ \text{ft}$



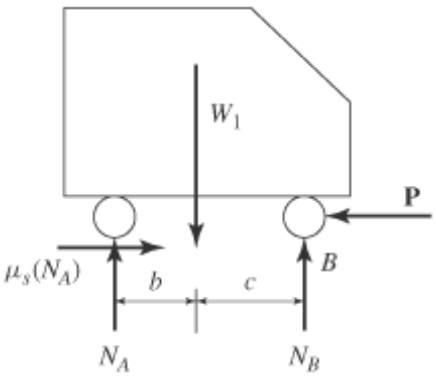
Solution:

Fork lift:

$\Sigma M_B = 0; \quad W_1\ c - N_A\ (b + c) = 0$

$$N_A = W_1\left(\frac{c}{b+c}\right) \qquad N_A = 1768.4\,\text{lb}$$

$\Sigma F_x = 0; \quad \mu_s\ N_A - P = 0$



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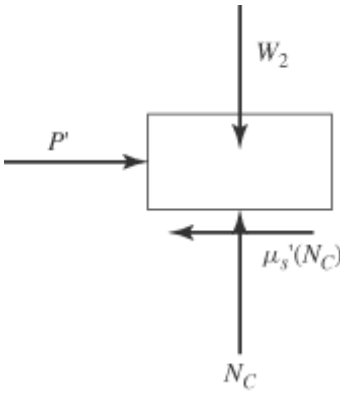
Crate: $P = \mu_s N_A$ $P = 707.37\,\text{lb}$

$\Sigma F_y = 0;$ $N_C - W_2 = 0$

$N_C = W_2$ $N_C = 300.00\,\text{lb}$

$\Sigma F_x = 0;$ $P' - \mu'_s N_C = 0$

$P' = \mu'_s N_C$ $P' = 105.00\,\text{lb}$



Thus $n = \dfrac{P}{P'}$ $n = 6.74$ $n = \text{floor}(n)$ $n = 6.00$

Problem 8-29

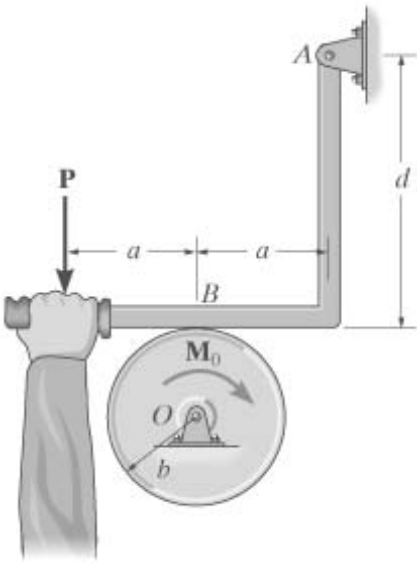
The brake is to be designed to be self locking, that is, it will not rotate when no load P is applied to it when the disk is subjected to a clockwise couple moment $\mathbf{M_O}$. Determine the distance d of the lever that will allow this to happen. The coefficient of static friction at B is μ_s.

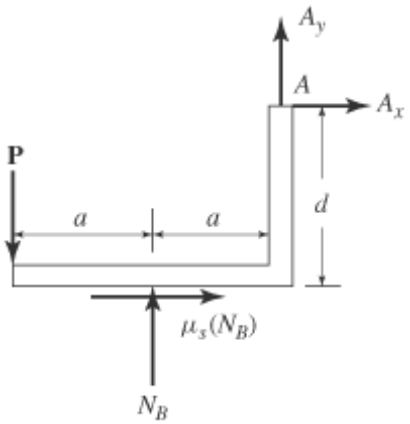
Given:

$a = 1.5$ ft

$b = 1$ ft

$\mu_s = 0.5$

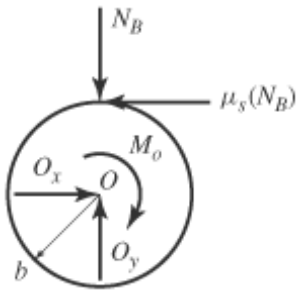




Solution:

$\Sigma M_0 = 0;$ $M_0 - \mu_s N_B\, b = 0$

$$N_B = \frac{M_0}{\mu_s\, b}$$



$\Sigma M_A = 0;$ $P\ 2\ a - N_B\ a + \mu_s\ N_B\ d = 0$

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$$P = 0$$

$$d = \frac{a}{\mu_s}$$

$$d = 3.00\ \text{ft}$$

Problem 8-30

The concrete pipe of weight W is being lowered from the truck bed when it is in the position shown. If the coefficient of static friction at the points of support A and B is μ_s determine where it begins to slip first: at A or B, or both at A and B.

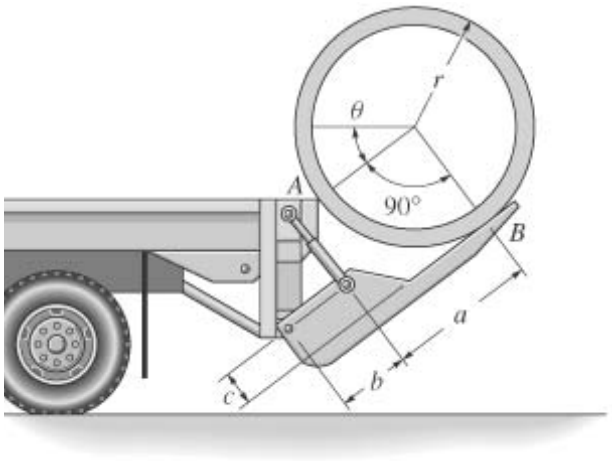
Given:

$W = 800$ lb $\quad a = 30$ in

$\mu_s = 0.4 \quad b = 18$ in

$\theta = 30$ deg $\quad c = 5$ in

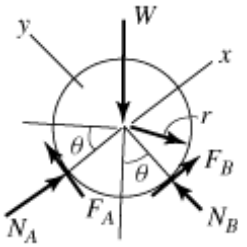
$r = 15$ in



Solution:

initial guesses are

$N_A = 10$ lb $\quad N_B = 10$ lb $\quad F_A = 10$ lb $\quad F_B = 10$ lb



Given *Assume slipping at A:*

$\Sigma F_x = 0; \quad N_A + F_B - W\sin(\theta) = 0$

$\Sigma F_y = 0; \quad F_A + N_B - W\cos(\theta) = 0$

$+\circlearrowleft \Sigma M_0 = 0; \quad F_B\, r - F_A\, r = 0$

$F_A = \mu_s\, N_A$

$$\begin{pmatrix} N_A \\ N_B \\ F_A \\ F_B \end{pmatrix} = \text{Find}(N_A, N_B, F_A, F_B) \qquad \begin{pmatrix} N_A \\ N_B \\ F_A \\ F_B \end{pmatrix} = \begin{pmatrix} 285.71 \\ 578.53 \\ 114.29 \\ 114.29 \end{pmatrix} \text{lb}$$

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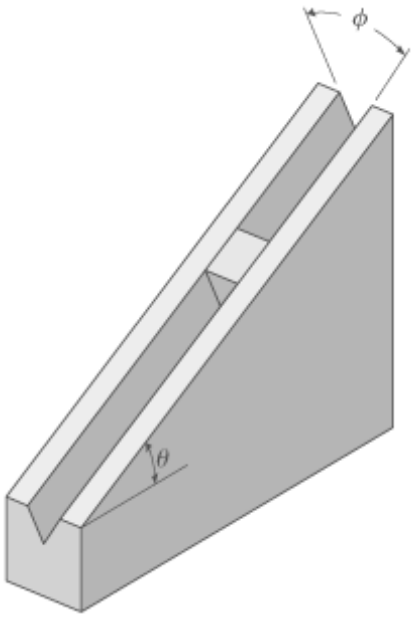
At B,

$$F_{Bmax} = \mu_S N_B$$

Since $F_B = 114.29\,\text{lb} < F_{Bmax} = 231.41\,\text{lb}$ then we conclude that slipping begins at A.

Problem 8-31

A wedge of mass M is placed in the grooved slot of an inclined plane. Determine the maximum angle θ for the incline without causing the wedge to slip. The coefficient of static friction between the wedge and the surfaces of contact is μ_s.



Given:

$M = 5$ kg

$\mu_S = 0.2$

$\phi = 60$ deg

$g = 9.81 \dfrac{\text{m}}{\text{s}^2}$

Solution:

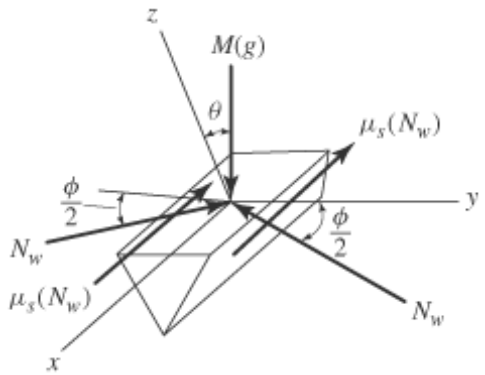
Initial guesses:

$N_W = 10$ N $\quad \theta = 10$ deg

Given

$\Sigma F_x = 0; \quad M\ g\ \sin(\theta) - 2\ \mu_S\ N_W = 0$

$\Sigma F_z = 0; \quad 2\ N_W \sin\left(\dfrac{\phi}{2}\right) - M\ g\ \cos(\theta) = 0$



Solving,

$$\begin{pmatrix} N_W \\ \theta \end{pmatrix} = \text{Find}(N_W, \theta)$$

$N_W = 45.5$ N

$\theta = 21.8\,\text{deg}$

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Problem 8-32

A roll of paper has a uniform weight W and is suspended from the wire hanger so that it rests against the wall. If the hanger has a negligible weight and the bearing at O can be considered frictionless, determine the force **P** needed to start turning the roll. The coefficient of static friction between the wall and the paper is μ_s.

Given:

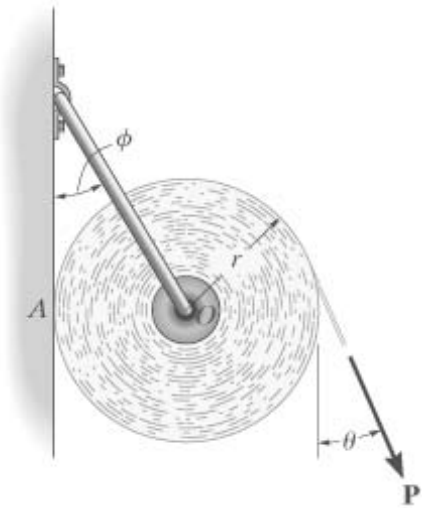
$W = 0.75$ lb

$\theta = 30$ deg

$\phi = 30$ deg

$\mu_s = 0.25$

$a = 3$ in



Solution:

Initial guesses:

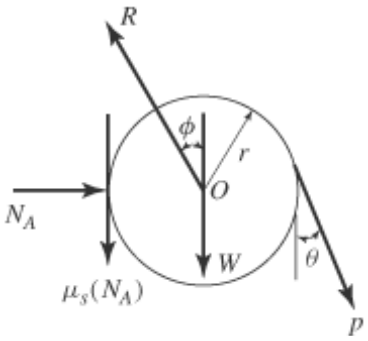
$R = 100$ lb $\quad N_A = 100$ lb $\quad P = 100$ lb

Given

$\Sigma F_x = 0; \quad N_A - R\sin(\phi) + P\sin(\theta) = 0$

$\Sigma F_y = 0; \quad R\cos(\phi) - W - P\cos(\theta) - \mu_s N_A = 0$

$\Sigma M_0 = 0; \quad \mu_s N_A a - P a = 0$



Solving for P,

$$\begin{pmatrix} R \\ N_A \\ P \end{pmatrix} = \text{Find}(R, N_A, P)$$

$R = 1.14$ lb $\quad N_A = 0.51$ lb $\quad P = 0.13$ lb

Problem 8-33

A roll of paper has a uniform weight W and is suspended from the wire hanger so that it rests against the wall. If the hanger has a negligible weight and the bearing at O can be considered frictionless, determine the minimum force **P** and the associated angle θ needed to start turning the roll. The coefficient of static friction between the wall and the paper is μ_s.

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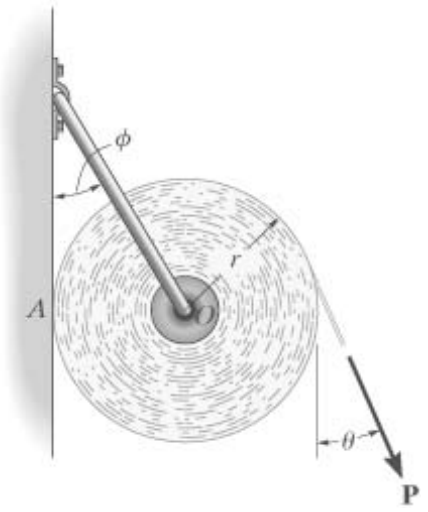
Given:

$W = 0.75$ lb

$\phi = 30$ deg

$\mu_S = 0.25$

$r = 3$ in

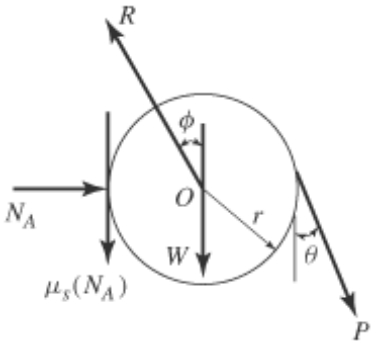


Solution:

$\Sigma F_x = 0;$ $\quad N_A - R\,\sin(\phi) + P\,\sin(\theta) = 0$

$\Sigma F_y = 0;$ $\quad R\,\cos(\phi) - W - P\,\cos(\theta) - \mu_S\,N_A = 0$

$\Sigma M_0 = 0;$ $\quad \mu_S\,N_A\,r - P\,r = 0$



Solving for P,

$$P = \frac{\mu_S\,W\,\sin(\phi)}{\cos(\phi) + \mu\,\sin(\theta - \phi) - \mu\,\sin(\phi)}$$

For minimum P we must have

$$\frac{dP}{d\theta} = \frac{-\mu_S^{\,2}\,W\,\sin(\phi)\cos(\theta - \phi)}{\left(\cos(\phi) + \mu_S\,\sin(\theta - \phi) - \mu_S\,\sin(\phi)\right)^2} = 0 \qquad \text{Implies} \qquad \cos(\theta - \phi) = 0$$

One answer is $\theta = \phi + 90$ deg $\qquad \theta = 120.00\,\text{deg}$

$$P = \frac{\mu_S\,W\sin(\phi)}{\cos(\phi) + \mu_S\,\sin(\theta - \phi) - \mu_S\,\sin(\phi)} \qquad P = 0.0946\,\text{lb}$$

Problem 8-34

The door brace *AB* is to be designed to prevent opening the door. If the brace forms a pin connection under the doorknob and the coefficient of static friction with the floor is μ_s determine the largest length *L* the brace can have to prevent the door from being opened. Neglect the weight of the brace.

Given:

$\mu_S = 0.5$

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$a = 3$ ft

Solution:

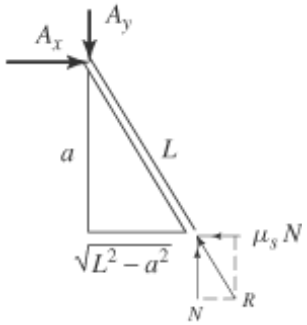
The brace is a two-force member.

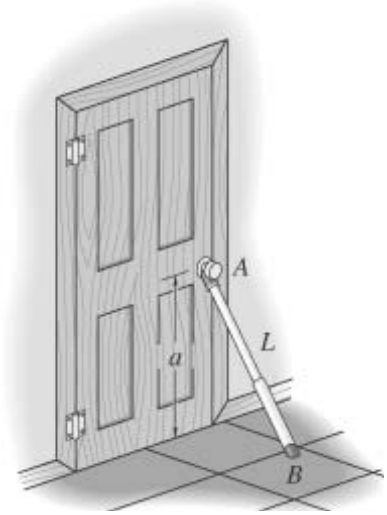
$$\frac{\mu_s N}{N} = \frac{\sqrt{L^2 - a^2}}{a}$$

$$\mu_s a = \sqrt{L^2 - a^2}$$

$$L = a\sqrt{1 + \mu_s^2}$$

$L = 3.35$ ft





Problem 8-35

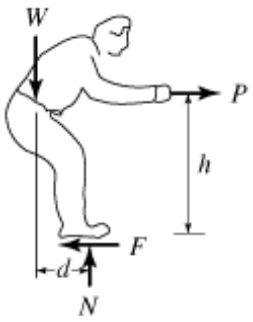
The man has a weight W, and the coefficient of static friction between his shoes and the floor is μ_s. Determine where he should position his center of gravity G at d in order to exert the maximum horizontal force on the door. What is this force?

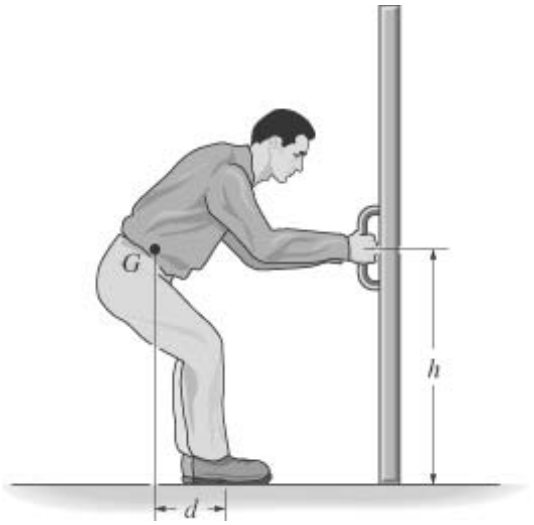
Given:

$W = 200$ lb

$\mu_s = 0.5$

$h = 3$ ft





Solution:

$N - W = 0 \quad N = W \quad N = 200.00$ lb

$F_{max} = \mu_s N \quad F_{max} = 100$ lb

$\xrightarrow{+} \Sigma F_x = 0; \quad P - F_{max} = 0$

$P = F_{max} \quad P = 100$ lb

$\circlearrowleft + \Sigma M_O = 0; \quad W\,d - P\,h = 0 \quad d = P\,\frac{h}{W} \quad d = 1.50$ ft

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Problem 8-36

In an effort to move the two crates, each of weight W, which are stacked on top of one another, the man pushes horizontally on them at the bottom of crate A as shown. Determine the smallest force **P** that must be applied in order to cause impending motion. Explain what happens. The coefficient of static friction between the crates is μ_s and between the bottom crate and the floor is μ_s'.

Given:

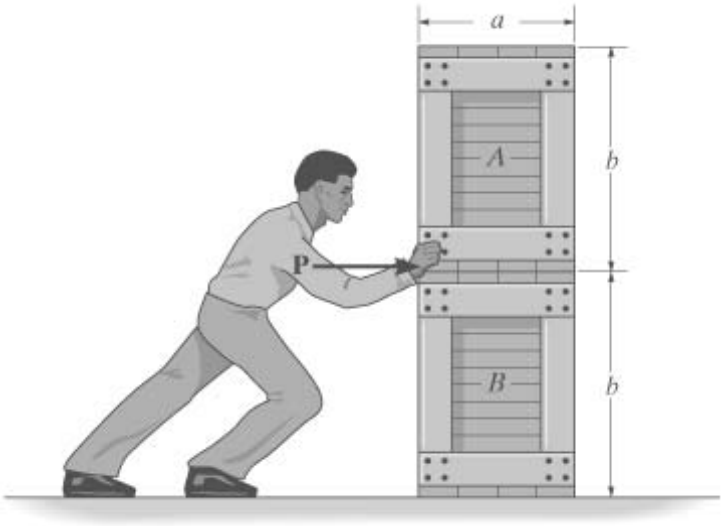
$W = 100$ lb

$\mu_s = 0.8$

$\mu'_s = 0.3$

$a = 2$ ft

$b = 3$ ft



Solution:

Assume crate A slips:

$\Sigma F_y = 0;\quad N_A - W = 0 \qquad N_A = W \qquad N_A = 100.00\,\text{lb}$

$\Sigma F_x = 0;\quad P - \mu_s N_A = 0 \qquad P_1 = \mu_s N_A \qquad P_1 = 80.00\,\text{lb}$

Assume crate B slips:

$\Sigma F_y = 0;\quad N_B - 2W = 0 \qquad N_B = 2W \qquad N_B = 200.00\,\text{lb}$

$\Sigma F_x = 0;\quad P - \mu'_s N_B = 0 \qquad P_2 = \mu'_s N_B \qquad P_2 = 60.00\,\text{lb}$

Assume both crates A and B tip:

$\Sigma M = 0;\quad 2W\left(\frac{a}{2}\right) - P\,b = 0 \qquad P_3 = W\left(\frac{a}{b}\right) \qquad P_3 = 66.7\,\text{lb}$

$P = \min(P_1, P_2, P_3) \qquad P = 60.00\,\text{lb}$

Problem 8-37

The man having a weight of W_1 pushes horizontally on the bottom of crate A, which is stacked on top of crate B. Each crate has a weight W_2. If the coefficient of static friction between each crate is μ_s and between the bottom crate, his shoes, and the floor is μ_s', determine if he can cause impending motion.

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Given:

$W_1 = 150$ lb

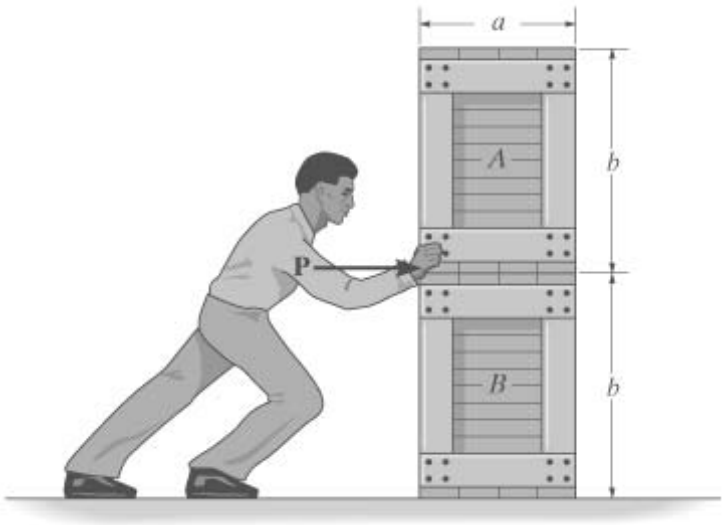
$W_2 = 100$ lb

$a = 2$ ft

$b = 3$ ft

$\mu_s = 0.8$

$\mu'_s = 0.3$



Assume crate A slips:

$\Sigma F_y = 0;\quad N_A - W_2 = 0 \qquad N_A = W_2 \qquad N_A = 100.00\,\text{lb}$

$\Sigma F_x = 0;\quad P - \mu_s N_A = 0 \qquad P_1 = \mu_s N_A \qquad P_1 = 80.00\,\text{lb}$

Assume crate B slips:

$\Sigma F_y = 0;\quad N_B - 2\,W_2 = 0 \qquad N_B = 2\,W_2 \qquad N_B = 200.00\,\text{lb}$

$\Sigma F_x = 0;\quad P - \mu'_s N_B = 0 \qquad P_2 = \mu'_s N_B \qquad P_2 = 60.00\,\text{lb}$

Assume both crates A and B tip:

$\Sigma M = 0;\quad 2\,W_2\left(\frac{a}{2}\right) - P\,b = 0 \qquad P_3 = W_2\,\frac{a}{b} \qquad P_3 = 66.7\,\text{lb}$

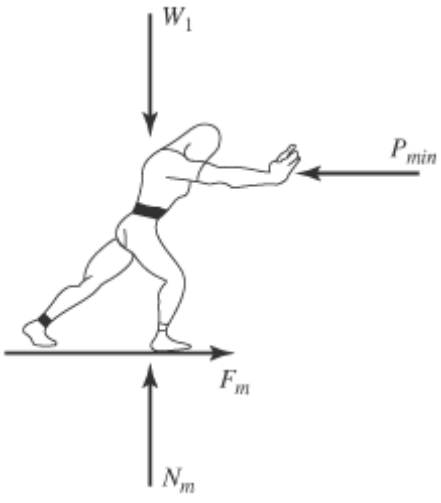
$P_{min} = \min\left(P_1, P_2, P_3\right) \qquad P_{min} = 60.00\,\text{lb}$

Now check to see if he can create this force

$\Sigma F_y = 0;\quad N_m - W_1 = 0 \qquad N_m = W_1$

$\Sigma F_x = 0;\quad F_m - P_{min} = 0 \qquad F_m = P_{min}$

$F_{mmax} = \mu'_s N_m$



Since $F_m = 60.00\,\text{lb} > F_{mmax} = 45.00\,\text{lb}$ then the man cannot create the motion.

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Problem 8-38

The crate has a weight W and a center of gravity at G. Determine the horizontal force **P** required to tow it.Also, determine the location of the resultant normal force measured from A.

Given:

$a = 3.5$ ft

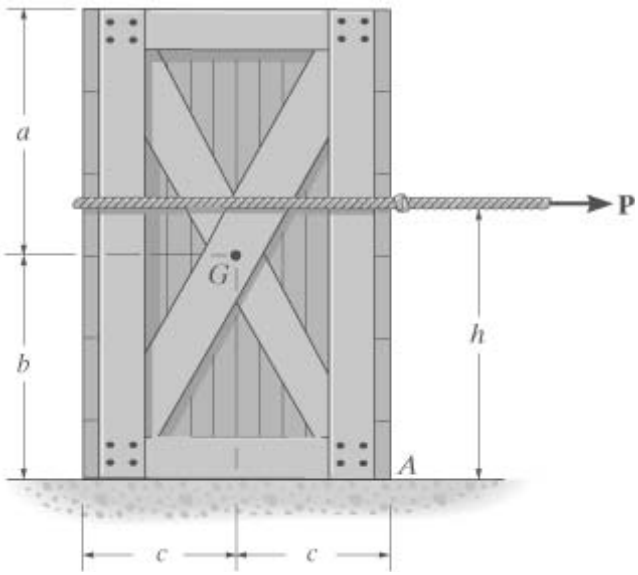
$b = 3$ ft

$c = 2$ ft

$W = 200$ lb

$h = 4$ ft

$\mu_s = 0.4$



Solution:

$\Sigma F_x = 0;$ $\quad P = F_O$

$\Sigma F_y = 0;$ $\quad N_O = W$

$N_O = 200.00$ lb

$\Sigma M_o = 0;$ $\quad -P\,h + W\,x = 0$

$F_O = \mu_s\, N_O$

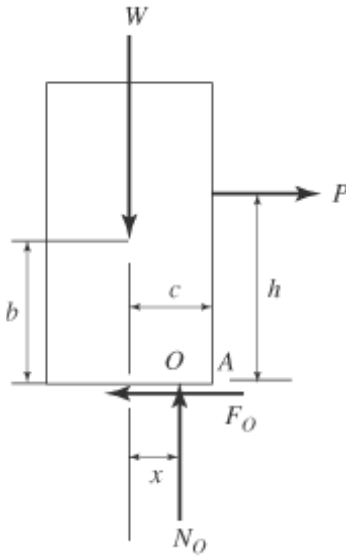
$F_O = 80.00$ lb

$P = F_O$

$P = 80.00$ lb

$x = P\,\dfrac{h}{W}$

$x = 1.60$ ft



The distance of N_O from A is

$c - x = 0.40$ ft

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Problem 8-39

The crate has a weight W and a center of gravity at G. Determine the height h of the tow rope so that the crate slips and tips at the same time. What horizontal force **P** is required to do this?

Given:

$a = 3.5$ ft

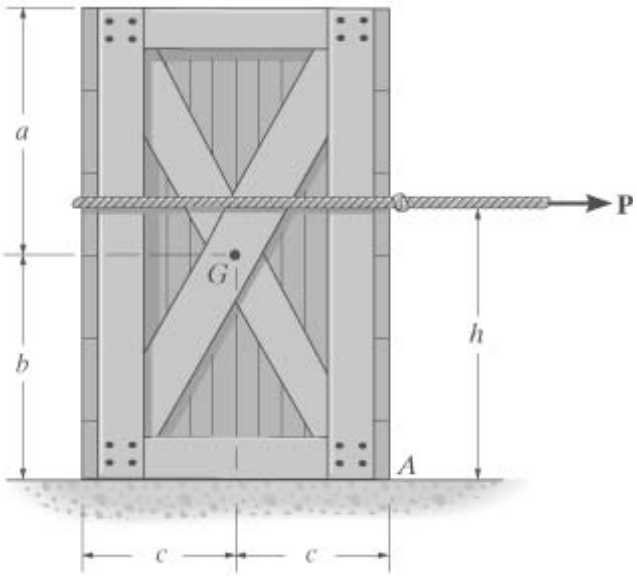
b = 3 ft

$c = 2$ ft

$W = 200$ lb

$h = 4$ ft

$\mu_s = 0.4$



Solution:

$\Sigma F_y = 0;$ $N_A = W$

$N_A = 200.00$ lb

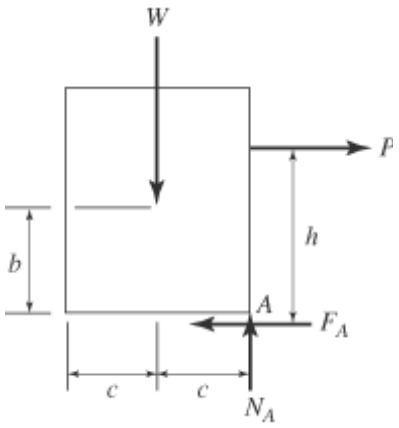
$\Sigma F_x = 0;$ $P = F_A$

$F_s = \mu_s N;$ $F_A = \mu_s W$

$F_A = 80.00$ lb $P = 80$ lb

$\Sigma M_A = 0;$ $-P\,h + W\,c = 0$

$h = W\,\dfrac{c}{P}$ $h = 5.00$ ft



Problem 8-40

Determine the smallest force the man must exert on the rope in order to move the crate of mass M. Also, what is the angle θ at this moment? The coefficient of static friction between the crate and the floor is μ_s.

Given:

$M = 80$ kg

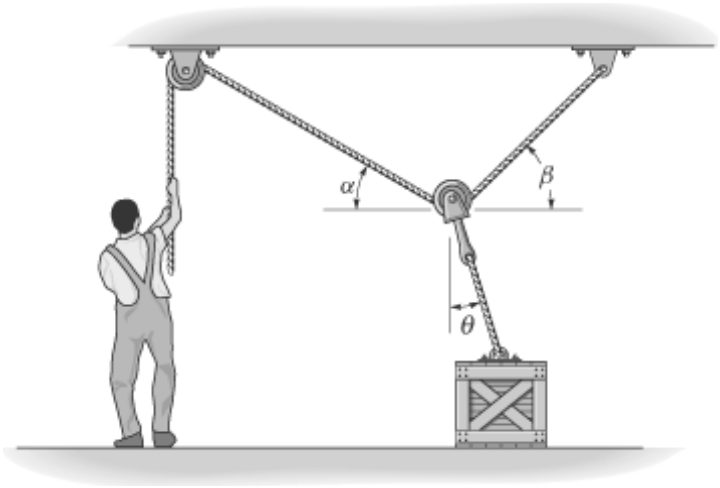
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$\mu_s = 0.3$

$\alpha = 30 \text{ deg}$

$\beta = 45 \text{ deg}$

$g = 9.81 \dfrac{\text{m}}{\text{s}^2}$



Solution : The initial guesses are $T_1 = 1 \text{ N}$ $N_C = 1 \text{ N}$ $\theta = 30 \text{ deg}$ $T = 1 \text{ N}$

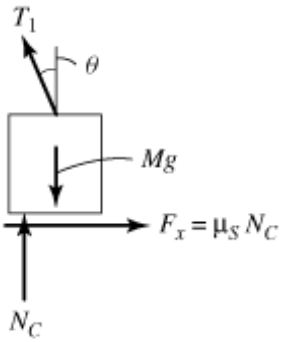
Given

$N_C - M\,g + T_1 \cos(\theta) = 0$

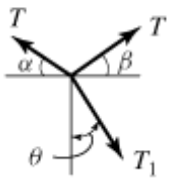
$\mu_s\, N_C - T_1 \sin(\theta) = 0$

$T \cos(\beta) - T \cos(\alpha) + T_1 \sin(\theta) = 0$

$T \sin(\beta) + T \sin(\alpha) - T_1 \cos(\theta) = 0$



$\begin{pmatrix} T \\ T_1 \\ N_C \\ \theta \end{pmatrix} = \text{Find}(T, T_1, N_C, \theta)$ $T = 451.86 \text{ N}$ $\theta = 7.50 \text{ deg}$



Problem 8-41

The symmetrical crab hook is used to lift packages by means of friction developed between the shoes *A*and *B* and a package. Determine the smallest coefficient of static friction at the shoes so that the package of weight *W* can be lifted.

Given:

$a = 1 \text{ ft}$

$b = 2 \text{ ft}$

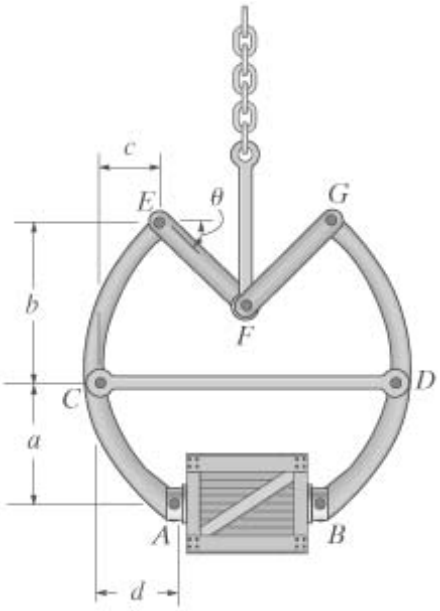
$c = 0.8 \text{ ft}$

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$d = 1$ ft

$\theta = 45$ deg

Solution:



From FBD (a)

$\Sigma F_y = 0; \quad W = 2\ F \sin(\theta)$

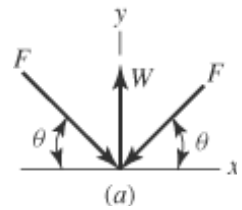
$$F = \frac{W}{2\sin(\theta)}$$

From FBD (b)

$\Sigma M_D = 0; \quad \mu\ N_B\ d + N_B\ a - F\ \sin(\theta)\ c - F\ \cos(\theta)\ b = 0$

$$\mu\ N_B\ d + N_B\ a - \frac{W}{2\ \sin(\theta)}\ \sin(\theta)\ c - \frac{W}{2\ \sin(\theta)}\cos(\theta)\ b = 0$$

$$N_B = \frac{W}{2\ (\mu\ d + a)}\left(c + \cot(\theta)\ b\right)$$



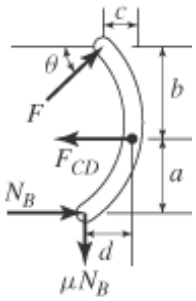
(a)

From FBD (c)

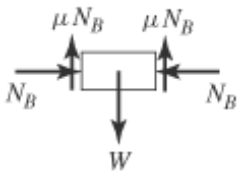
$\Sigma F_y = 0; \quad 2\ \mu\ N_B - W = 0$

$$2\ \mu\left[\frac{W}{2(\mu\ d + a)}\left(c + \cot(\theta)\ b\right)\right] - W = 0$$

$$\mu = \frac{a}{c + b\ \cot(\theta) - d} \qquad \mu = 0.56$$



(b)



(c)

Problem 8-42

The friction hook is made from a fixed frame which is shown colored and a cylinder of negligible weight. A piece of paper is placed between the smooth wall and the cylinder. Determine the smallest coefficient of static friction μ at all points of contact so that any weight W of paper p

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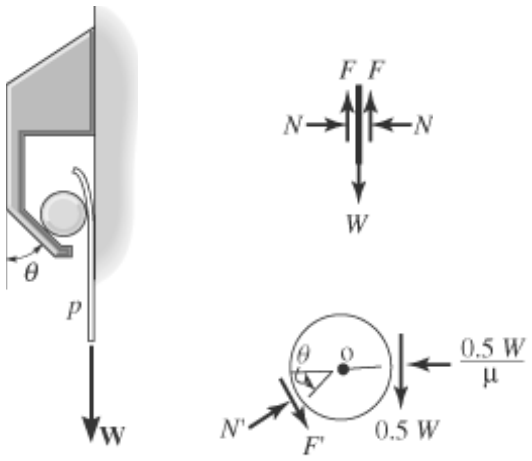
can be held.

Given:

$\theta = 20$ deg

Solution:

Paper:



$+\uparrow \Sigma F_y = 0;$

$F + F - W = 0 \qquad F = \frac{W}{2}$

$F = \mu N \qquad N = \frac{W}{2\mu}$

Cylinder:

$F' r - \left(\frac{W}{2}\right) r = 0 \qquad F' = \frac{W}{2}$

$N' - \left(\frac{W}{2}\right) \sin(\theta) - \frac{W}{2\mu} \cos(\theta) = 0 \qquad N' = \frac{W}{2}\left(\sin(\theta) + \frac{1}{\mu}\cos(\theta)\right)$

$F' = \mu N' \qquad \frac{W}{2} = \mu \left(\frac{W}{2}\right)\left(\sin(\theta) + \frac{1}{\mu}\cos(\theta)\right)$

$1 = \mu \sin(\theta) + \cos(\theta)$

$\mu = \frac{1 - \cos(\theta)}{\sin(\theta)}$

$\mu = 0.176$

Problem 8-43

The crate has a weight W_1 and a center of gravity at G. If the coefficient of static friction between the crate and the floor is μ_s, determine if the man of weight W_2 can push the crate to the left. The coefficient of static friction between his shoes and the floor is μ'_s. Assume the man exerts only a horizontal force on the crate.

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Given:

$W_1 = 300$ lb

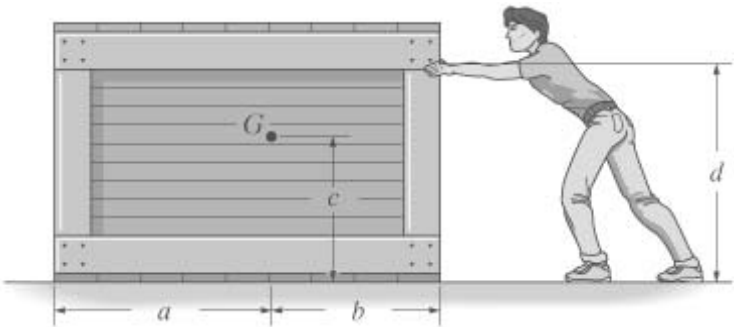
$W_2 = 200$ lb

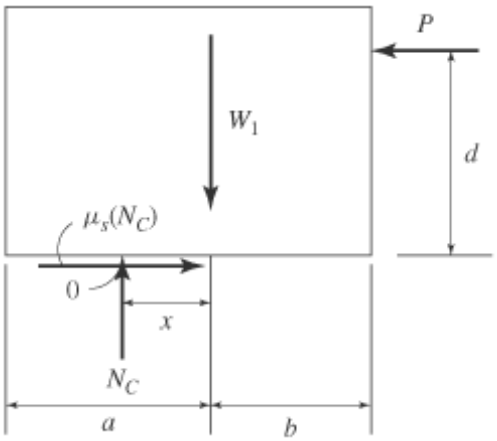
$\mu_s = 0.2$

$\mu'_s = 0.35$

$a = 4.5$ ft $\quad c = 3$ ft

$b = 3.5$ ft $\quad d = 4.5$ ft





Solution:

$\Sigma F_y = 0; \quad N_C - W_1 = 0 \quad N_C = W_1$

$\Sigma F_x = 0; \quad \mu_s N_C - P = 0 \quad P = \mu_s N_C$

$\Sigma M_O = 0; \quad -W_1 x + P d = 0 \quad x = \dfrac{P d}{W_1}$

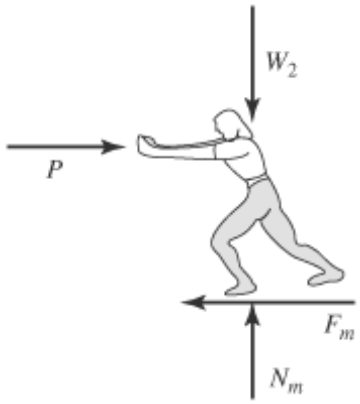
Since $x = 0.90\,\text{ft} < a = 4.50\,\text{ft}$ there will not be any tipping.

$\Sigma F_y = 0; \quad N_m - W_2 = 0 \quad N_m = W_2 \quad N_m = 200.00\,\text{lb}$

$\Sigma F_x = 0; \quad P - F_m = 0 \quad F_m = P \quad F_m = 60.00\,\text{lb}$

$F_{mmax} = \mu'_s N_m \quad F_{mmax} = 70.00\,\text{lb}$

Since $F_m = 60.00\,\text{lb} < F_{mmax} = 70.00\,\text{lb}$ then the man can push the crate.



Problem 8-44

The crate has a weight W_1 and a center of gravity at G. If the coefficient of static friction between the crate and the floor is μ_s, determine the smallest weight of the man so that he can push the crate to the left. The coefficient of static friction between his shoes and the floor is μ'_s. Assume the man exerts only a horizontal force on the crate.

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Given:

$W_1 = 300$ lb

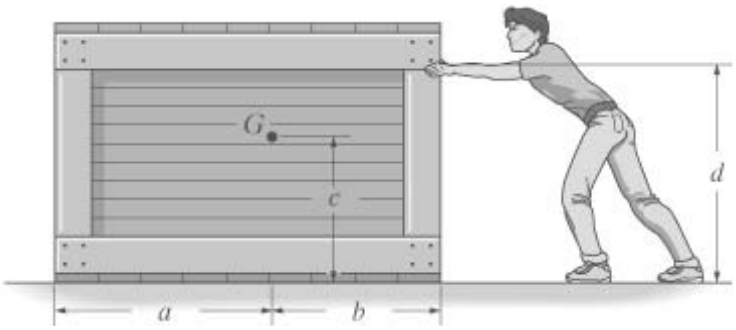
$W_2 = 200$ lb

$\mu_s = 0.2$

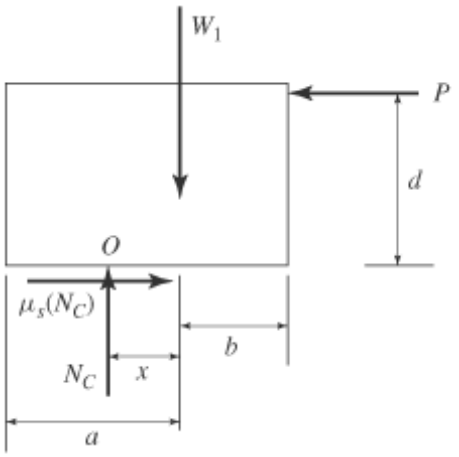
$\mu'_s = 0.35$

$a = 4.5$ ft $\quad c = 3$ ft

$b = 3.5$ ft $\quad d = 4.5$ ft



Solution:

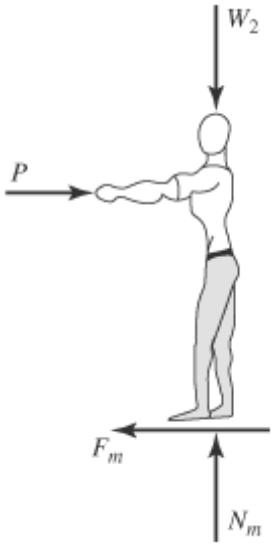


$\Sigma F_y = 0; \quad N_C - W_1 = 0 \quad N_C = W_1$

$\Sigma F_x = 0; \quad \mu_s N_C - P = 0 \quad P = \mu_s N_C$

$\Sigma M_O = 0; \quad -W_1 x + P d = 0 \quad x = \dfrac{P d}{W_1}$

Since $x = 0.90\,\text{ft} < a = 4.50\,\text{ft}$
there will not be any tipping.



$\Sigma F_x = 0; \quad P - F_m = 0 \quad F_m = P \quad F_m = 60.00\,\text{lb}$

$F_m = \mu'_s N_m \quad N_m = \dfrac{F_m}{\mu'_s} \quad N_m = 171.4\,\text{lb}$

$\Sigma F_y = 0; \quad N_m - W_2 = 0 \quad W_2 = N_m \quad W_2 = 171.4\,\text{lb}$

Problem 8-45

The wheel has weight W_A and rests on a surface for which the coefficient of friction is μ_B. A cord wrapped around the wheel is attached to the top of the homogeneous block of weight W_C.

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If the coefficient of static friction at D is μ_D determine the smallest vertical force that can be applied tangentially to the wheel which will cause motion to impend.

Given:

$W_A = 20$ lb

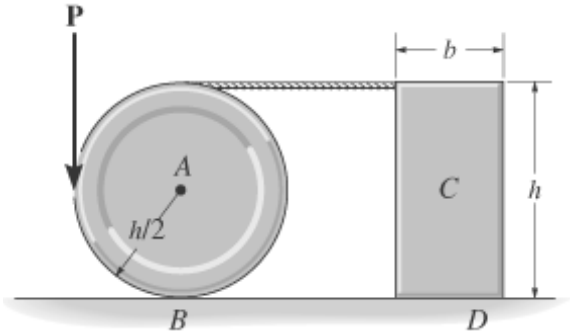
$\mu_B = 0.2$

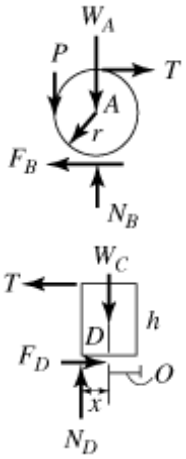
$W_C = 30$ lb

$\mu_D = 0.3$

$h = 3$ ft

$b = 1.5$ ft





Solution: Assume that slipping occurs at B, but that the block does not move.

Guesses $P = 1$ lb $N_B = 1$ lb $F_B = 1$ lb $T = 1$ lb $N_D = 1$ lb $F_D = 1$ lb

$x = 1$ ft

Given $N_B - W_A - P = 0$ $T - F_B = 0$ $\left(P - T - F_B\right)\frac{h}{2} = 0$

$F_B = \mu_B N_B$ $-T + F_D = 0$ $N_D - W_C = 0$

$T h - N_D x = 0$

$$\begin{pmatrix} P \\ N_B \\ F_B \\ T \\ N_D \\ F_D \\ x \end{pmatrix} = \text{Find}\left(P, N_B, F_B, T, N_D, F_D, x\right) \qquad \begin{pmatrix} P \\ N_B \\ F_B \\ T \\ N_D \\ F_D \end{pmatrix} = \begin{pmatrix} 13.33 \\ 33.33 \\ 6.67 \\ 6.67 \\ 30.00 \\ 6.67 \end{pmatrix} \text{lb} \qquad x = 0.67 \text{ ft}$$

Now checke the assumptions $F_{Dmax} = \mu_D N_D$

Since $F_D = 6.67$ lb $< F_{Dmax} = 9.00$ lb then the block does not slip

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Since $x = 0.67\,\text{ft} < \frac{b}{2} = 0.75\,\text{ft}$ then the block does not tip.

So our original assumption is correct.

$P = 13.33\,\text{lb}$

Problem 8-46

Determine the smallest couple moment which can be applied to the wheel of weight W_1 that will cause impending motion. The cord is attached to the block of weight W_2, and the coefficients of static friction are μ_B and μ_D.

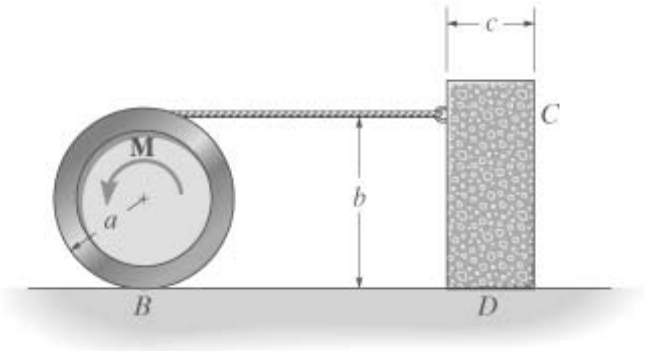
Given:

$W_1 = 20\ \text{lb}$ $a = 1.5\ \text{ft}$

$W_2 = 30\ \text{lb}$ $b = 3\ \text{ft}$

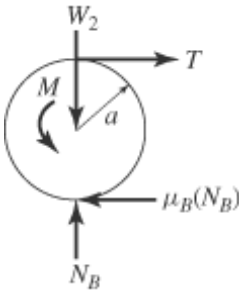
$\mu_B = 0.2$ $c = 1.5\ \text{ft}$

$\mu_D = 0.3$



Solution:

For the wheel : Assume slipping occurs,

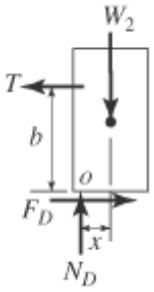


$\Sigma F_y = 0;$ $N_B - W_1 = 0$ $N_B = W_1$ $N_B = 20.00\,\text{lb}$

$\Sigma F_x = 0;$ $T - \mu_B N_B = 0$ $T = \mu_B N_B$ $T = 4.00\,\text{lb}$

$\Sigma M_B = 0;$ $M - T2a = 0$ $M = T2a$ $M = 12.00\,\text{lb}\cdot\text{ft}$

For block



$\Sigma F_y = 0;$ $N_D - W_2 = 0$ $N_D = W_2$ $N_D = 30.00\,\text{lb}$

$\Sigma F_x = 0;$ $F_D - T = 0$ $F_D = T$ $F_D = 4.00\,\text{lb}$

$\Sigma M_O = 0;$ $Tb - N_D x = 0$ $x = T\frac{b}{N_D}$ $x = 0.40\,\text{ft}$

$F_{Dmax} = \mu_D N_D$ $F_{Dmax} = 9.00\,\text{lb}$

Since $F_D = 4.00\,\text{lb} < F_{Dmax} = 9.00\,\text{lb}$ then the block doesn't slip

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Since $x = 0.40\,\text{ft} < \frac{c}{2} = 0.75\,\text{ft}$ then the block doesn't tip.

Thus neither slipping nor tipping occurs for the block, and our assumption and answer are correct.

Problem 8-47

The beam *AB* has a negligible mass and thickness and is subjected to a triangular distributed loading. It is supported at one end by a pin and at the other end by a post having a mass m_p and negligible thickness. Determine the minimum force **P** needed to move the post. The coefficients of static friction at *B* and *C* are μ_B and μ_C respectively.

Given:

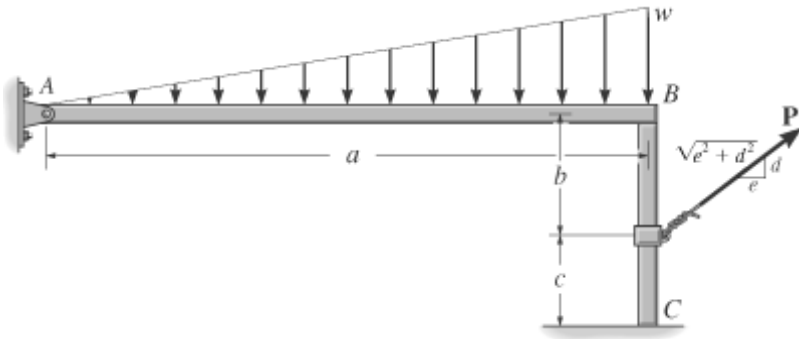
$m_p = 50\ \text{kg}$ $a = 2\ \text{m}$

$\mu_B = 0.4$ $b = 400\ \text{mm}$

$\mu_C = 0.2$ $c = 300\ \text{mm}$

$w = 800\ \frac{\text{N}}{\text{m}}$ $d = 3$

$g = 9.81\ \frac{\text{m}}{\text{s}^2}$ $e = 4$



Solution:

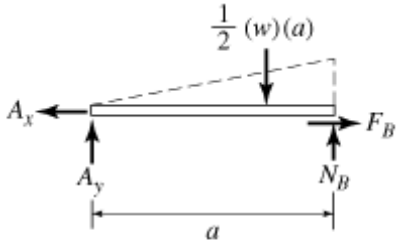
Member *AB*:

$\curvearrowleft + \Sigma M_A = 0; \quad -\left(\frac{1}{2} w a\right)\left(\frac{2a}{3}\right) + N_B a = 0 \quad N_B = \frac{1}{3} w a \quad N_B = 533.33\ \text{N}$

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Post:

Assume slipping occurs at C: $F_C = \mu_C N_C$

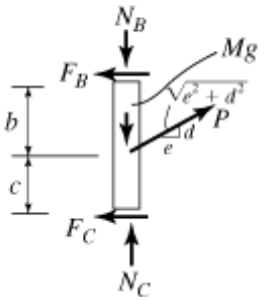


The initial guesses are

$P = 1$ N $\quad N_C = 1$ N $\quad F_B = 1$ N

Given

$$\frac{-e}{\sqrt{d^2 + e^2}} P\,c + F_B(b + c) = 0$$



$$\frac{e}{\sqrt{d^2 + e^2}} P - F_B - \mu_C N_C = 0$$

$$\frac{d}{\sqrt{e^2 + d^2}} P + N_C - N_B - m_p\, g = 0$$

$$\begin{pmatrix} P \\ N_C \\ F_B \end{pmatrix} = \text{Find}\left(P, N_C, F_B\right) \qquad P = 354.79 \text{ N}$$

Now check to see if the post slips at B. $F_{Bmax} = \mu_B N_B$

Since $F_B = 122$ N $<$ $F_{Bmax} = 213$ N then our assumptions are correct

$P = 355$ N

Problem 8-48

The beam AB has a negligible mass and thickness and is subjected to a triangular distributed loading. It is supported at one end by a pin and at the other end by a post having a mass m_p and negligible thickness. Determine the two coefficients of static friction at B and at C so that when the magnitude of the applied force is increased to **P** the post slips at both B and C simultaneously.

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Given:

$m_p = 50$ kg

$P = 150$ N

$w = 800 \dfrac{\text{N}}{\text{m}}$

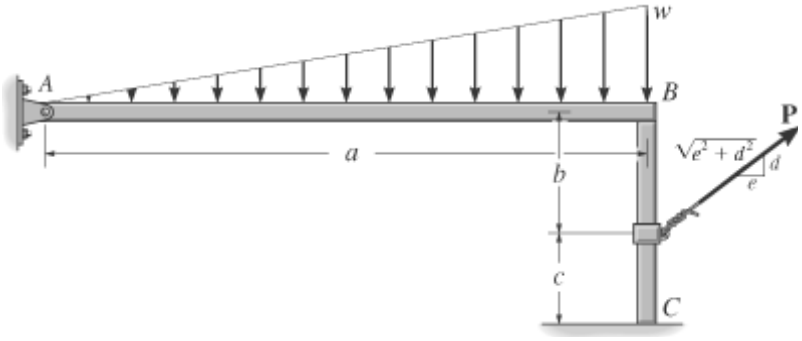
$a = 2$ m

$b = 400$ mm

$c = 300$ mm

$d = 3$

$e = 4$



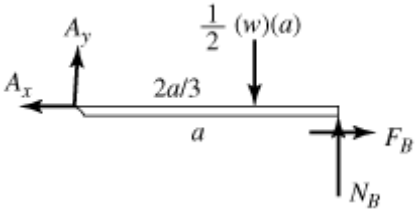
Solution: Member *AB*:

$$-\frac{1}{2} w a \frac{2a}{3} + N_B a = 0$$

$$N_B = \frac{1}{3} w a \qquad N_B = 533.33 \text{ N}$$

Post:

Guesses $N_C = 1$ N $\mu_B = 0.2$ $\mu_C = 0.2$

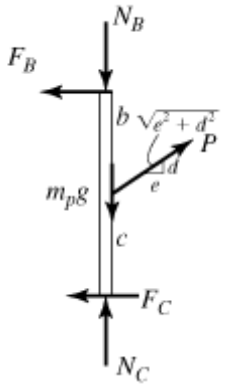


Given

$$N_C - N_B + P\left(\frac{d}{\sqrt{d^2 + e^2}}\right) - m_p g = 0$$

$$\left(\frac{e}{\sqrt{d^2 + e^2}}\right) P - \mu_C N_C - \mu_B N_B = 0$$

$$\left(\frac{-e}{\sqrt{d^2 + e^2}}\right) P c + \mu_B N_B (b + c) = 0$$

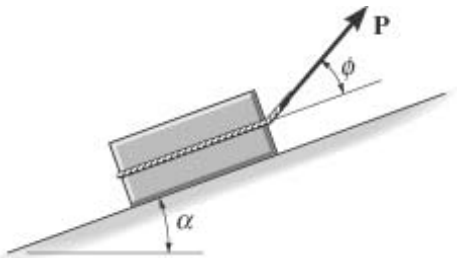


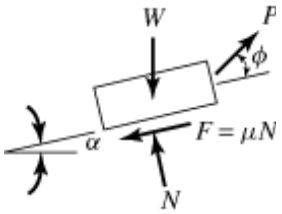
$$\begin{pmatrix} N_C \\ \mu_B \\ \mu_C \end{pmatrix} = \text{Find}\left(N_C, \mu_B, \mu_C\right) \qquad \begin{pmatrix} \mu_B \\ \mu_C \end{pmatrix} = \begin{pmatrix} 0.0964 \\ 0.0734 \end{pmatrix}$$

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Problem 8-49

The block of weight W is being pulled up the inclined plane of slope α using a force **P**. If **P** acts at the angle ϕ as shown, show that for slipping to occur, $P = W \sin(\alpha + \theta)/ \cos(\phi - \theta)$ where θ is the angle of friction; $\theta = \tan^{-1} \mu$..





Solution: Let $\mu = \tan(\theta)$

$\Sigma F_x = 0;$ $P\cos(\phi) - W\sin(\alpha) - \mu N = 0$

$\Sigma F_y = 0;$ $N - W\cos(\alpha) + P\sin(\phi) = 0$

$$P\cos(\phi) - W\sin(\alpha) - \mu\left(W\cos(\alpha) - P\sin(\phi)\right) = 0$$

$$P = W\left(\frac{\sin(\alpha) + \mu\cos(\alpha)}{\cos(\phi) + \mu\sin(\phi)}\right) = W\left(\frac{\sin(\alpha) + \tan(\theta)\cos(\alpha)}{\cos(\phi) + \tan(\theta)\sin(\phi)}\right)$$

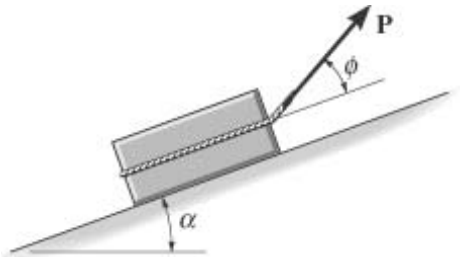
$$P = W\left(\frac{\sin(\alpha)\cos(\theta) + \sin(\theta)\cos(\alpha)}{\cos(\phi)\cos(\theta) + \sin(\theta)\sin(\phi)}\right)$$

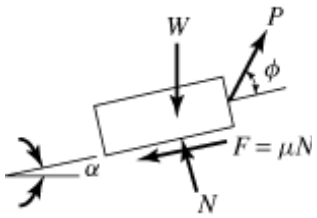
$$P = W\left(\frac{\sin(\alpha + \theta)}{\cos(\phi - \theta)}\right) \qquad \text{(QED)}$$

Problem 8-50

Determine the angle ϕ at which **P** should act on the block so that the magnitude of **P** is as small as possible to begin pulling the block up the incline. What is the corresponding value of P? The block has weight W and the slope α is known.

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Solution: Let $\mu = \tan(\theta)$

$\Sigma F_x = 0;$ $\quad P\cos(\phi) - W\sin(\alpha) - \mu N = 0$

$\Sigma F_y = 0;$ $\quad N - W\cos(\alpha) + P\sin(\phi) = 0$

$$P\cos(\phi) - W\sin(\alpha) - \mu\left(W\cos(\alpha) - P\sin(\phi)\right) = 0$$

$$P = W\left(\frac{\sin(\alpha) + \mu\cos(\alpha)}{\cos(\phi) + \mu\sin(\phi)}\right) = W\left(\frac{\sin(\alpha) + \tan(\theta)\cos(\alpha)}{\cos(\phi) + \tan(\theta)\sin(\phi)}\right)$$

$$P = W\left(\frac{\sin(\alpha)\cos(\theta) + \sin(\theta)\cos(\alpha)}{\cos(\phi)\cos(\theta) + \sin(\theta)\sin(\phi)}\right) = W\frac{\sin(\alpha + \theta)}{\cos(\phi - \theta)}$$

$$\frac{dP}{d\phi} = W\left[\frac{\sin(\alpha + \theta)\sin(\phi - \theta)}{\cos^2(\phi - \theta)}\right] = 0$$

$$\sin(\alpha + \theta)\sin(\phi - \theta) = 0$$

$\sin(\phi - \theta) = 0$ $\quad \phi = \theta$ $\quad P = W\sin(\alpha + \phi)$

Problem 8-51

Two blocks *A* and *B*, each having a mass *M*, are connected by the linkage shown. If the coefficient of static friction at the contacting surfaces is μ_s determine the largest vertical force **P** that may be applied to pin *C* of the linkage without causing the blocks to move. Neglect the weight of the links.

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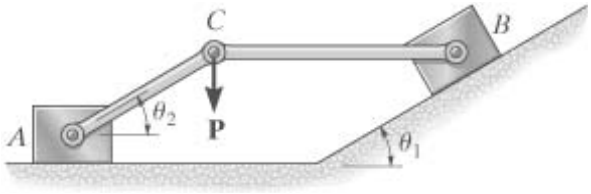
Given:

$M = 6$ kg

$\mu_s = 0.5$

$\theta_1 = 30$ deg

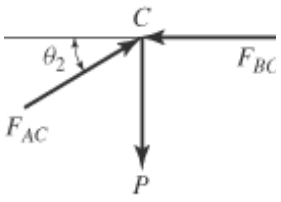
$\theta_2 = 30$ deg



Solution:

Guesses $N_A = 1$ N $F_A = 1$ N $F_{AC} = 1$ N

$P = 1$ N $N_B = 1$ N $F_B = 1$ N $F_{BC} = 1$ N



Assume that A slips first

Given

$F_{AC}\cos(\theta_2) - F_{BC} = 0$

$F_{AC}\sin(\theta_2) - P = 0$

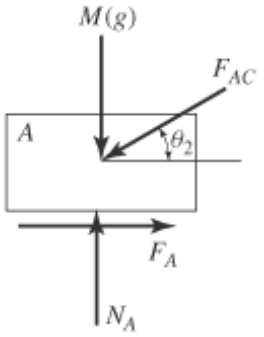
$N_A - Mg - F_{AC}\sin(\theta_2) = 0$

$F_A - F_{AC}\cos(\theta_2) = 0$

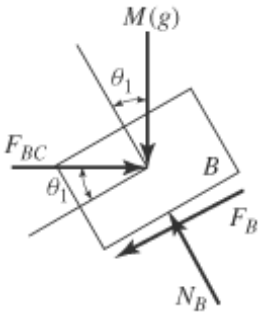
$F_{BC}\cos(\theta_1) - Mg\sin(\theta_1) - F_B = 0$

$-F_{BC}\sin(\theta_1) - Mg\cos(\theta_1) + N_B = 0$

$F_A = \mu_s N_A$



$$\begin{pmatrix} P_1 \\ N_A \\ F_A \\ N_B \\ F_B \\ F_{AC} \\ F_{BC} \end{pmatrix} = \text{Find}(P, N_A, F_A, N_B, F_B, F_{AC}, F_{BC}) \qquad P_1 = 23.9 \text{ N}$$



Assume that B slips first

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Given

$$F_{AC}\cos(\theta_2) - F_{BC} = 0$$

$$F_{AC}\sin(\theta_2) - P = 0$$

$$N_A - M g - F_{AC}\sin(\theta_2) = 0$$

$$F_A - F_{AC}\cos(\theta_2) = 0$$

$$F_{BC}\cos(\theta_1) - M g\sin(\theta_1) - F_B = 0$$

$$-F_{BC}\sin(\theta_1) - M g\cos(\theta_1) + N_B = 0$$

$$F_B = \mu_s N_B$$

$$\begin{pmatrix} P_2 \\ N_A \\ F_A \\ N_B \\ F_B \\ F_{AC} \\ F_{BC} \end{pmatrix} = \text{Find}(P, N_A, F_A, N_B, F_B, F_{AC}, F_{BC}) \qquad P_2 = 51.5 \text{ N}$$

$$P = \min(P_1, P_2) \qquad P = 23.9 \text{ N}$$

Problem 8-52

Block C has a mass m_c and is confined between two walls by smooth rollers. If the block rests on top of the spool of mass m_s, determine the minimum cable force **P** needed to move the spool. The cable is wrapped around the spool's inner core. The coefficients of static friction at μ_A and μ_B.

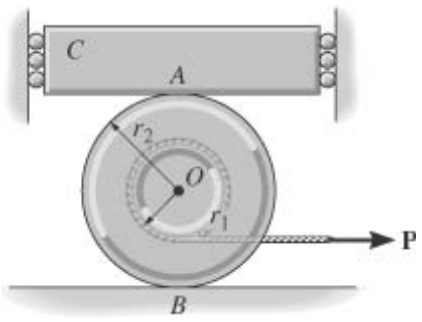
Given:

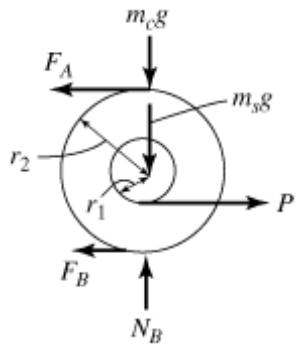
$$g = 9.81 \frac{\text{m}}{\text{s}^2}$$

$$m_c = 50 \text{ kg}$$

$$m_s = 40 \text{ kg}$$

$$\mu_A = 0.3$$





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$\mu_B = 0.6$ $r_1 = 0.2$ m $r_2 = 0.4$ m

Solution: Assume that the spool slips at A but not at B.

The initial guesses are $F_B = 2$ N $P = 3$ N $N_B = 1$ N

Given $P - F_B - \mu_A m_c g = 0$

$N_B - m_s g - m_c g = 0$

$P(r_1 + r_2) - F_B 2r_2 = 0$

$$\begin{pmatrix} F_B \\ P \\ N_B \end{pmatrix} = \text{Find}(F_B, P, N_B) \qquad \begin{pmatrix} F_B \\ P \\ N_B \end{pmatrix} = \begin{pmatrix} 441.45 \\ 588.60 \\ 882.90 \end{pmatrix} \text{N}$$

Now check the no slip assumption at B $F_{Bmax} = \mu_B N_B$ $F_{Bmax} = 529.74$ N

Since $F_B = 441$ N $<$ $F_{Bmax} = 530$ N then our assumptions are correct.

$P = 589$ N

Problem 8-53

A board of weight W_1 is placed across the channel and a boy of weight W_2 attempts to walk across. If the coefficient of static friction at A and B μ_s, determine if he can make the crossing; and if not, how far will he get from A before the board slips?

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Given:

$W_1 = 50$ lb

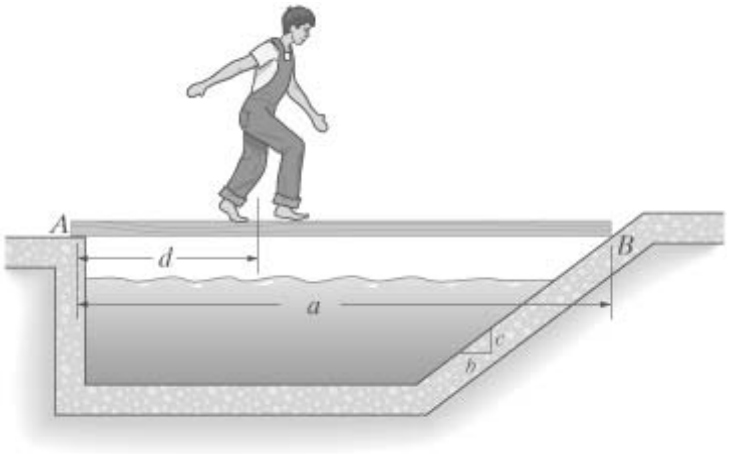
$W_2 = 100$ lb

$\mu_s = 0.4$

$a = 10$ ft

$b = 4$

$c = 3$



Solution:

Initial guesses: $N_A = 10$ lb $N_B = 20$ lb $d = 3$ ft

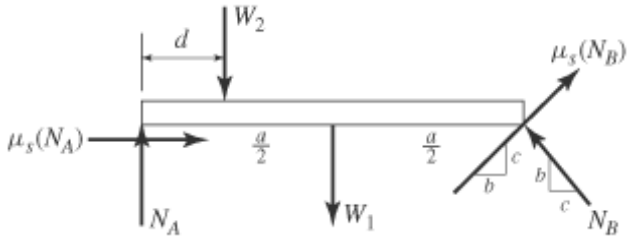
Given

$$\Sigma F_x = 0; \qquad \mu_s N_A + \mu_s N_B\left(\frac{b}{\sqrt{b^2 + c^2}}\right) - N_B\left(\frac{c}{\sqrt{b^2 + c^2}}\right) = 0$$

$$\Sigma F_y = 0; \qquad N_A - W_2 - W_1 + N_B\left(\frac{b}{\sqrt{b^2 + c^2}}\right) + \mu_s N_B\left(\frac{c}{\sqrt{b^2 + c^2}}\right) = 0$$

$$\Sigma M_B = 0; \qquad W_1\left(\frac{a}{2}\right) + W_2(a - d) - N_A a = 0$$

$$\begin{pmatrix} N_A \\ N_B \\ d \end{pmatrix} = \text{Find}\left(N_A, N_B, d\right)$$



$$\begin{pmatrix} N_A \\ N_B \end{pmatrix} = \begin{pmatrix} 60.34 \\ 86.21 \end{pmatrix} \text{lb} \qquad d = 6.47 \text{ ft}$$

Since $d = 6.47$ ft $< a = 10.00$ ft then the board will slip

---

Problem 8-54

Determine the minimum force P needed to push the tube E up the incline. The tube has a mass of M_1 and the roller D has a mass of M_2. The force acts parallel to the plane, and the coefficients of static friction at the contacting surfaces are μ_A, μ_B and μ_C. Each cylinder has a radius of r.

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Given:

$M_1 = 75$ kg

$M_2 = 100$ kg

$\mu_A = 0.3$

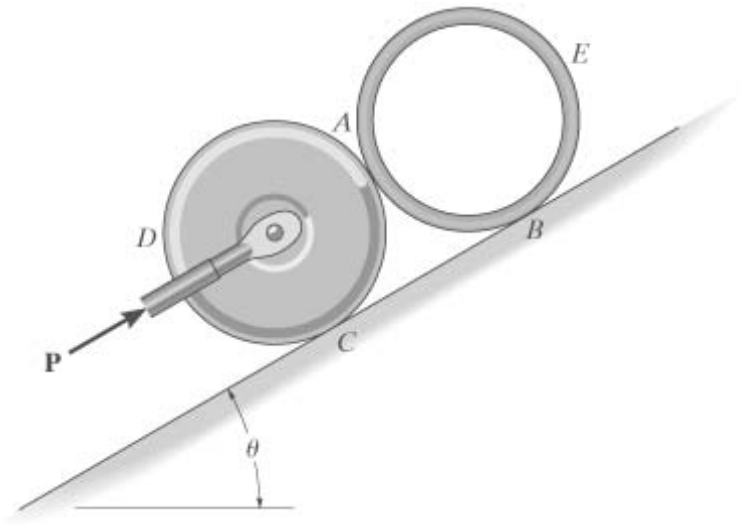
$\mu_B = 0.25$

$\mu_C = 0.4$

$\theta = 30$ deg

$r = 150$ mm

$g = 9.81 \dfrac{\text{m}}{\text{s}^2}$



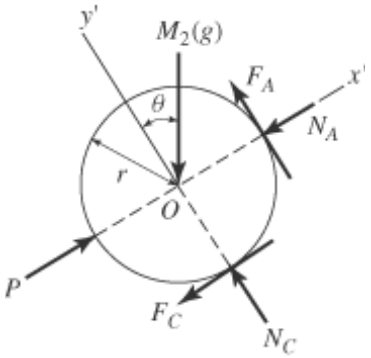
Solution:

Initial guesses:

$N_A = 100$ N $\quad F_A = 10$ N $\quad P = 100$ N

$N_B = 200$ N $\quad F_B = 20$ N

$N_C = 300$ N $\quad F_C = 30$ N



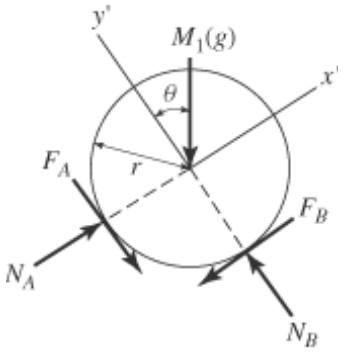
Given

For roller D

$\Sigma F_{x'} = 0; \quad P - N_A - F_C - M_2 g \sin(\theta) = 0$

$\Sigma F_{y'} = 0; \quad N_C + F_A - M_2 g \cos(\theta) = 0$

$\Sigma M_{0'} = 0; \quad F_A r - F_C r = 0$



For tube E

$\Sigma F_{x'} = 0; \quad N_A - F_B - M_1 g \sin(\theta) = 0$

$\Sigma F_{y'} = 0; \quad N_B - F_A - M_1 g \cos(\theta) = 0$

$\Sigma M_{0'} = 0; \quad F_A r - F_B r = 0$

Assuming slipping occurs only at A. $\quad F_A = \mu_A N_A$

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$$\begin{pmatrix} N_A \\ N_B \\ N_C \\ F_A \\ F_B \\ F_C \\ P \end{pmatrix} = \text{Find}\left(N_A, N_B, N_C, F_A, F_B, F_C, P\right) \qquad \begin{pmatrix} N_A \\ N_B \\ N_C \\ F_A \\ F_B \\ F_C \\ P \end{pmatrix} = \begin{pmatrix} 526 \\ 795 \\ 692 \\ 158 \\ 158 \\ 158 \\ 1174 \end{pmatrix} \text{ N}$$

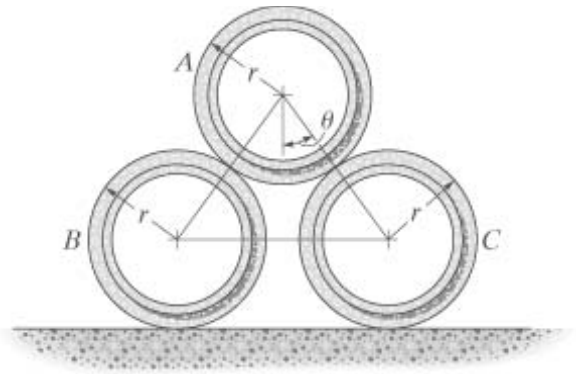
$$F_{Bmax} = \mu_B N_B \qquad F_{Cmax} = \mu_C N_C \qquad \begin{pmatrix} F_{Bmax} \\ F_{Cmax} \end{pmatrix} = \begin{pmatrix} 199 \\ 277 \end{pmatrix} \text{ N}$$

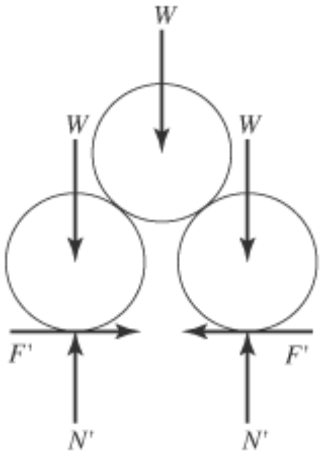
Since $F_B = 158$ N $< F_{Bmax} = 199$ N and $F_C = 158$ N $< F_{Cmax} = 277$ N then our assumption is correct

$P = 1174$ N

Problem 8-55

The concrete pipe at *A* rests on top of *B* and *C*. If the coefficient of static friction between the pipes is μ_s and at the ground μ'_s, determine their smallest values so that the pipes will not slip. Each pipe has a radius *r* and weight *W*, and the angle between the centers as indicated is θ.

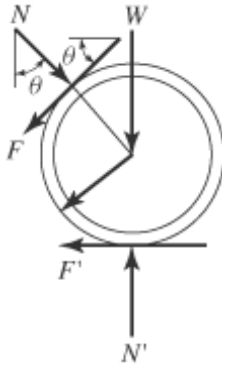




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Solution:

Total System :



$\Sigma F_y = 0; \qquad 2N' - 3W = 0 \qquad N' = \frac{3}{2}W$

Pipe C:

$\Sigma F_x = 0; \qquad -F\cos(\theta) + N\sin(\theta) - F' = 0$

$\Sigma F_y = 0; \qquad N' - W - N\cos(\theta) - F\sin(\theta) = 0$

$\Sigma M_O = 0; \qquad F\,r - F'\,r = 0$

Solving, $\qquad F = F' = \frac{1}{2}\left(\frac{W\sin(\theta)}{\cos(\theta)+1}\right) \qquad N = \frac{W}{2}$

For Pipe A:

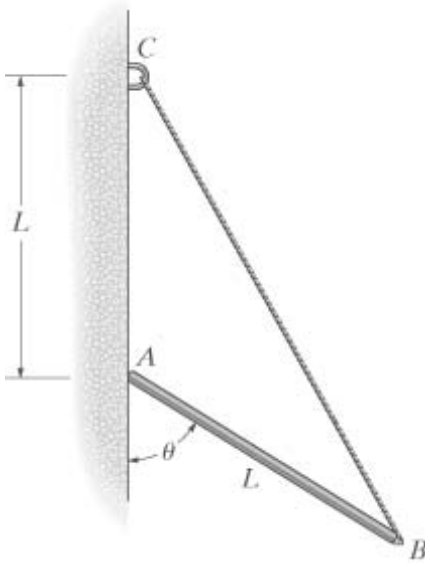
$$\mu_s N \geq \frac{1}{2}\left(\frac{W\sin(\theta)}{\cos(\theta)+1}\right) \qquad \mu_s \geq \frac{\sin(\theta)}{\cos(\theta)+1}$$

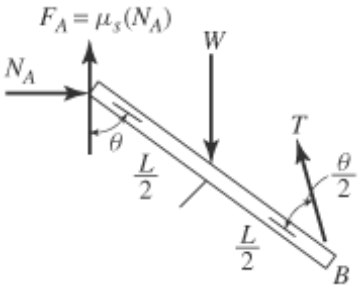
For Pipe C:

$$\mu'_s N' \geq \frac{1}{2}\left(\frac{W\sin(\theta)}{\cos(\theta)+1}\right) \qquad \mu'_s \geq \frac{1}{3}\frac{\sin(\theta)}{\cos(\theta)+1}$$

Problem 8-56

The uniform pole has a weight W and length L. Its end B is tied to a supporting cord, and end A is placed against the wall, for which the coefficient of static friction is μ_s. Determine the largest angle θ at which the pole can be placed without slipping.





Solution:

$\Sigma F_x = 0; \qquad N_A - T\sin\left(\frac{\theta}{2}\right) = 0$

$\Sigma F_y = 0; \qquad \mu_s N_A - W + T\cos\left(\frac{\theta}{2}\right) = 0$

$\Sigma M_B = 0; \qquad -N_A L\cos(\theta) - \mu_s N_A L\sin(\theta) + W\left(\frac{L}{2}\right)\sin(\theta) = 0$

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Solving we find

$$N_A = T\sin\left(\frac{\theta}{2}\right)$$

$$\mu_s T\sin\left(\frac{\theta}{2}\right) - W + T\cos\left(\frac{\theta}{2}\right) = 0 \qquad W = T\left(\cos\left(\frac{\theta}{2}\right) + \mu_s\sin\left(\frac{\theta}{2}\right)\right)$$

$$T\sin\left(\frac{\theta}{2}\right)\left(L\cos(\theta) + \mu_s L\sin(\theta)\right) = T\left(\cos\left(\frac{\theta}{2}\right) + \mu_s\sin\left(\frac{\theta}{2}\right)\right)\frac{L}{2}\sin(\theta)$$

$$\sin\left(\frac{\theta}{2}\right)\left(\cos(\theta) + \mu_s\sin(\theta)\right) = \frac{1}{2}\sin(\theta)\left(\cos\left(\frac{\theta}{2}\right) + \mu_s\sin\left(\frac{\theta}{2}\right)\right)$$

$$\frac{\mu_s}{2}\sin\left(\frac{\theta}{2}\right)\sin(\theta) = \frac{1}{2}\sin(\theta)\cos\left(\frac{\theta}{2}\right) - \sin\left(\frac{\theta}{2}\right)\cos(\theta)$$

$$\mu_s = \frac{\sin(\theta)\cos\left(\frac{\theta}{2}\right) - 2\sin\left(\frac{\theta}{2}\right)\cos(\theta)}{\sin\left(\frac{\theta}{2}\right)\sin(\theta)}$$

$$\mu_s = \frac{\cos\left(\frac{\theta}{2}\right)}{\sin\left(\frac{\theta}{2}\right)} - \frac{2\left(\cos\left(\frac{\theta}{2}\right)^2 - \sin\left(\frac{\theta}{2}\right)^2\right)}{2\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)} = \cot\left(\frac{\theta}{2}\right) - \cos\left(\frac{\theta}{2}\right) + \tan\left(\frac{\theta}{2}\right)$$

$$\mu_s = \tan\left(\frac{\theta}{2}\right) \qquad \theta = 2\,\mathrm{atan}(\mu)$$

Problem 8-57

The carpenter slowly pushes the uniform board horizontally over the top of the saw horse. The board has a uniform weight density γ and the saw horse has a weight W and a center of gravity at G. Determine if the saw horse will stay in position, slip, or tip if the board is pushed forward at the given distance d. The coefficients of static friction are shown in the figure.

Given:

$$\gamma = 3\ \frac{\mathrm{lb}}{\mathrm{ft}}$$

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$L = 18$ ft

$W = 15$ lb

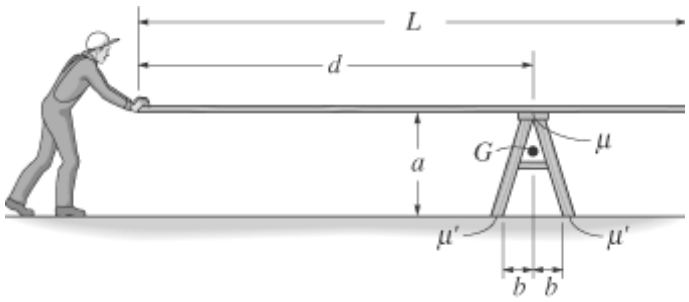
$a = 3$ ft

$b = 1$ ft

$\mu = 0.5$

$\mu' = 0.3$

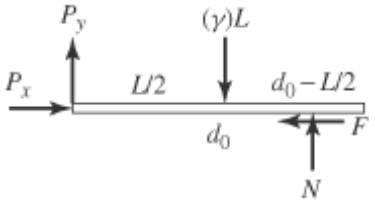
$d = 10$ ft



Solution:

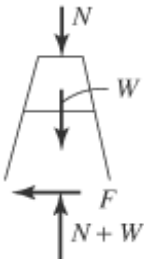
Board:

$$-L\gamma\left(\frac{L}{2}\right) + Nd = 0 \quad N = \frac{1}{2}L^2\left(\frac{\gamma}{d}\right) \quad N = 48.60\,\text{lb}$$



To cause slipping of the board on the saw horse:

$$P_{xb} = \mu N \qquad P_{xb} = 24.30\,\text{lb}$$



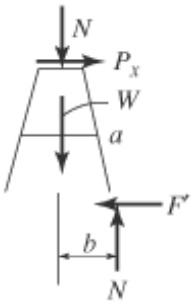
To cause slipping at the ground:

$$P_{xg} = \mu'(N + W) \qquad P_{xg} = 19.08\,\text{lb}$$

To cause tipping

$$(N + W)b - P_{xt}a = 0$$

$$P_{xt} = \frac{(N + W)b}{a} \qquad P_{xt} = 21.20\,\text{lb}$$



Choose the critical case

$$P_x = \min\left(P_{xb}, P_{xg}, P_{xt}\right) \qquad P_x = 19.08\,\text{lb}$$

Problem 8-58

The carpenter slowly pushes the uniform board horizontally over the top of the saw horse. The board has a uniform weight density γ and the saw horse has a weight W and a center of gravity at G. Determine if the saw horse will stay in position, slip, or tip if the board is pushed forward at the given distance d. The coefficients of static friction are shown in the figure.

Given:

$$\gamma = 3\,\frac{\text{lb}}{\text{ft}}$$

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$L = 18$ ft

$W = 15$ lb

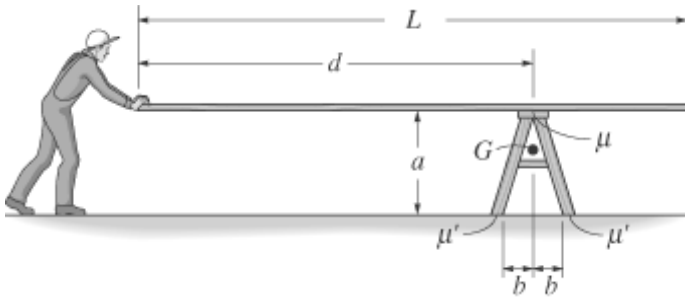
$a = 3$ ft

$b = 1$ ft

$\mu = 0.5$

$\mu' = 0.3$

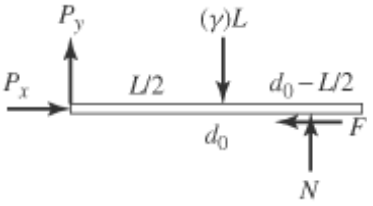
$d = 14$ ft



Solution:

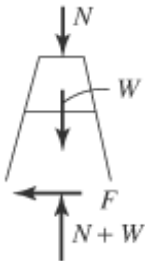
Board:

$$-L\,\gamma\left(\frac{L}{2}\right) + N d = 0 \quad N = \frac{1}{2}\,L^2\left(\frac{\gamma}{d}\right) \quad N = 34.71\text{ lb}$$



To cause slipping of the board on the saw horse:

$$P_{xb} = \mu N \qquad P_{xb} = 17.36\text{ lb}$$



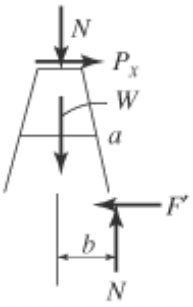
To cause slipping at the ground:

$$P_{xg} = \mu'(N + W) \qquad P_{xg} = 14.91\text{ lb}$$

To cause tipping

$$(N + W)b - P_{xt}a = 0$$

$$P_{xt} = \frac{(N + W)b}{a} \qquad P_{xt} = 16.57\text{ lb}$$



Choose the critical case

$$P_x = \min\left(P_{xb}, P_{xg}, P_{xt}\right) \qquad P_x = 14.91\text{ lb}$$

Problem 8-59

The disk of mass m_o rests on the surface for which the coefficient of static friction is μ_A
Determine the largest couple moment M that can be applied to the bar without causing motion.

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Given:

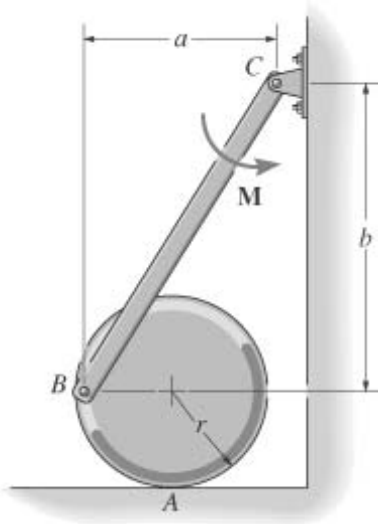
$m_o = 45$ kg

$\mu_A = 0.2$

$a = 300$ mm

$b = 400$ mm

$r = 125$ mm



Solution:

Guesses

$M = 1$ N m $\quad N_A = 1$ N

$B_x = 1$ N $\quad B_y = 1$ N

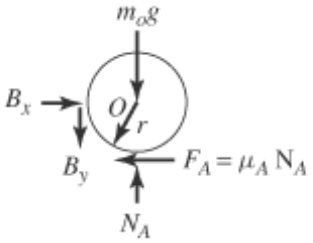
Given

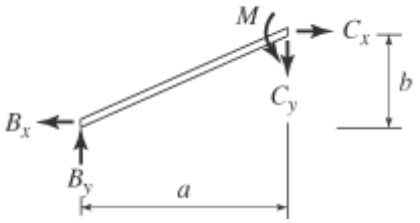
$M - B_x b - B_y a = 0$

$B_x - \mu_A N_A = 0$

$N_A - m_o g - B_y = 0$

$B_y r - \mu_A N_A r = 0$





$$\begin{pmatrix} M \\ N_A \\ B_x \\ B_y \end{pmatrix} = \text{Find}(M, N_A, B_x, B_y)$$

$$\begin{pmatrix} N_A \\ B_x \\ B_y \end{pmatrix} = \begin{pmatrix} 551.81 \\ 110.36 \\ 110.36 \end{pmatrix} \text{ N} \qquad M = 77.3\,\text{N}\cdot\text{m}$$

Problem 8-60

The disk of mass m_0 rests on the surface for which the coefficient of static friction is μ_A Determine the friction force at A.

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Given:

$M = 50\ \text{N}\cdot\text{m}$

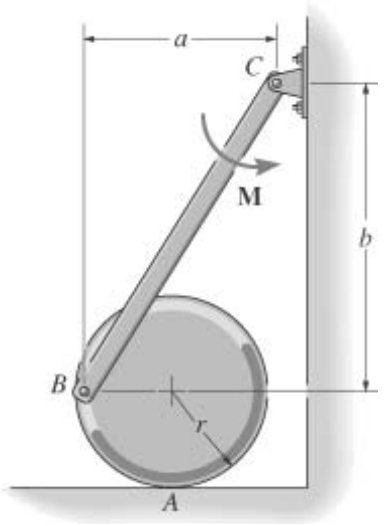
$m_o = 45\ \text{kg}$

$\mu_A = 0.2$

$a = 300\ \text{mm}$

$b = 400\ \text{mm}$

$r = 125\ \text{mm}$



Solution: Assume no motion

Guesses

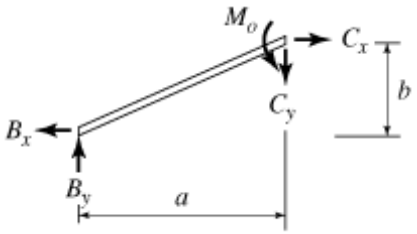
$B_x = 1\ \text{N} \quad B_y = 1\ \text{N} \quad N_A = 1\ \text{N} \quad F_A = 1\ \text{N}$

Given $M - B_y a - B_x b = 0$

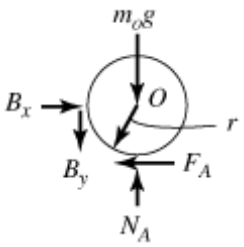
$N_A - B_y - m_o g = 0$

$B_x - F_A = 0$

$B_y r - F_A r = 0$



$$\begin{pmatrix} B_x \\ B_y \\ N_A \\ F_A \end{pmatrix} = \text{Find}(B_x, B_y, N_A, F_A) \qquad \begin{pmatrix} N_A \\ F_A \end{pmatrix} = \begin{pmatrix} 512.88 \\ 71.43 \end{pmatrix}\ \text{N}$$



Check assumption: $F_{Amax} = \mu_A N_A$

Since $F_A = 71.4\ \text{N} < F_{Amax} = 102.6\ \text{N}$ then our assumption is good.

$F_A = 71.4\ \text{N}$

Problem 8-61

A block of weight *W* is attached to a light rod *AD* that pivots at pin *A*. If the coefficient of static friction between the plane and the block is μ_s, determine the minimum angle θ at which the block may be placed on the plane without slipping. Neglect the *size* of the block in the

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calculation.

Given:

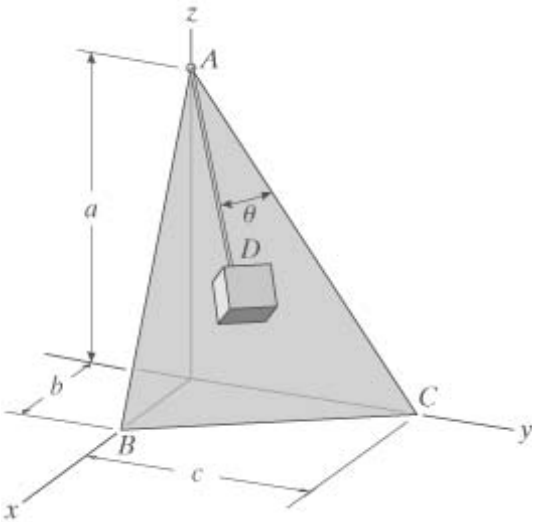
$a = 4$ ft

$b = 2$ ft

$c = 3$ ft

$\mu_s = 0.4$

$W = 6$ lb

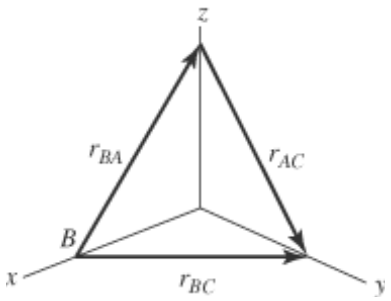


Solution:

Establish a unit vector perpendicular to the plane.

$$\mathrm{r_{BC}} = \begin{pmatrix} -b \\ c \\ 0 \end{pmatrix} \qquad \mathrm{r_{BA}} = \begin{pmatrix} -b \\ 0 \\ a \end{pmatrix}$$

$$\mathrm{n_1} = \frac{\mathrm{r_{BC}} \times \mathrm{r_{BA}}}{\left|\mathrm{r_{BC}} \times \mathrm{r_{BA}}\right|} \qquad \mathrm{n_1} = \begin{pmatrix} 0.77 \\ 0.51 \\ 0.38 \end{pmatrix}$$



Establish 2 other unit vectors in this plane, and one in the z direction

$$\mathrm{k} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \qquad \mathrm{r_{CA}} = \begin{pmatrix} 0 \\ -c \\ a \end{pmatrix} \qquad \mathrm{n_2} = \frac{\mathrm{r_{CA}}}{\left|\mathrm{r_{CA}}\right|} \qquad \mathrm{n_3} = \mathrm{n_1} \times \mathrm{n_2}$$

Guess $T = 1$ lb $N = 1$ lb $\theta = 10$ deg

Given $N\mathrm{n_1} + T\left(\cos(\theta)\,\mathrm{n_2} - \sin(\theta)\,\mathrm{n_3}\right) - \mu_s N\left(\sin(\theta)\,\mathrm{n_2} + \cos(\theta)\,\mathrm{n_3}\right) - W\mathrm{k} = 0$

$$\begin{pmatrix} N \\ T \\ \theta \end{pmatrix} = \mathrm{Find}(N, T, \theta) \qquad \begin{pmatrix} N \\ T \end{pmatrix} = \begin{pmatrix} 2.30 \\ 5.46 \end{pmatrix} \mathrm{lb} \qquad \theta = 20.37\,\mathrm{deg}$$

Problem 8-62

Determine the force **P** needed to lift the load of weight W. Smooth rollers are placed between

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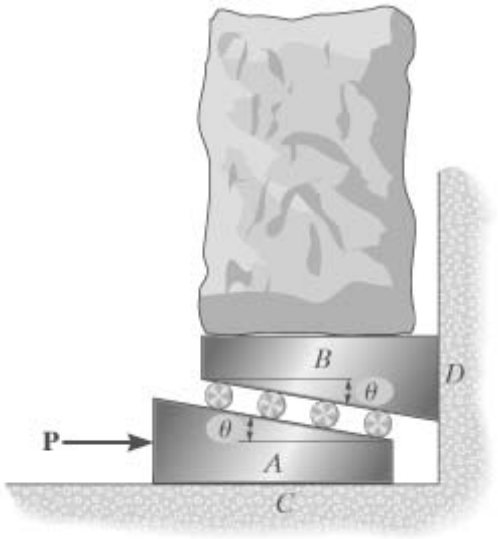
the wedges. The coefficient of static friction between A and C and between B and D is μ_s. Neglect the weight of each wedge.

Given:

$\theta = 10$ deg

$W = 100$ lb

$\mu_s = 0.3$



Solution:

Initial guesses:

$N' = 10$ lb $\quad N_A = 15$ lb

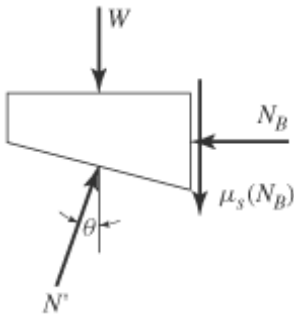
$N_B = 20$ lb $\quad P = 5$ lb

Given

Wedge B:

$$\Sigma F_x = 0; \quad N'\sin(\theta) - N_B = 0$$

$$\Sigma F_y = 0; \quad N'\cos(\theta) - W - \mu_s N_B = 0$$



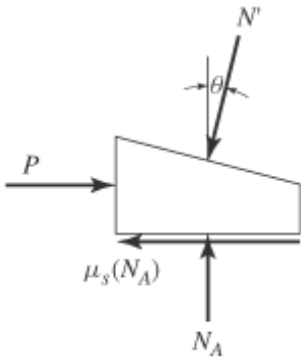
Wedge A:

$$\Sigma F_x = 0; \quad P - N'\sin(\theta) - \mu_s N_A = 0$$

$$\Sigma F_y = 0; \quad N_A - N'\cos(\theta) = 0$$

$$\begin{pmatrix} N' \\ N_B \\ N_A \\ P \end{pmatrix} = \text{Find}(N', N_B, N_A, P) \qquad \begin{pmatrix} N' \\ N_B \\ N_A \end{pmatrix} = \begin{pmatrix} 107.2 \\ 18.6 \\ 105.6 \end{pmatrix} \text{lb}$$

$P = 50.3\,\text{lb}$



Problem 8-63

The wedge is used to level the floor of a building. For the floor loading shown, determine the horizontal force **P** that must be applied to move the wedge forward. The coefficient of static friction between the wedge and the two surfaces of contact is μ_s. Neglect the size and weight

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of the wedge and the thickness of the beam.

Units Used: $kN = 10^3\ N$

Given:

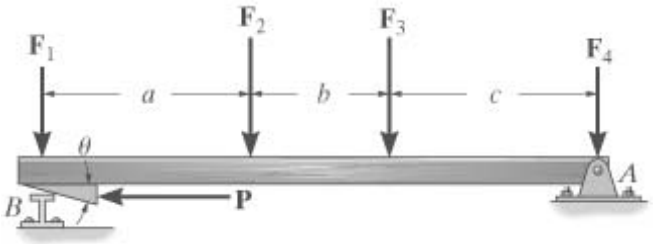
$F_1 = 2\ kN \quad a = 3\ m$

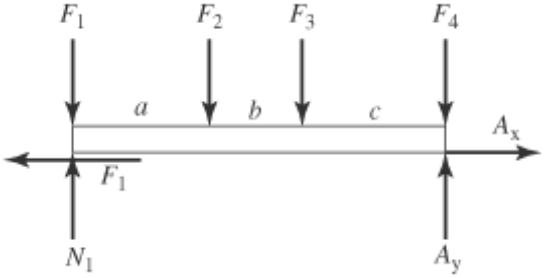
$F_2 = 4\ kN \quad b = 2\ m$

$F_3 = 4\ kN \quad c = 3\ m$

$F_4 = 2\ kN \quad \theta = 15\ deg$

$\mu_s = 0.25$





Solution:

Guesses

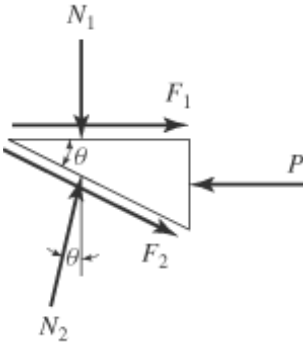
$N_1 = 1\ kN \quad N_2 = 1\ kN \quad P = 1\ kN$

Given

$(F_1 - N_1)(a + b + c) + F_2(b + c) + F_3 c = 0$

$N_2 \cos(\theta) - \mu_s N_2 \sin(\theta) - N_1 = 0$

$\mu_s N_1 + \mu_s N_2 \cos(\theta) + N_2 \sin(\theta) - P = 0$



$\begin{pmatrix} N_1 \\ N_2 \\ P \end{pmatrix} = \text{Find}(N_1, N_2, P) \qquad \begin{pmatrix} N_1 \\ N_2 \end{pmatrix} = \begin{pmatrix} 6 \\ 6.66 \end{pmatrix} kN$

$P = 4.83\,kN$

Problem 8-64

The three stone blocks have weights W_A, W_B, and W_C. Determine the smallest horizontal force **P** that must be applied to block *C* in order to move this block.The coefficient of static friction between the blocks is μ_s, and between the floor and each block μ'_s.

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Given:

$W_A = 600$ lb

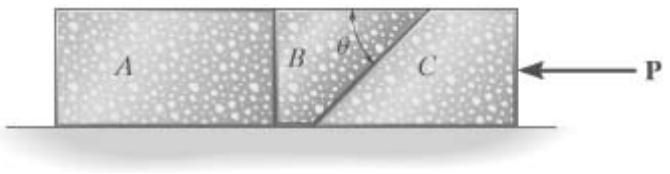
$W_B = 150$ lb

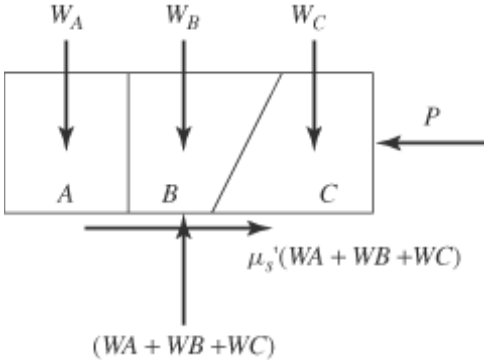
$W_C = 500$ lb

$\mu_s = 0.3$

$\mu'_s = 0.5$

$\theta = 45$ deg





Solution:

Assume all blocks slip together

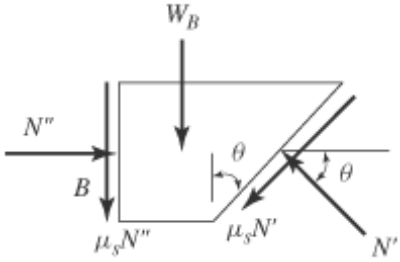
$N_1 = W_A + W_B + W_C$

$P_1 = \mu'_s N_1 \qquad P_1 = 625.00\,\text{lb}$

Assume that block A does not move and block B moves up

Guesses

$N_C = 1$ lb $\quad N' = 1$ lb $\quad N'' = 1$ lb $\quad P = 1$ lb



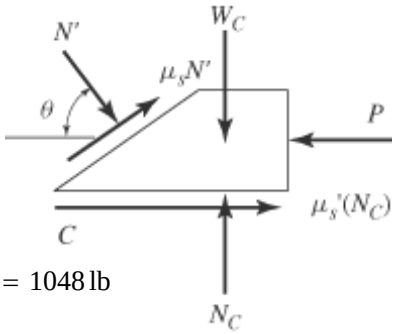
Given

$N'' + \mu'_s N_C - P = 0$

$N_C - W_C - W_B - \mu_s N'' = 0$

$N' \sin(\theta) - \mu_s N' \cos(\theta) - \mu_s N'' - W_B = 0$

$N'' - \mu_s N' \sin(\theta) - N' \cos(\theta) = 0$



$$\begin{pmatrix} N_C \\ N' \\ N'' \\ P_2 \end{pmatrix} = \text{Find}(N_C, N', N'', P) \qquad \begin{pmatrix} N_C \\ N' \\ N'' \end{pmatrix} = \begin{pmatrix} 838.71 \\ 684.30 \\ 629.03 \end{pmatrix} \text{lb} \qquad P_2 = 1048\,\text{lb}$$

Choose the critical case $\quad P = \min(P_1, P_2) \qquad P = 625.00\,\text{lb}$

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Problem 8-65

If the spring is compressed a distance δ and the coefficient of static friction between the tapered stub *S* and the slider *A* is μ_{sA}, determine the horizontal force **P** needed to move the slider forward. The stub is free to move without friction within the fixed collar *C*. The coefficient of static friction between *A* and surface *B* is μ_{AB}. Neglect the weights of the slider and stub.

Given:

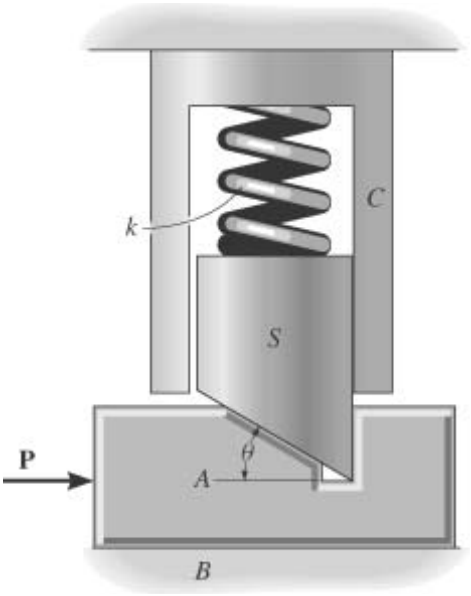
$\delta = 60 \text{ mm}$

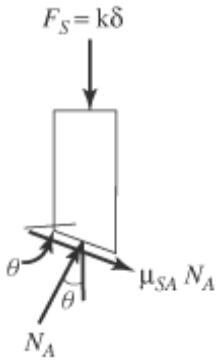
$\mu_{sA} = 0.5$

$\mu_{AB} = 0.4$

$k = 300 \dfrac{\text{N}}{\text{m}}$

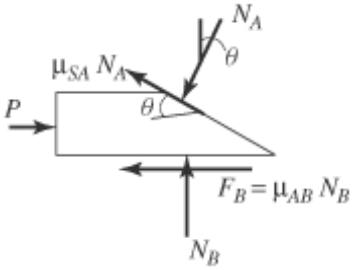
$\theta = 30 \text{ deg}$





Solution:

Stub:



$+\uparrow \Sigma F_y = 0; \qquad N_A \cos(\theta) - \mu_{sA} N_A \sin(\theta) - k\delta = 0$

$$N_A = \frac{k\delta}{\cos(\theta) - \mu_{sA}\sin(\theta)} \qquad N_A = 29.22 \text{ N}$$

Slider:

$+\uparrow \Sigma F_y = 0; \qquad N_B - N_A \cos(\theta) + \mu_{sA} N_A \sin(\theta) = 0$

$$N_B = N_A \cos(\theta) - \mu_{sA} N_A \sin(\theta) \qquad N_B = 18 \text{ N}$$

$\xrightarrow{+} \Sigma F_x = 0; \qquad P - \mu_{AB} N_B - N_A \sin(\theta) - \mu_{sA} N_A \cos(\theta) = 0$

$$P = \mu_{AB} N_B + N_A \sin(\theta) + \mu_{sA} N_A \cos(\theta) \qquad P = 34.5 \text{ N}$$

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Problem 8-66

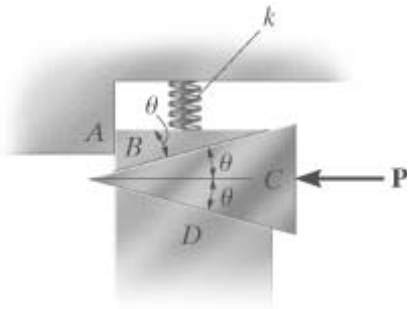
The coefficient of static friction between wedges B and C is μ_{s1} and between the surfaces of contact B and A and C and D, μ_{s2}. If the spring is compressed to a distance δ when in the position shown, determine the smallest force P needed to move wedge C to the left. Neglect the weight of the wedges.

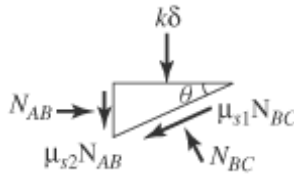
Given:

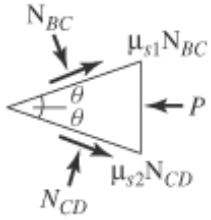
$\mu_{s1} = 0.6 \quad \theta = 15 \text{ deg}$

$\mu_{s2} = 0.4 \quad k = 500 \dfrac{\text{N}}{\text{m}}$

$\delta = 200 \text{ mm}$







Solution:

The initial guesses:

$N_{AB} = 10 \text{ N} \quad N_{BC} = 20 \text{ N} \quad N_{CD} = 30 \text{ N} \quad P = 40 \text{ N}$

Given

Wedge B:

$$N_{AB} - \mu_{s1} N_{BC} \cos(\theta) - N_{BC} \sin(\theta) = 0$$

$$N_{BC} \cos(\theta) - \mu_{s1} N_{BC} \sin(\theta) - \mu_{s2} N_{AB} - k\delta = 0$$

Wedge C:

$$N_{CD} \cos(\theta) - \mu_{s2} N_{CD} \sin(\theta) + \mu_{s1} N_{BC} \sin(\theta) - N_{BC} \cos(\theta) = 0$$

$$N_{CD} \sin(\theta) + \mu_{s2} N_{CD} \cos(\theta) + N_{BC} \sin(\theta) + \mu_{s1} N_{BC} \cos(\theta) - P = 0$$

$$\begin{pmatrix} N_{AB} \\ N_{BC} \\ N_{CD} \\ P \end{pmatrix} = \text{Find}(N_{AB}, N_{BC}, N_{CD}, P) \qquad \begin{pmatrix} N_{AB} \\ N_{BC} \\ N_{CD} \end{pmatrix} = \begin{pmatrix} 176.39 \\ 210.40 \\ 197.77 \end{pmatrix} \text{ N} \quad P = 303.99 \text{ N}$$

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Problem 8-67

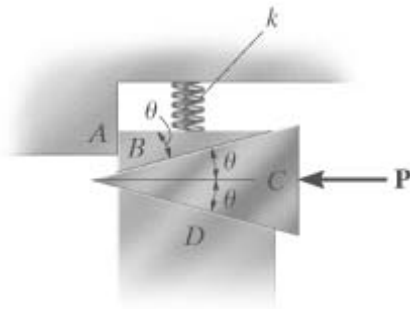
The coefficient of static friction between the wedges B and C is μ_{s1} and between the surfaces of contact B and A and C and D, μ_{s2}. Determine the smalles allowable compression of the spring δ without causing wedge C to move to the left. Neglect the weight of the wedges.

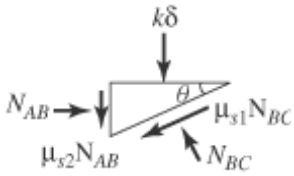
Given:

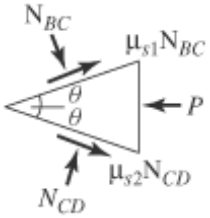
$\mu_{s1} = 0.6 \quad \theta = 15 \text{ deg}$

$\mu_{s2} = 0.4 \quad k = 500 \dfrac{\text{N}}{\text{m}}$

$P = 50 \text{ N}$







Solution:

The initial guesses:

$N_{AB} = 10 \text{ N} \quad N_{BC} = 20 \text{ N} \quad N_{CD} = 30 \text{ N} \quad \delta = 10 \text{ mm}$

Given

Wedge B:

$$N_{AB} - \mu_{s1} N_{BC} \cos(\theta) - N_{BC} \sin(\theta) = 0$$

$$N_{BC} \cos(\theta) - \mu_{s1} N_{BC} \sin(\theta) - \mu_{s2} N_{AB} - k\delta = 0$$

Wedge C:

$$N_{CD} \cos(\theta) - \mu_{s2} N_{CD} \sin(\theta) + \mu_{s1} N_{BC} \sin(\theta) - N_{BC} \cos(\theta) = 0$$

$$N_{CD} \sin(\theta) + \mu_{s2} N_{CD} \cos(\theta) + N_{BC} \sin(\theta) + \mu_{s1} N_{BC} \cos(\theta) - P = 0$$

$$\begin{pmatrix} N_{AB} \\ N_{BC} \\ N_{CD} \\ \delta \end{pmatrix} = \text{Find}(N_{AB}, N_{BC}, N_{CD}, \delta) \qquad \begin{pmatrix} N_{AB} \\ N_{BC} \\ N_{CD} \end{pmatrix} = \begin{pmatrix} 29.01 \\ 34.61 \\ 32.53 \end{pmatrix} \text{ N} \qquad \delta = 32.90 \text{ mm}$$

Problem 8-68

The wedge blocks are used to hold the specimen in a tension testing machine. Determine the design angle θ of the wedges so that the specimen will not slip regardless of the applied load. The coefficients of static friction are μ_A at A and μ_B at B. Neglect the weight of the blocks.

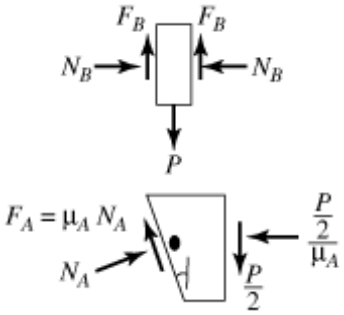
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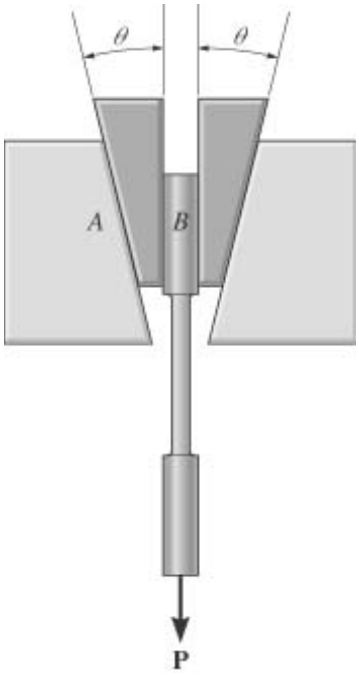
Given:

$\mu_A = 0.1$

$\mu_B = 0.6$

Solution:





Specimen:

$$+\uparrow \Sigma F_y = 0; \qquad 2F_B - P = 0 \qquad F_B = \frac{P}{2}$$

Wedge:

$$\xrightarrow{+} \Sigma F_x = 0; \qquad N_A\cos(\theta) - \mu_A N_A \sin(\theta) - \frac{P}{2\mu_B} = 0$$

$$+\uparrow \Sigma F_y = 0; \qquad \mu_A N_A \cos(\theta) + N_A \sin(\theta) - \frac{P}{2} = 0$$

If we eliminate P we have

$$\mu_B\left(N_A\cos(\theta) - \mu_A N_A \sin(\theta)\right) = \mu_A N_A \cos(\theta) + N_A \sin(\theta)$$

$$\left(1 + \mu_A\mu_B\right)\sin(\theta) = \left(\mu_B - \mu_A\right)\cos(\theta)$$

$$\theta = \operatorname{atan}\left(\frac{\mu_B - \mu_A}{1 + \mu_A\mu_B}\right) \qquad \theta = 25.3\,\text{deg}$$

Problem 8-69

The wedge is used to level the member. Determine the reversed horizontal force **-P** that must be applied to pull the wedge out to the left. The coefficient of static friction between the wedge and the two surfaces of contact is μ_s. Neglect the weight of the wedge.

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Units Used:

$kN = 10^3 N$

Given:

$\mu_s = 0.15$

$\theta = 5 \text{ deg}$

$a = 600 \text{ mm}$

$b = 500 \text{ mm}$

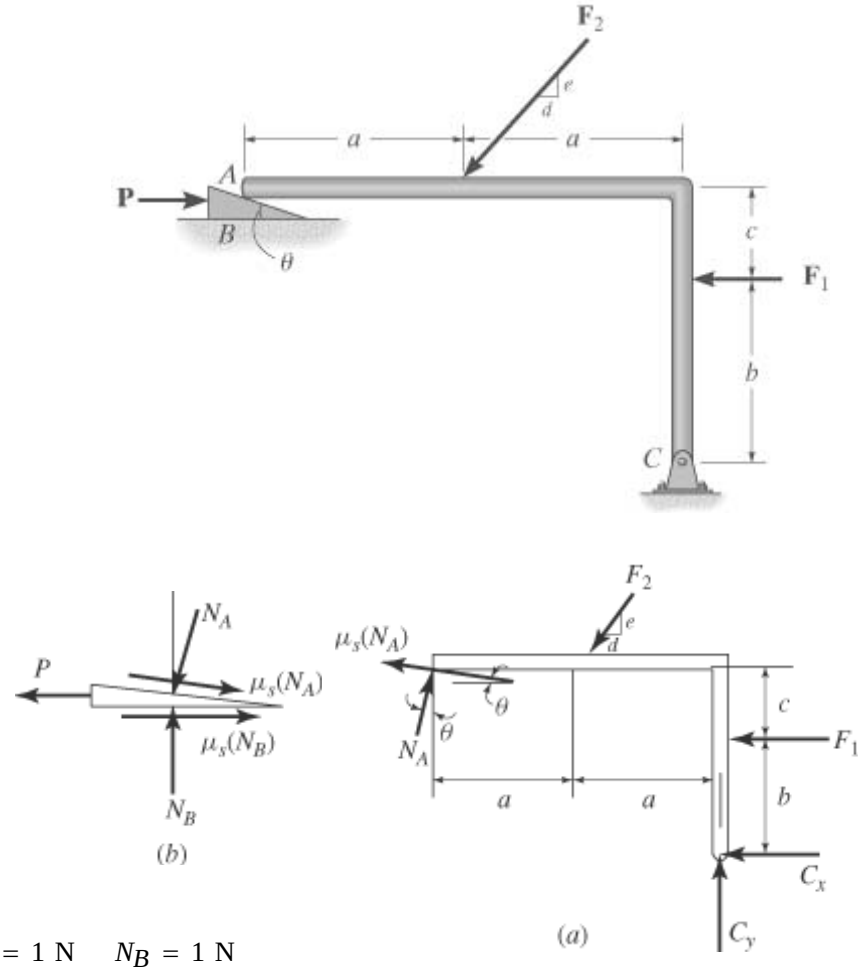
$c = 250 \text{ mm}$

$d = 3$

$e = 4$

$F_1 = 8 \text{ kN}$

$F_2 = 15 \text{ kN}$



Solution:

Guesses

$P = 1 \text{ N} \quad N_A = 1 \text{ N} \quad N_B = 1 \text{ N}$

Given

$-P + \mu_s N_B + \mu_s N_A \cos(\theta) - N_A \sin(\theta) = 0$

$N_B - N_A \cos(\theta) - \mu_s N_A \sin(\theta) = 0$

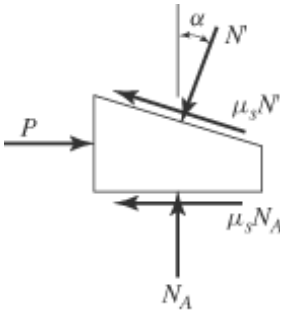
$F_1 b + \left(\frac{e}{\sqrt{e^2 + d^2}}\right) F_2 a + \left(\frac{d}{\sqrt{e^2 + d^2}}\right) F_2 (b + c) - \left(N_A \cos(\theta) + \mu_s N_A \sin(\theta)\right) 2a \ldots = 0$
$+ \left(\mu_s N_A \cos(\theta) - N_A \sin(\theta)\right)(b + c)$

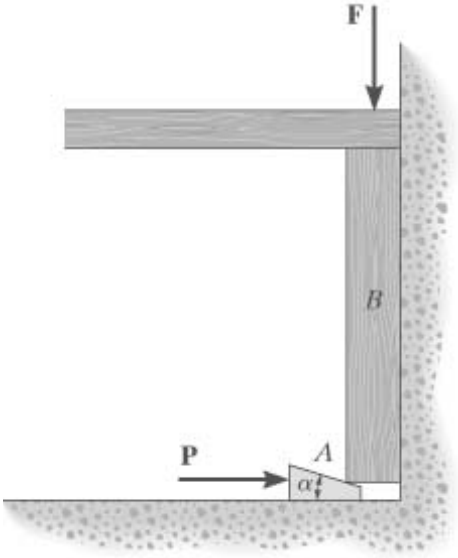
$\begin{pmatrix} P \\ N_A \\ N_B \end{pmatrix} = \text{Find}(P, N_A, N_B) \qquad \begin{pmatrix} N_A \\ N_B \end{pmatrix} = \begin{pmatrix} 15.42 \\ 15.56 \end{pmatrix} \text{kN} \qquad P = 3.29 \text{ kN}$

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Problem 8-70

If the coefficient of static friction between all the surfaces of contact is μ_s, determine the force **P** that must be applied to the wedge in order to lift the brace that supports the load **F**.





Solution:

System:

$\Sigma F_x = 0; \quad P - N_B - \mu_s N_A = 0 \qquad (1)$

$\Sigma F_y = 0; \quad N_A - \mu_s N_B - F = 0 \qquad (2)$

Wedge *A*:

$\Sigma F_x = 0; \quad P - \mu_s N_A - \mu_s N' \cos(\alpha) - N' \sin(\alpha) = 0 \qquad (3)$

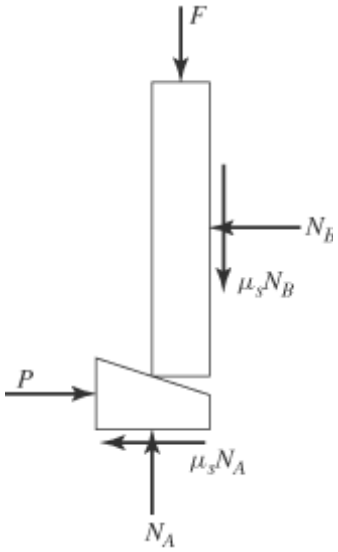
$\Sigma F_y = 0; \quad N_A - N' \cos(\alpha) + \mu_s N' \sin(\alpha) = 0 \qquad (4)$

From Eqs. (3) and (4):

$$N' = \frac{P - \mu_s N_A}{\mu_s \cos(\alpha) + \sin(\alpha)} \qquad (5)$$

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$$N_A = \frac{P\left(1 - \mu_s \tan(\alpha)\right)}{2\mu_s + \left(1 - \mu_s^2\right)\tan(\alpha)} \qquad (6)$$



From Eqs. (1) and (2):

$$N_B = \frac{P - \mu_s N_A}{\mu_s}$$

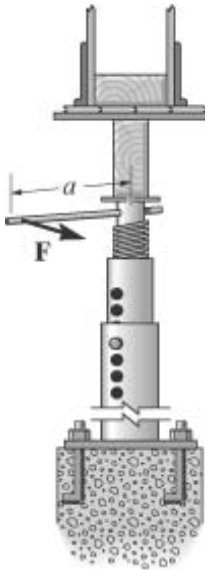
$$P = \frac{\left(1 + \mu_s^2\right)N_A - F}{\mu_s}$$

Combining Eqs. (5) and (6) :

$$P = F\left[\frac{\left(1 - \mu_s^2\right)\tan(\alpha) + 2\mu_s}{1 - 2\mu_s \tan(\alpha) - \mu_s^2}\right]$$

Problem 8-71

The column is used to support the upper floor. If a force **F** is applied perpendicular to the handle to tighten the screw, determine the compressive force in the column. The square-threaded screw on the jack has a coefficient of static friction μ_s, mean diameter d, and a lead h.



Units Used:

$\text{kN} = 10^3\ \text{N}$

Given:

$F = 80\ \text{N}$ $\mu_s = 0.4$ $d = 25\ \text{mm}$

$h = 3\ \text{mm}$ $a = 0.5\ \text{m}$

Solution:

$\phi_s = \text{atan}(\mu_s)$ $\phi_s = 21.80\,\text{deg}$

$\theta_p = \text{atan}\left(\frac{h}{\pi\, d}\right)$ $\theta_p = 2.187\,\text{deg}$

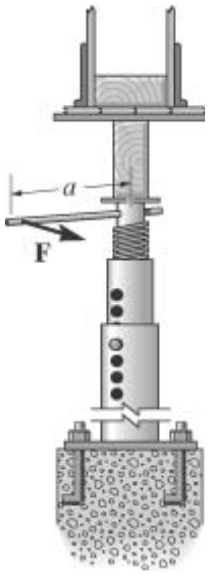
$M = F\,a = W\left(\frac{d}{2}\right)\tan(\phi_s + \theta_p)$

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$$W = 2F\left(\frac{a}{d\,\tan\left(\phi_s + \theta_p\right)}\right) \qquad W = 7.19\,\text{kN}$$

Problem 8-72

The column is used to support the upper floor. If the force **F** is removed from the handle of the jack, determine if the screw is self-locking.. The square-threaded screw on the jack has a coefficient of static friction μ_s, mean diameter d, and a lead h.



Given:

$F = 80\ \text{N} \quad \mu_s = 0.4 \quad d = 25\ \text{mm}$

$h = 3\ \text{mm} \quad a = 0.5\ \text{m}$

Solution:

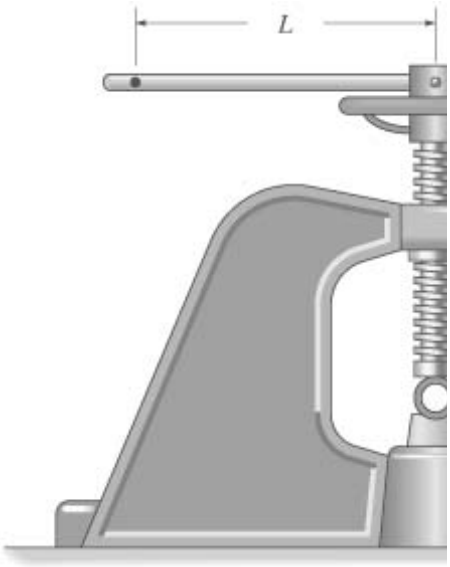
$$\phi_s = \text{atan}\left(\mu_s\right) \qquad \phi_s = 21.8\,\text{deg}$$

$$\theta_p = \text{atan}\left(\frac{h}{\pi\, d}\right) \qquad \theta_p = 2.19\,\text{deg}$$

Since $\phi_s = 21.8\,\text{deg} > \theta_p = 2.19\,\text{deg}$, screw is self locking.

Problem 8-73

The vise is used to grip the pipe. If a horizontal force $\mathbf{F_1}$ is applied perpendicular to the end of the handle of length l, determine the compressive force **F** developed in the pipe. The square threads have a mean diameter d and a lead a. How much force must be applied perpendicular to the handle to loosen the vise?



Given:

$F_1 = 25\ \text{lb}$

$d = 1.5\ \text{in}$

$\mu_s = 0.3$

$L = 10\ \text{in}$

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$a = 0.2$ in

Solution:

$r = \frac{d}{2}$

$\theta = \text{atan}\left(\frac{a}{2\pi r}\right)$ $\theta = 2.43\,\text{deg}$

$\phi = \text{atan}(\mu_s)$ $\phi = 16.70\,\text{deg}$

$F_1 L = F\, r \tan(\theta + \phi)$

$F = F_1 \left(\frac{L}{r \tan(\theta + \phi)}\right)$ $F = 961\,\text{lb}$

To loosen screw,

$P L = F\, r \tan(\phi - \theta)$

$P = F\, r \frac{\tan(\phi - \theta)}{L}$ $P = 18.3\,\text{lb}$

Problem 8-74

Determine the couple forces **F** that must be applied to the handle of the machinist's vise in order to create a compressive force F_A in the block. Neglect friction at the bearing A. The guide at B is smooth so that the axial force on the screw is F_A. The single square-threaded screw has a mean radius b and a lead c, and the coefficient of static friction is μ_s.

Given:

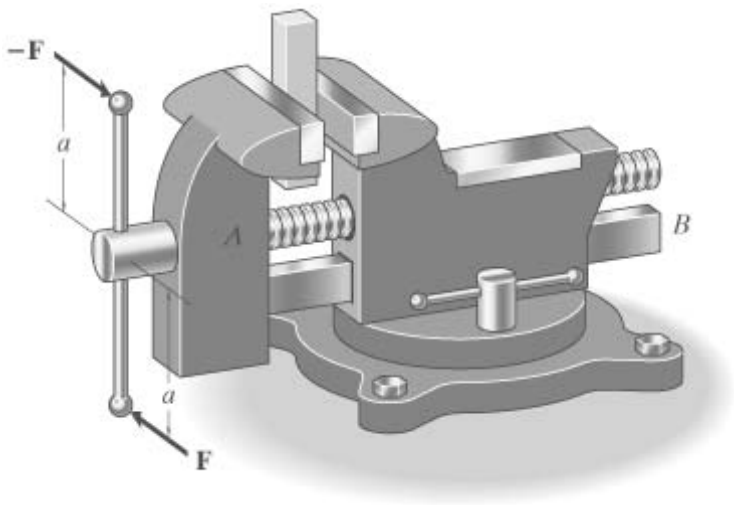
$a = 125$ mm

$F_A = 400$ N

$b = 6$ mm

$c = 8$ mm

$\mu_s = 0.27$



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Solution:

$\phi = \operatorname{atan}(\mu_s)$ $\phi = 15.11\,\text{deg}$

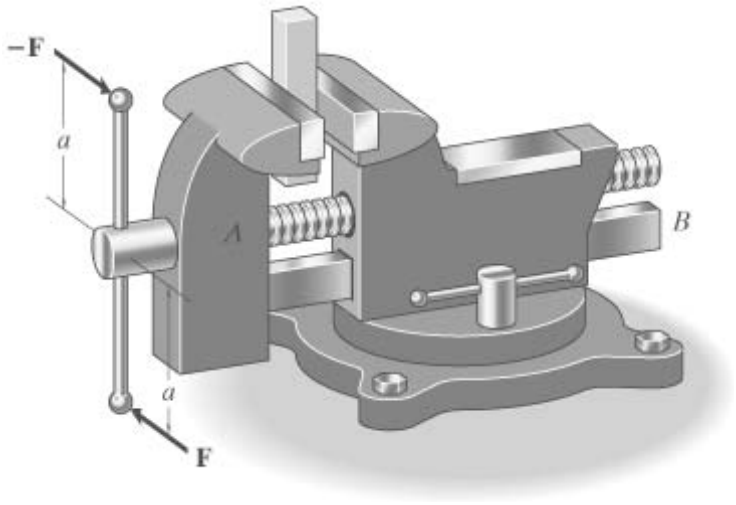
$\theta = \operatorname{atan}\left(\frac{c}{2\ \pi\ b}\right)$ $\theta = 11.98\,\text{deg}$

$F\ 2\ a = F_A\ b\ \tan(\theta + \phi)$

$F = F_A \left(\frac{b}{2a}\right) \tan(\theta + \phi)$ $F = 4.91\ \text{N}$

Problem 8-75

If couple forces **F** are applied to the handle of the machinist's vise, determine the compressive force developed in the block. Neglect friction at the bearing *A*. The guide at *B* is smooth. The single square-threaded screw has a mean radius of r_1 and a lead of r_2, and the coefficient of static friction is μ_s.



Units Used:

$\text{kN} = 10^3\ \text{N}$

Given:

$F = 35\ \text{N}$

$a = 125\ \text{mm}$

$r_1 = 6\ \text{mm}$

$r_2 = 8\ \text{mm}$

$\mu_s = 0.27$

Solution:

$\phi = \operatorname{atan}(\mu_s)$ $\phi = 15.11\,\text{deg}$

$\theta = \operatorname{atan}\left(\frac{r_2}{2\ \pi\ r_1}\right)$ $\theta = 11.98\,\text{deg}$

$F\ 2\ a = P\ r_1\ \tan(\theta + \phi)$

$P = 2\ F \left(\frac{a}{r_1\ \tan(\theta + \phi)}\right)$ $P = 2.85\,\text{kN}$

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Problem 8-76

The machine part is held in place using the double-end clamp.The bolt at B has square threads with a mean radius r and a lead r_1, and the coefficient of static friction with the nut is μ_s. If a torque M is applied to the nut to tighten it, determine the normal force of the clamp at the smooth contacts A and C.

Given:

$a = 260$ mm

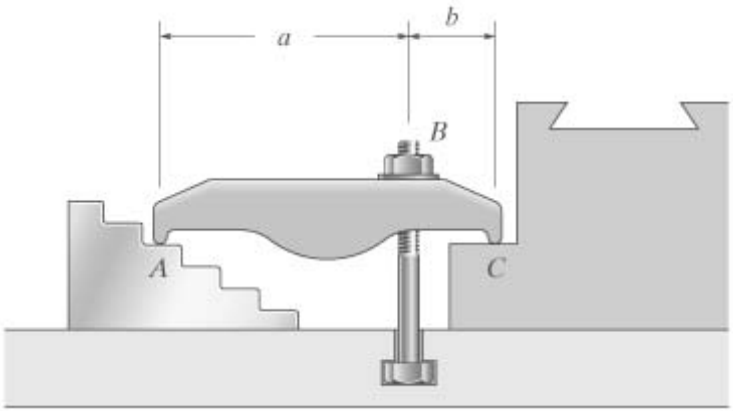
$b = 90$ mm

$r = 4$ mm

$r_l = 2$ mm

$\mu_S = 0.5$

$M = 0.4$ N·m



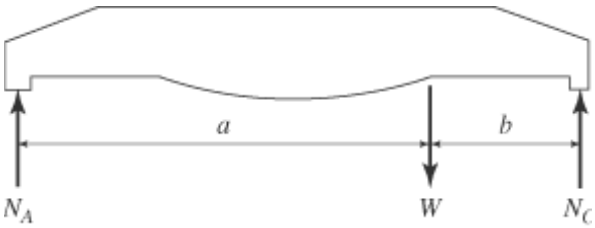
Solution:

$\phi = \operatorname{atan}(\mu_S)$

$\phi = 26.57\,\text{deg}$

$\theta = \operatorname{atan}\left(\dfrac{r_l}{2\pi r}\right)$

$\theta = 4.55\,\text{deg}$



$M = W r \tan(\theta + \phi)$

$W = \dfrac{M}{r\tan(\theta + \phi)}$

$W = 165.67$ N

$\Sigma M_A = 0;\quad N_C(a + b) - W a = 0$

$N_C = W\dfrac{a}{a + b}$ $\quad N_C = 123$ N

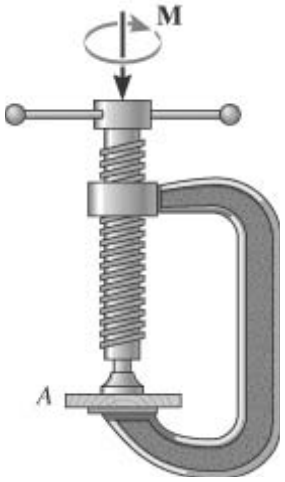
$\Sigma F_y = 0;\quad N_A - W + N_C = 0$

$N_A = W - N_C$ $\quad N_A = 42.6$ N

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Problem 8-77

Determine the clamping force on the board *A* if the screw of the "*C*" clamp is tightened with a twist **M**. The single square-threaded screw has a mean radius *r*, a lead *h*, and the coefficient of static friction is μ_s.



Units Used:

$kN = 10^3\ N$

Given:

$M = 8\ N{\cdot}m$

$r = 10\ mm$

$h = 3\ mm$

$\mu_s = 0.35$

Solution:

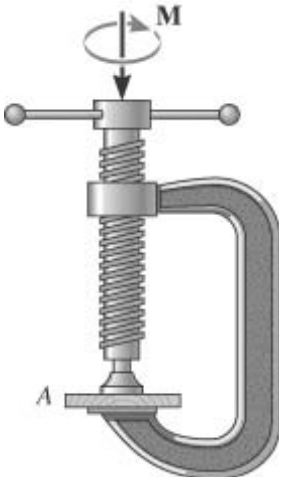
$\phi_s = \operatorname{atan}(\mu_s)$ $\quad \phi_s = 19.29\,deg$

$\theta_p = \operatorname{atan}\left(\frac{1}{2}\,\frac{h}{\pi\, r}\right)$ $\quad \theta_p = 2.734\,deg$

$M = P\, r \tan(\phi_s + \theta_p)$ $\quad P = \frac{M}{r\, \tan(\phi_s + \theta_p)}$ $\quad P = 1.98\,kN$

Problem 8-78

If the required clamping force at the board *A* is to be *P*, determine the torque **M** that must be applied to the handle of the "*C*" clamp to tighten it down. The single square-threaded screw has a mean radius *r*, a lead *h*, and the coefficient of static friction is μ_s.



Given:

$P = 50\ N$

$r = 10\ mm$

$h = 3\ mm$

$\mu_s = 0.35$

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Solution:

$\phi_S = \operatorname{atan}(\mu_S)$ $\phi_S = 19.29\,\text{deg}$

$\theta_P = \operatorname{atan}\left(\frac{1}{2}\,\frac{h}{\pi\, r}\right)$ $\theta_P = 2.73\,\text{deg}$

$M = P\, r \tan(\phi_S + \theta_P)$ $M = 0.202\,\text{N}\cdot\text{m}$

Problem 8-79

Determine the clamping force on the board at *A* if the screw of the hold-down clamp is tightened with a twist **M**. The single square-threaded screw has a mean radius of r and a lead of r_l, and the coefficient of static friction is μ_s.

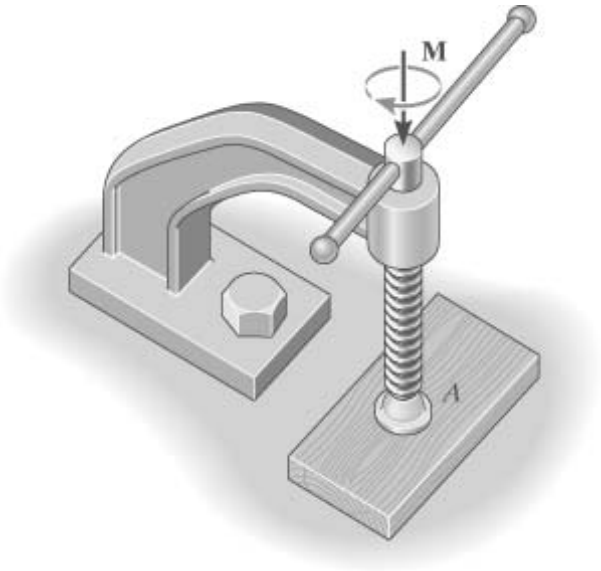
Given:

$M = 0.2$ N m

$r = 8$ mm

$r_l = 2$ mm

$\mu_S = 0.38$



Solution:

$\phi = \operatorname{atan}(\mu_S)$ $\phi = 20.81\,\text{deg}$

$\theta = \operatorname{atan}\left(\frac{r_l}{2\pi r}\right)$ $\theta = 2.28\,\text{deg}$

$M = F\, r \tan(\theta + \phi)$

$F = \frac{M}{r \tan(\theta + \phi)}$ $F = 58.7$ N

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Problem 8-80

If the required clamping force at the board *A* is to be *F*, determine the torque *M* that must be applied to the handle of the hold-down clamp to tighten it down.The single square-threaded screw has a mean radius *r* and a lead r_1, and the coefficient of static friction is μ_s.

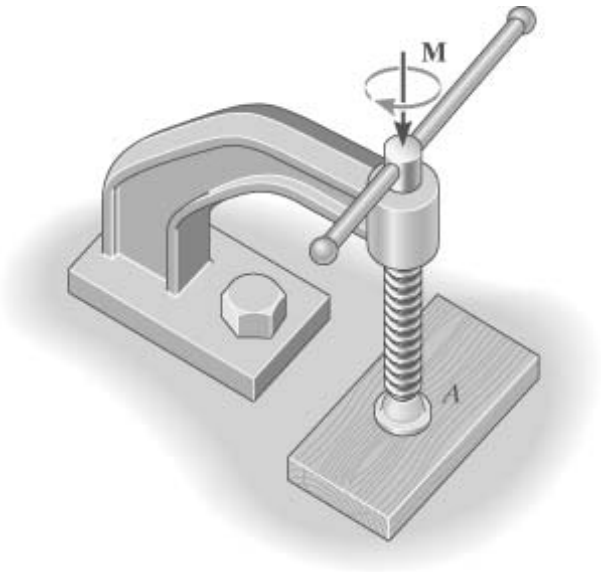
Given:

$F = 70$ N

$r = 8$ mm

$r_l = 2$ mm

$\mu_s = 0.38$



Solution:

$\phi = \operatorname{atan}(\mu_s)$ $\phi = 20.81\,\text{deg}$

$\theta = \operatorname{atan}\left(\frac{r_l}{2\pi r}\right)$ $\theta = 2.2785\,\text{deg}$

$M = F\,r\tan(\theta + \phi)$ $M = 0.24\,\text{N}\cdot\text{m}$

Problem 8-81

The fixture clamp consist of a square-threaded screw having a coefficient of static friction μ_s mean diameter *d*, and a lead *h*. The five points indicated are pin connections. Determine the clamping force at the smooth blocks *D* and *E* when a torque *M* is applied to the handle of the screw.

Given:

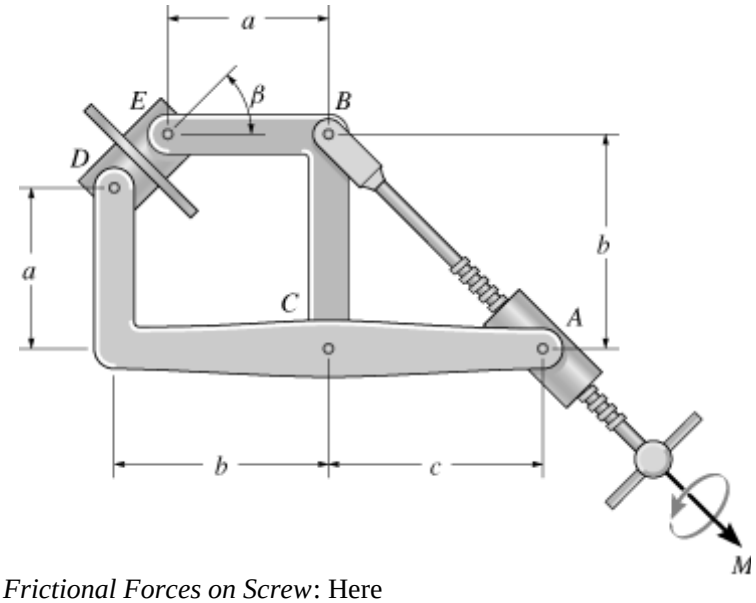
$\mu_s = 0.3$ $a = 30$ mm

$d = 3$ mm $b = 40$ mm

$h = 1$ mm $c = 40$ mm

$M = 0.08$ N·m $\beta = 45$ deg

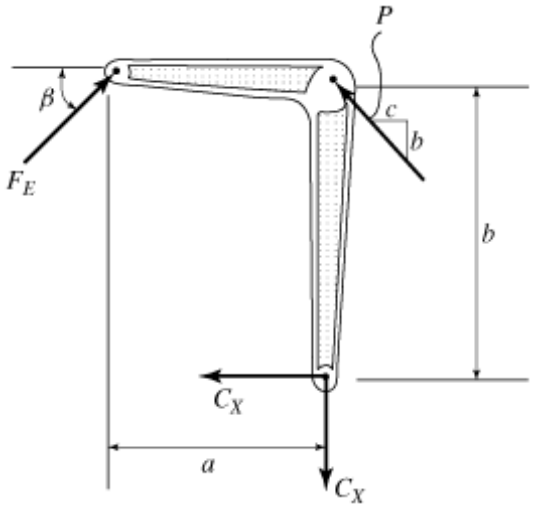
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Frictional Forces on Screw: Here

$$\theta = \operatorname{atan}\left[\frac{h}{2\,\pi\left(\frac{d}{2}\right)}\right] \qquad \theta = 6.06\,\text{deg}$$

$$\phi_S = \operatorname{atan}(\mu_S) \qquad \phi_S = 16.70\,\text{deg}$$



Applying Eq.8-3, we have

$$M = P\left(\frac{d}{2}\right)\tan(\theta + \phi_S)$$

$$P = 2\left(\frac{M}{d\,\tan(\theta + \phi_S)}\right) \qquad P = 127.15\ \text{N}$$

Note since $\phi_S = 16.70\,\text{deg} > \theta = 6.06\,\text{deg}$, the screw is self-locking. It will not unscrew even if the moment *M* is removed.

Equations of Equilibrium and Friction:

$$\circlearrowleft + \quad \Sigma M_c = 0;$$

$$\left(\frac{c}{\sqrt{b^2 + c^2}}\right) P\,b - F_E\cos(\beta)b - F_E\sin(\beta)a = 0$$

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$$F_E = \frac{P\,c\,b}{\sqrt{b^2 + c^2}\left(\cos(\beta)\,b + \sin(\beta)\,a\right)} \qquad F_E = 72.7 \text{ N}$$

The equilibrium of clamped block requires that

$$F_D = F_E \qquad F_D = 72.7 \text{ N}$$

Problem 8-82

The clamp provides pressure from several directions on the edges of the board. If the square-threaded screw has a lead *h,* radius *r,* and the coefficient of static friction is μ_s, determine the horizontal force developed on the board at *A* and the vertical forces developed at *B* and *C* if a torque **M** is applied to the handle to tighten it further. The blocks at *B* and *C* are pin-connected to the board.

Given:

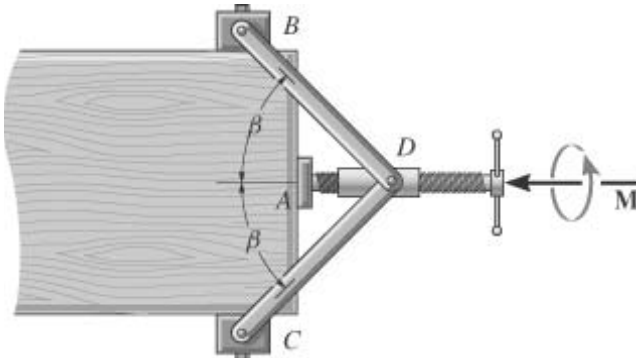
$h = 3$ mm

$r = 10$ mm

$\mu_s = 0.4$

$M = 1.5$ N·m

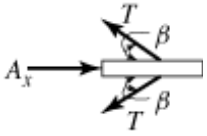
$\beta = 45$ deg



Solution:

$\phi_s = \operatorname{atan}(\mu_s)$ $\qquad \phi_s = 21.801 \text{ deg}$

$\theta = \operatorname{atan}\left(\frac{h}{2\pi r}\right)$ $\qquad \theta = 2.734 \text{ deg}$



$M = A_x\, r \tan(\phi_s + \theta)$ $\qquad A_x = \frac{M}{r\,\tan(\phi_s + \theta)}$ $\qquad A_x = 329$ N

$\xrightarrow{+} \Sigma F_x = 0;$

$A_x - 2T\cos(\beta) = 0$ $\qquad T = \frac{1}{2}\left(\frac{A_x}{\cos(\beta)}\right)$ $\qquad T = 232.36$ N

$C_y = T\sin(\beta)$ $\qquad B_y = C_y$ $\qquad \begin{pmatrix} B_y \\ C_y \end{pmatrix} = \begin{pmatrix} 164.3 \\ 164.3 \end{pmatrix}$ N

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Problem 8-83

The two blocks under the double wedge are brought together using a left and right square-threaded screw. If the mean diameter is d, the lead is r_l, and the coefficient of static friction is μ_s, determine the torque needed to draw the blocks together. The coefficient of static friction between each block and its surfaces of contact is μ'_s.

Units Used:

$\text{kN} = 10^3\ \text{N}$

Given:

$F = 5\ \text{kN}$

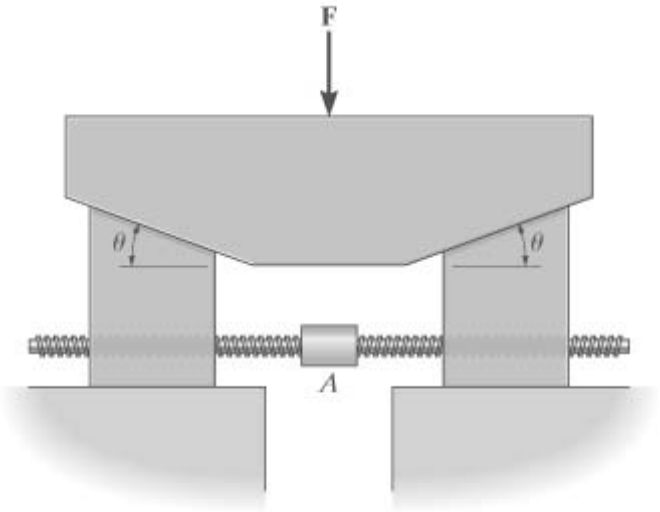
$\theta = 20\ \text{deg}$

$d = 20\ \text{mm}$

$r_l = 5\ \text{mm}$

$\mu_s = 0.4$

$\mu'_s = 0.4$



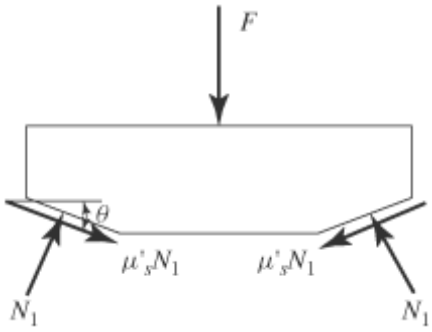
Solution:

Top block:

$$-F + 2N_1\cos(\theta) - 2\mu'_s N_1 \sin(\theta) = 0$$

$$N_1 = \frac{F}{2\left(\cos(\theta) - \mu'_s \sin(\theta)\right)}$$

$N_1 = 3.1138\ \text{kN}$



Bottom block:

$$N' - N_1\cos(\theta) + \mu'_s N_1 \sin(\theta) = 0$$

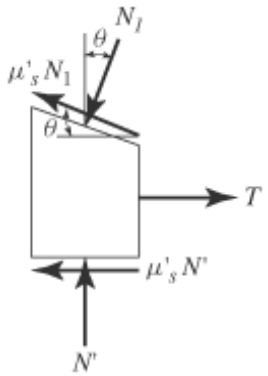
$$N' = N_1\cos(\theta) - \mu'_s N_1 \sin(\theta)$$

$N' = 2.50\ \text{kN}$

$$-N_1\sin(\theta) - \mu'_s N_1\cos(\theta) + T - \mu'_s N' = 0$$

$$T = N_1\sin(\theta) + \mu'_s N_1\cos(\theta) + \mu'_s N'$$

$T = 3.2354\ \text{kN}$



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$\phi = \operatorname{atan}(\mu_s)$ $\phi = 21.80\,\text{deg}$

$\theta = \operatorname{atan}\left(\frac{r_l}{\pi d}\right)$ $\theta = 4.55\,\text{deg}$

Since there are two blocks,

$M = 2T\frac{d}{2}\tan(\theta + \phi)$ $M = 32\,\text{N}\cdot\text{m}$

Problem 8-84

The two blocks under the double wedge are brought together using a left and right square-threaded screw. If the mean diameter is d, the lead is r_l, and the coefficient of static friction is μ_s, determine the torque needed to spread the blocks apart. The coefficient of static friction between each block and its surfaces of contact is μ'_s.

Units Used:

$\text{kN} = 10^3\ \text{N}$

Given:

$F = 5\ \text{kN}$

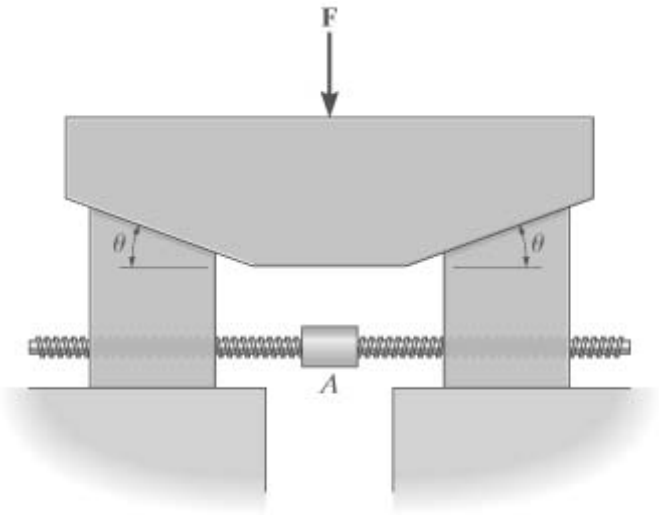
$\theta = 20\ \text{deg}$

$d = 20\ \text{mm}$

$r_l = 5\ \text{mm}$

$\mu_s = 0.4$

$\mu'_s = 0.4$



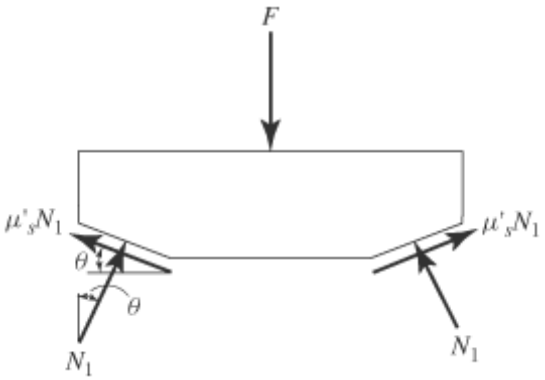
Solution:

Top block:

$-F + 2N_1\cos(\theta) + 2\mu'_s N_1\sin(\theta) = 0$

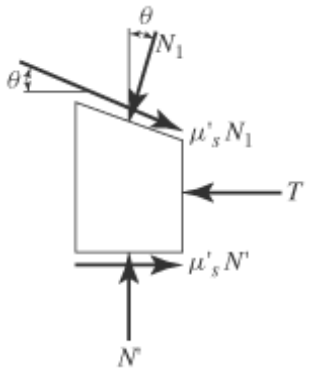
$N_1 = \frac{F}{2(\cos(\theta) + \mu'_s\sin(\theta))}$

$N_1 = 2.3223\,\text{kN}$



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Bottom block :



$$N' - N_1 \cos(\theta) - \mu'_s N_1 \sin(\theta) = 0$$

$$N' = N_1 \cos(\theta) + \mu'_s N_1 \sin(\theta)$$

$$N' = 2.50\,\text{kN}$$

$$-N_1 \sin(\theta) + \mu'_s N_1 \cos(\theta) - T + \mu'_s N' = 0$$

$$T = -N_1 \sin(\theta) + \mu'_s N_1 \cos(\theta) + \mu'_s N'$$

$$T = 1.0786\,\text{kN}$$

$$\phi = \text{atan}(\mu_s) \qquad \phi = 21.80\,\text{deg}$$

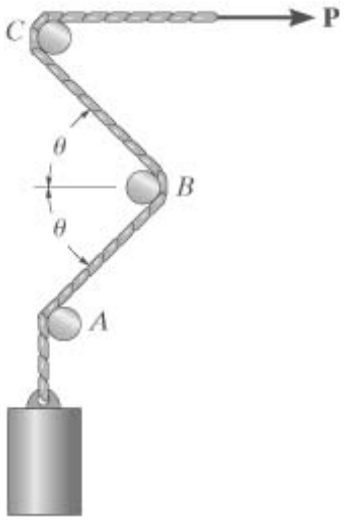
$$\theta = \text{atan}\left(\frac{r_l}{\pi d}\right) \qquad \theta = 4.55\,\text{deg}$$

Since there are two blocks,

$$M = 2T\frac{d}{2}\tan(\phi - \theta) \qquad M = 6.7\,\text{N}\cdot\text{m}$$

Problem 8-85

The cord supporting the cylinder of mass *M* passes around three pegs, *A*, *B*, *C*, where the coefficient of friction is μ_s. Determine the range of values for the magnitude of the horizontal force **P** for which the cylinder will not move up or down.



Given:

$$M = 6\ \text{kg}$$

$$\theta = 45\ \text{deg}$$

$$\mu_s = 0.2$$

$$g = 9.81\ \frac{\text{m}}{\text{s}^2}$$

Solution:

Total angle $\beta = \frac{5}{2}\pi - 4\theta$ $\beta = 270.00\,\text{deg}$

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Forces $P_{min} = M g e^{-\mu_s \beta}$ $P_{max} = M g e^{\mu_s \beta}$

Answer $P_{min} = 15.9\ \text{N} < P < P_{max} = 217.4\ \text{N}$

Problem 8-86

The truck, which has mass m_t, is to be lowered down the slope by a rope that is wrapped around a tree. If the wheels are free to roll and the man at *A* can resist a pull *P*, determine the minimum number of turns the rope should be wrapped around the tree to lower the truck at a constant speed. The coefficient of kinetic friction between the tree and rope is μ_k.

Units Used:

$\text{Mg} = 1000\ \text{kg}$

Given:

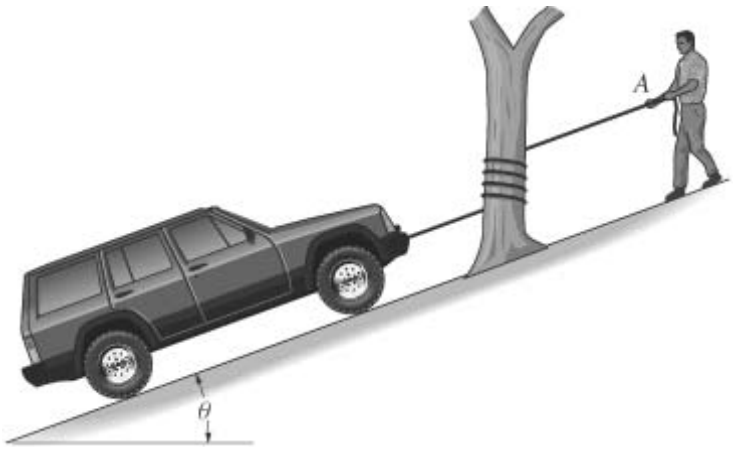
$m_t = 3.4\ \text{Mg}$

$P = 300\ \text{N}$

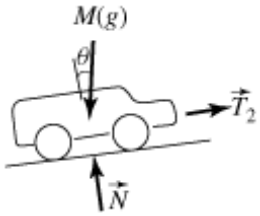
$\theta = 20\ \text{deg}$

$\mu_k = 0.3$

$g = 9.81\ \frac{\text{m}}{\text{s}^2}$



Solution:



$\nearrow \Sigma F_x = 0;$

$T_2 - m_t g \sin(\theta) = 0$ $T_2 = m_t g \sin(\theta)$ $T_2 = 11407.74\ \text{N}$

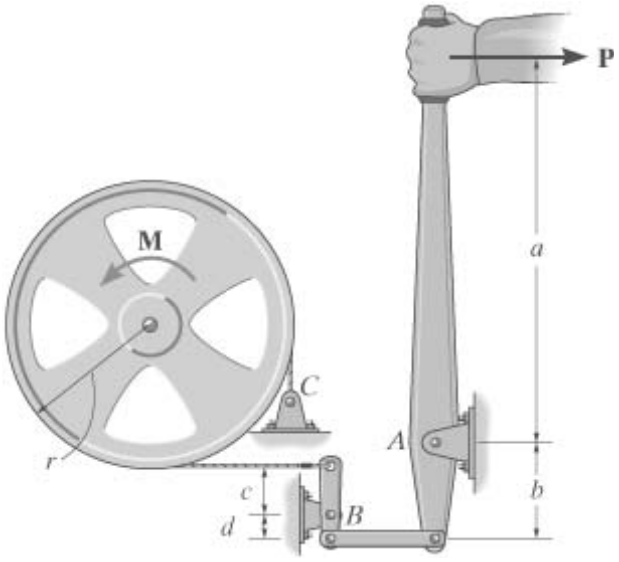
$T_2 = P e^{\mu_k \beta}$ $\beta = \frac{\ln\left(\frac{T_2}{P}\right)}{\mu_k}$ $\beta = 694.86\ \text{deg}$

$\text{Use}\ \ \text{ceil}\left(\frac{\beta}{360\ \text{deg}}\right) = 2.00\ \ \text{turns}$

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Problem 8-87

The wheel is subjected to a torque **M**. If the coefficient of kinetic friction between the band brake and the rim of the wheel is μ_k, determine the smallest horizontal force **P** that must be applied to the lever to stop the wheel.



Given:

$a = 400$ mm $d = 25$ mm

$b = 100$ mm $r = 150$ mm

$c = 50$ mm $M = 50$ N·m

$\mu_k = 0.3$

Solution:

Initial guesses: $T_1 = 5$ N $T_2 = 10$ N

Given

Wheel:

$\Sigma M_0 = 0;$ $-T_2 r + T_1 r + M = 0$ $T_2 = T_1 e^{\mu_k\left(\frac{3\pi}{2}\right)}$

$\begin{pmatrix} T_1 \\ T_2 \end{pmatrix} = \text{Find}(T_1, T_2)$ $T_1 = 54.66$ N

Link:

$\Sigma M_B = 0;$ $T_1 c - F d = 0$ $F = T_1\left(\frac{c}{d}\right)$ $F = 109.32$ N

Lever:

$\Sigma M_A = 0;$ $-P a + F b = 0$ $P = F\left(\frac{b}{a}\right)$ $P = 27.3$ N

Problem 8-88

A cylinder *A* has a mass *M*. Determine the smallest force **P** applied to the handle of the lever required for equilibrium. The coefficient of static friction between the belt and the wheel is μ_s. The drum is pin connected at its center, *B*.

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Given:

$M = 75$ kg

$a = 700$ mm

$b = 25$ mm

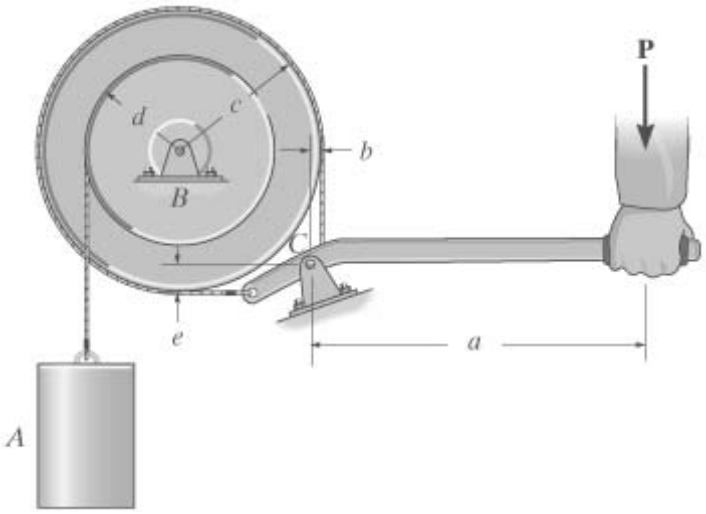
$c = 300$ mm

$d = 200$ mm

$e_1 = 60$ mm

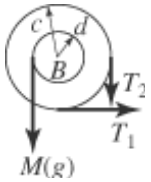
$\mu_s = 0.3$

$e = 2.718$



Solution:

Initial guesses: $T_1 = 1$ N $T_2 = 1$ N $P = 1$ N

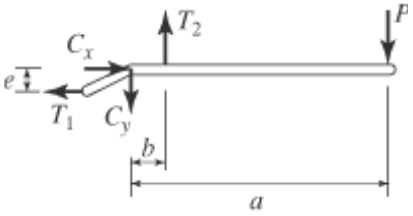


Given

Drum:

$$T_2 = T_1 e^{\mu_s\left(\frac{3\pi}{2}\right)}$$

$$-T_2 c + T_1 c + M g d = 0$$



Lever:

$$-T_1 e_1 + T_2 b - P a = 0$$

$$\begin{pmatrix} T_1 \\ T_2 \\ P \end{pmatrix} = \text{Find}(T_1, T_2, P) \qquad \begin{pmatrix} T_1 \\ T_2 \end{pmatrix} = \begin{pmatrix} 157.7 \\ 648.2 \end{pmatrix} \text{ N} \qquad P = 9.63 \text{ N}$$

Problem 8-89

Determine the largest mass of cylinder *A* that can be supported from the drum if a force **P** is applied to the handle of the lever. The coefficient of static friction between the belt and the wheel is μ_s. The drum is pin supported at its center, *B*.

Given:

$P = 20$ N

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$a = 700$ mm

$b = 25$ mm

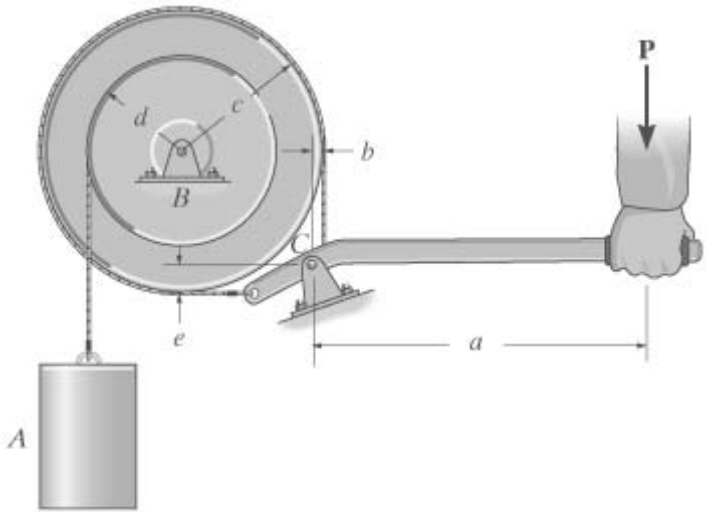
$c = 300$ mm

$d = 200$ mm

$e_1 = 60$ mm

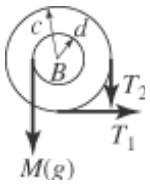
$\mu_s = 0.3$

$e = 2.718$



Solution:

Initial guesses: $T_1 = 1$ N $T_2 = 1$ N $M = 1$ kg

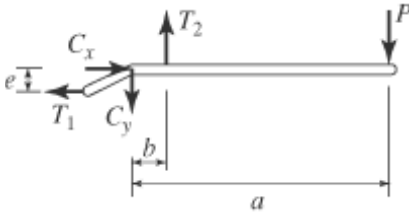


Given

Drum:

$$T_2 = T_1 e^{\mu_s\left(\frac{3\pi}{2}\right)}$$

$$-T_2 c + T_1 c + M g d = 0$$



Lever:

$$-T_1 e_1 + T_2 b - P a = 0$$

$$\begin{pmatrix} T_1 \\ T_2 \\ M \end{pmatrix} = \text{Find}(T_1, T_2, M) \qquad \begin{pmatrix} T_1 \\ T_2 \end{pmatrix} = \begin{pmatrix} 327.4 \\ 1.3 \times 10^3 \end{pmatrix} \text{ N} \qquad M = 155.7 \text{ kg}$$

Problem 8-90

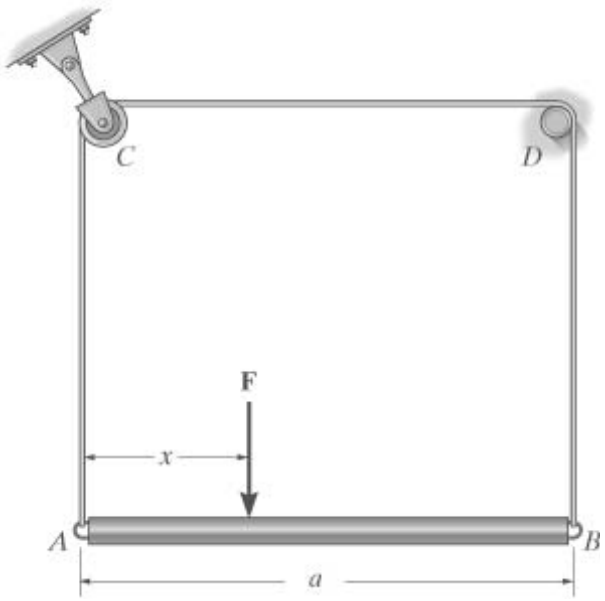
The uniform bar *AB* is supported by a rope that passes over a frictionless pulley at *C* and a fixed peg at *D*. If the coefficient of static friction between the rope and the peg is μ_D, determine the smallest distance *x* from the end of the bar at which a force **F** may be placed and not cause the bar to move.

Given:

$F = 20$ N $a = 1$ m

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$\mu_D = 0.3$



Solution:

Initial guesses:

$T_A = 5$ N $\quad T_B = 10$ N $\quad x = 10$ m

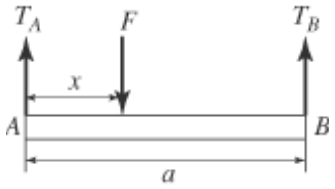
Given

$\Sigma M_A = 0; \qquad -F\,x + T_B\,a = 0$

$\Sigma F_y = 0; \qquad T_A + T_B - F = 0$

$$T_A = T_B e^{\mu_D\left(\frac{\pi}{2}\right)}$$

$$\begin{pmatrix} T_A \\ T_B \\ x \end{pmatrix} = \text{Find}\left(T_A, T_B, x\right)$$



$x = 0.38$ m

Problem 8-91

Determine the smallest lever force **P** needed to prevent the wheel from rotating if it is subjected to a torque **M**. The coefficient of static friction between the belt and the wheel is μ_s. The wheel is pin-connected at its center, *B*.

Given:

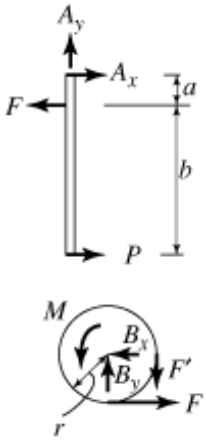
$M = 250$ N m

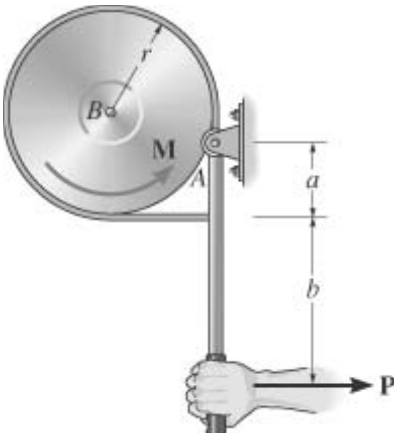
$\mu_s = 0.3$

$r = 400$ mm

$a = 200$ mm

$b = 750$ mm





Solution:

$\circlearrowleft + \Sigma M_A = 0; \qquad -F\,a + P(a + b) = 0$

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$$F = P\left(\frac{a+b}{a}\right)$$

$$\beta = \frac{3\pi}{2}$$

$$F' = F e^{\mu_s \beta}$$

$\curvearrowleft + \Sigma M_B = 0;$ $\quad -P\left(\frac{a+b}{a}\right)e^{\mu_s\beta} r + M + P\left(\frac{a+b}{a}\right)r = 0$

$$P = \frac{M a}{(a+b)r\left(e^{\mu_s \beta} - 1\right)} \qquad P = 42.3 \text{ N}$$

Problem 8-92

Determine the torque M that can be resisted by the band brake if a force **P** is applied to the handle of the lever. The coefficient of static friction between the belt and the wheel is μ_s. The wheel is pin-connected at its center, B.

Given:

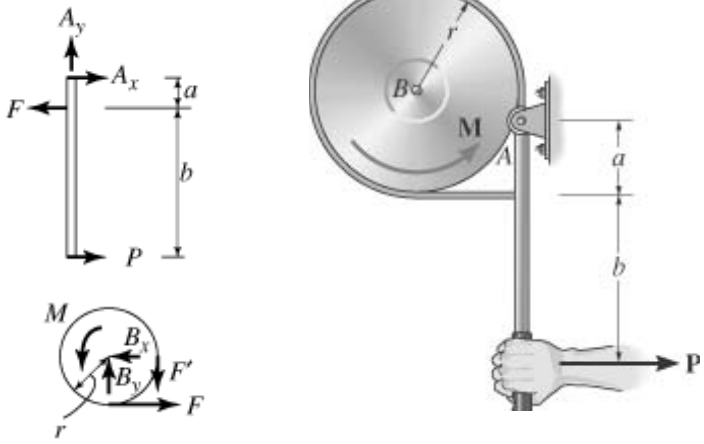
$P = 30$ N

$\mu_s = 0.3$

$r = 400$ mm

$a = 200$ mm

$b = 750$ mm



Solution:

$\curvearrowleft + \Sigma M_A = 0;$ $\quad -F a + P(a+b) = 0$

$$F = P\left(\frac{a+b}{a}\right) \qquad F = 142.5 \text{ N}$$

$$F' = F e^{\mu_s \frac{3\pi}{2}} \qquad F' = 585.8 \text{ N}$$

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$\circlearrowleft \Sigma M_B = 0; \qquad -F' r + F r + M = 0$

$M = F' r - F r \qquad M = 177\,\mathrm{N\cdot m}$

Problem 8-93

Blocks A and B weigh W_A and W_B and respectively. Using the coefficients of static friction indicated, determine the greatest weight of block D without causing motion.

Given:

$W_A = 50$ lb

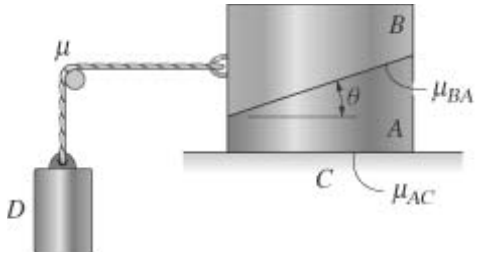
$W_B = 30$ lb

$\mu = 0.5$

$\mu_{BA} = 0.6$

$\mu_{AC} = 0.4$

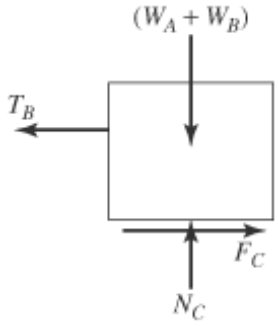
$\theta = 20$ deg



Assume that B slips on A, but A does not move.

Guesses $W_D = 1$ lb $T_B = 1$ lb

$N_B = 1$ lb $N_C = 1$ lb $F_C = 1$ lb



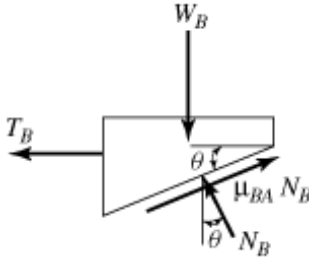
Given $W_D = T_B e^{\mu \frac{\pi}{2}}$

$-T_B + F_C = 0$

$N_C - W_A - W_B = 0$

$-T_B + \mu_{BA} N_B \cos(\theta) - N_B \sin(\theta) = 0$

$N_B \cos(\theta) + \mu_{BA} N_B \sin(\theta) - W_B = 0$



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$$\begin{pmatrix} W_D \\ T_B \\ N_B \\ N_C \\ F_C \end{pmatrix} = \text{Find}(W_D, T_B, N_B, N_C, F_C) \qquad \begin{pmatrix} W_D \\ T_B \\ N_B \\ N_C \\ F_C \end{pmatrix} = \begin{pmatrix} 12.75 \\ 5.81 \\ 26.20 \\ 80.00 \\ 5.81 \end{pmatrix} \text{lb}$$

Now check the assumption that A does not move

$F_{Cmax} = \mu_{AC} N_C \qquad F_{Cmax} = 32.00\,\text{lb}$

Since $F_C = 5.81\,\text{lb} < F_{Cmax} = 32.00\,\text{lb}$ then our assumption is good. $W_D = 12.75\,\text{lb}$

Problem 8-94

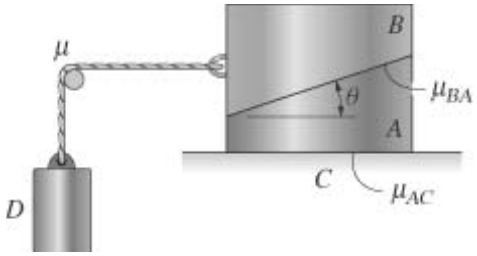
Blocks *A* and *B* have weight *W*, and *D* weighs W_D. Using the coefficients of static friction indicated, determine the frictional force between blocks *A* and *B* and between block *A* and the floor *C*.

Given:

$W = 75\ \text{lb} \qquad \mu_{BA} = 0.6$

$W_D = 30\ \text{lb} \qquad \mu_{AC} = 0.4$

$\mu = 0.5 \qquad \theta = 20\ \text{deg}$



Solution:

$W_D = T_B e^{\mu \frac{\pi}{2}} \qquad T_B = \dfrac{W_D}{e^{\frac{1}{2}\mu\pi}} \qquad T_B = 13.679\,\text{lb}$

$F_C = T_B \qquad F_C = 13.68\,\text{lb}$

Check $N_C - 2W = 0 \qquad N_C = 2W \qquad N_C = 150.00\,\text{lb}$

$F_{Cmax} = \mu_{AC} N_C \qquad F_{Cmax} = 60.00\,\text{lb}$

Since $F_C = 13.68\,\text{lb} < F_{Cmax} = 60.00\,\text{lb}$ then the system does not slip at C.

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For block B:

The initial guessess: $N_B = 1$ lb $F_B = 1$ lb

Given

$+\uparrow \Sigma F_y = 0;$ $N_B\cos(\theta) + F_B\sin(\theta) - W = 0$

$\xrightarrow{+} \Sigma F_x = 0;$ $F_B\cos(\theta) - N_B\sin(\theta) - T_B = 0$

$\begin{pmatrix} N_B \\ F_B \end{pmatrix} = \text{Find}(N_B, F_B)$ $N_B = 65.80$ lb $F_B = 38.51$ lb

Check

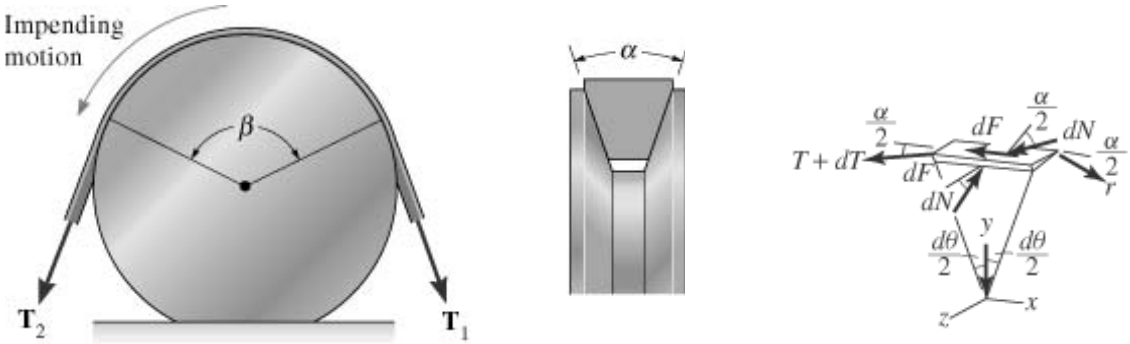
$F_{Bmax} = \mu_{BA} N_B$ $F_{Bmax} = 39.48$ lb

Since $F_B = 38.51$ lb $< F_{Bmax} = 39.48$ lb then no slipping occurs between the blocks

---

Problem 8-95

Show that the frictional relationship between the belt tensions, the coefficient of friction μ, and the angular contacts α and β for the V-belt is $T_2 = T_1 e^{\mu\beta/\sin(\alpha/2)}$ when the belt is on the verge of slipping.



Solution:

FBD of a section of the belt is shown.

Proceeding in the general manner:

$\Sigma F_x = 0;$ $-(T + \mathrm{d}T)\cos\left(\frac{\mathrm{d}\theta}{2}\right) + T\cos\left(\frac{\mathrm{d}\theta}{2}\right) + 2\mu \mathrm{d}N = 0$

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$$\Sigma F_y = 0; \qquad -(T + dT)\sin\left(\frac{d\theta}{2}\right) - T\sin\left(\frac{d\theta}{2}\right) + 2dN\sin\left(\frac{\alpha}{2}\right) = 0$$

Since $d\theta$, dN, and dT are small, these become

$$dT = 2\mu dN \qquad T d\theta = 2dN\sin\left(\frac{\alpha}{2}\right)$$

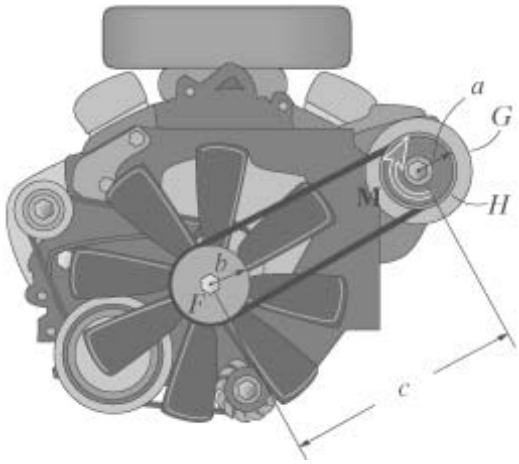
Combine $$\frac{dT}{T} = \mu\frac{d\theta}{\sin\left(\frac{\alpha}{2}\right)}$$

Integrate from $\theta = 0, T = T_1$ to $\theta = \beta, T = T_2$

We get, $$T_2 = T_1 e^{\frac{\mu\beta}{\sin\left(\frac{\alpha}{2}\right)}}$$ Q.E.D

Problem 8-96

A *V*-fan-belt (*V*-angle θ) of an automobile engine passes around the hub *H* of a generator *G* and over the housing *F* to a fan. If the generator locks, and the maximum tension the belt can sustain is T_{max}, determine the maximum possible torque **M** resisted by the axle as the belt slips over the hub. Assume that slipping of the belt occurs only at *H* and that the coefficient of kinetic friction for the hub is μ_s.



Given:

$\theta = 60$ deg $\quad b = 2$ in

$a = 2$ in $\quad c = 1.25$ ft

$T_{max} = 175$ lb $\quad \mu_s = 0.25$

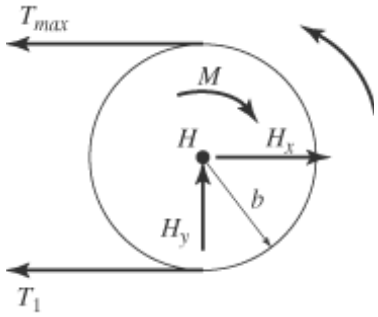
Solution:

$$-T_1 a + T_{max} a - M = 0$$

$$T_1 = T_{max} e^{-\mu_s\left(\frac{\pi}{\sin\left(\frac{1}{2}\theta\right)}\right)} \qquad T_1 = 36.4\,\text{lb}$$

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$M = -T_1 a + T_{max} a$ $M = 23.1\ \text{lb}\cdot\text{ft}$



Problem 8-97

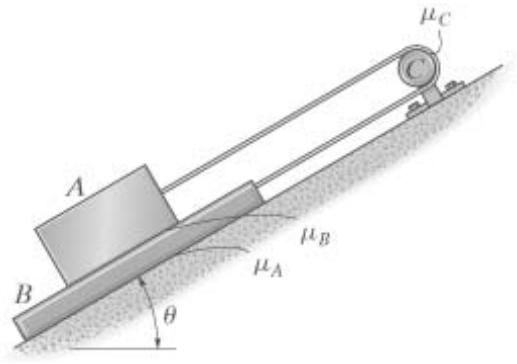
A cable is attached to the plate B of mass M_B, passes over a fixed peg at C, and is attached to the block at A. Using the coefficients of static friction shown, determine the smallest mass of block A so that it will prevent sliding motion of B down the plane.

Given:

$M_B = 20\ \text{kg}$ $\mu_A = 0.2$

$\theta = 30\ \text{deg}$ $\mu_B = 0.3$

$g = 9.81\ \frac{\text{m}}{\text{s}^2}$ $\mu_C = 0.3$



Solution:

Iniitial guesses: $T_1 = 1\ \text{N}$ $T_2 = 1\ \text{N}$ $N_A = 1\ \text{N}$ $N_B = 1\ \text{N}$ $M_A = 1\ \text{kg}$

Given

Block A:

$\Sigma F_x = 0;$ $T_1 - \mu_A N_A - M_A g \sin(\theta) = 0$

$\Sigma F_y = 0;$ $N_A - M_A g \cos(\theta) = 0$

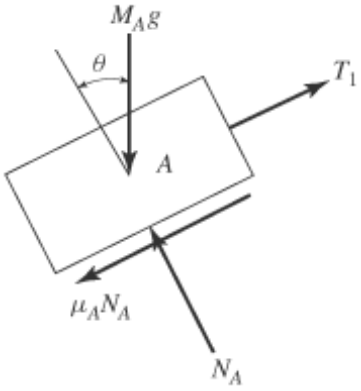


Plate B:

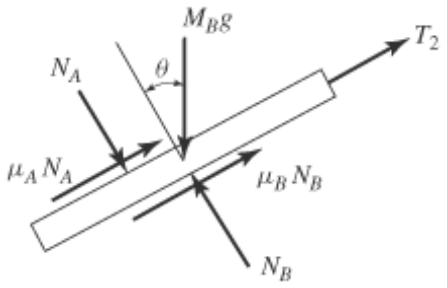
$\Sigma F_x = 0;$ $T_2 - M_B g \sin(\theta) + \mu_B N_B + \mu_A N_A = 0$

$\Sigma F_y = 0;$ $N_B - N_A - M_B g \cos(\theta) = 0$

Peg C: $T_2 = T_1 e^{\mu_C \pi}$

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$$\begin{pmatrix} T_1 \\ T_2 \\ N_A \\ N_B \\ M_A \end{pmatrix} = \text{Find}(T_1, T_2, N_A, N_B, M_A)$$



$M_A = 2.22$ kg

Problem 8-98

The simple band brake is constructed so that the ends of the friction strap are connected to the pin at *A* and the lever arm at *B*. If the wheel is subjected to a torque *M*, determine the smallest force **P** applied to the lever that is required to hold the wheel stationary. The coefficient of static friction between the strap and wheel is μ_s.

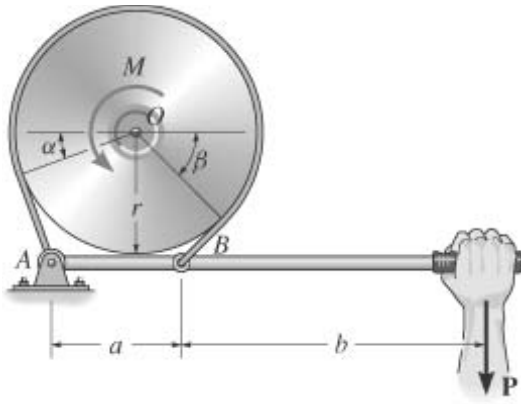
Given:

$M = 80$ lb·ft $\quad \beta = 45$ deg

$\mu_s = 0.5 \quad r = 1.25$ ft

$\alpha = 20$ deg $\quad a = 1.5$ ft

$b = 3$ ft



Solution:

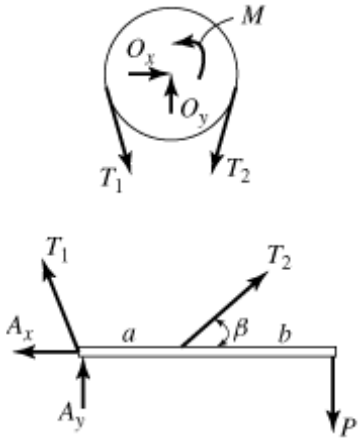
The initial guesses: $T_1 = 10$ lb $\quad T_2 = 20$ lb $\quad P = 30$ lb

Given

$$T_1 r + M - T_2 r = 0$$

$$T_2 = T_1 e^{\mu_s(\pi+\alpha+\beta)}$$

$$T_2 \sin(\beta) a - (a + b) P = 0$$



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$$\begin{pmatrix} T_1 \\ T_2 \\ P \end{pmatrix} = \text{Find}(T_1, T_2, P)$$

$$\begin{pmatrix} T_1 \\ T_2 \end{pmatrix} = \begin{pmatrix} 8.56 \\ 72.56 \end{pmatrix} \text{lb} \qquad P = 17.10\,\text{lb}$$

Problem 8-99

The uniform beam of weight W_1 is supported by the rope which is attached to the end of the beam, wraps over the rough peg, and is then connected to the block of weight W_2. If the coefficient of static friction between the beam and the block, and between the rope and the peg, is μ_s, determine the maximum distance that the block can be placed from A and still remain in equilibrium. Assume the block will not tip.

Given:

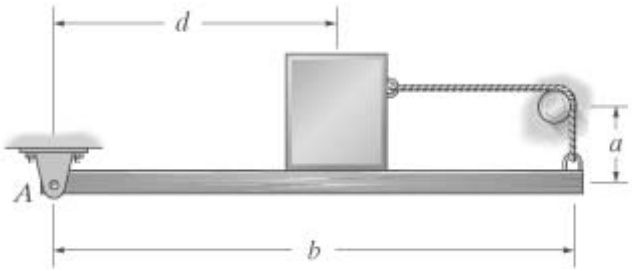
$W_1 = 50$ lb

$W_2 = 100$ lb

$\mu_s = 0.4$

$a = 1$ ft

$b = 10$ ft



Solution:

Block:

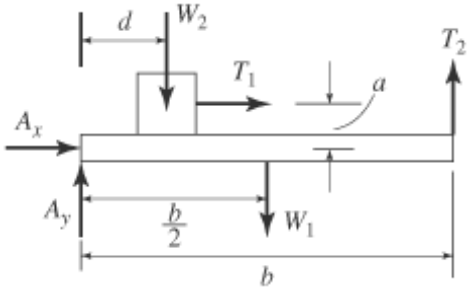
$\Sigma F_y = 0;\quad N - W_2 = 0$

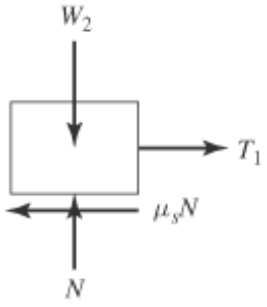
$N = W_2 \qquad N = 100.00\,\text{lb}$

$\Sigma F_x = 0;\quad T_1 - \mu_s N = 0$

$T_1 = \mu_s N \qquad T_1 = 40.00\,\text{lb}$

$T_2 = T_1 e^{\mu_s\left(\frac{\pi}{2}\right)} \qquad T_2 = 74.97\,\text{lb}$





System:

$\Sigma M_A = 0;\ -W_2 d - T_1 a - W_1\left(\frac{b}{2}\right) + T_2 b = 0$

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$$d = \frac{T_2 b - T_1 a - W_1\left(\frac{b}{2}\right)}{W_2}$$

$d = 4.6\,\text{ft}$

Problem 8-100

The uniform concrete pipe has weight W and is unloaded slowly from the truck bed using the rope and skids shown. If the coefficient of kinetic friction between the rope and pipe is μ_k, determine the force the worker must exert on the rope to lower the pipe at constant speed. There is a pulley at B, and the pipe does not slip on the skids. The lower portion of the rope is parallel to the skids.

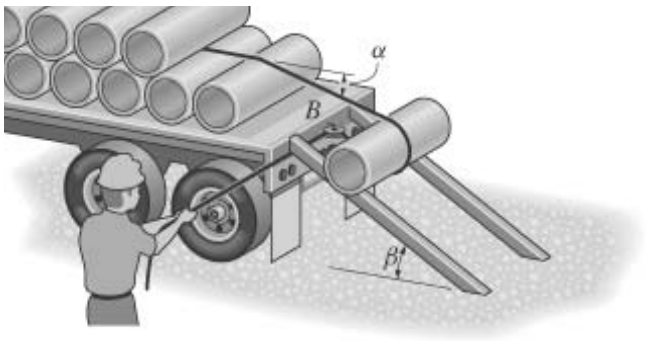
Given:

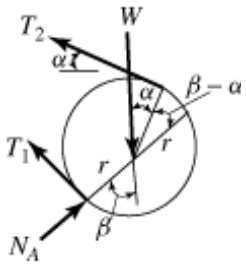
$W = 800\ \text{lb}$

$\mu_k = 0.3$

$\alpha = 15\ \text{deg}$

$\beta = 30\ \text{deg}$





Solution:

$$-W r \sin(\beta) + T_2 \cos(\alpha)\left(r\cos(\alpha) + r\cos(\beta)\right) + T_2 \sin(\alpha)\left(r\sin(\alpha) + r\sin(\beta)\right) = 0$$

$$T_2 = \frac{W\sin(\beta)}{1 + \cos(\alpha)\cos(\beta) + \sin(\alpha)\sin(\beta)} \qquad T_2 = 203.47\,\text{lb}$$

$$T_1 = T_2 e^{-\mu_k(\pi+\beta-\alpha)} \qquad T_1 = 73.3\,\text{lb}$$

Problem 8-101

A cord having a weight density γ and a total length L is suspended over a peg P as shown. If the coefficient of static friction between the peg and cord is μ_s, determine the longest length h which one side of the suspended cord can have without causing motion. Neglect the size of the peg and the length of cord draped over it.

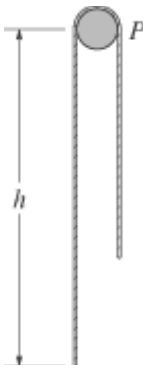
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Given:

$\gamma = 0.5 \frac{\text{lb}}{\text{ft}}$

$L = 10 \text{ ft}$

$\mu_s = 0.5$



Solution:

$T_2 = T_1 e^{\mu\beta}$ $T_1 = \gamma(L - h)$ $T_2 = \gamma h$

$\gamma h = \gamma(L - h)\, e^{\mu_s \pi}$ $h = L\left(\frac{e^{\mu_s \pi}}{1 + e^{\mu_s \pi}}\right)$ $h = 8.28 \text{ ft}$

Problem 8-102

Granular material, having a density ρ is transported on a conveyor belt that slides over the fixed surface, having a coefficient of kinetic friction of μ_k. Operation of the belt is provided by a motor that supplies a torque **M** to wheel *A*.The wheel at *B* is free to turn, and the coefficient of static friction between the wheel at *A* and the belt is μ_A. If the belt is subjected to a pretension *T* when no load is on the belt, determine the greatest volume *V* of material that is permitted on the belt at any time without allowing the belt to stop. What is the torque **M** required to drive the belt when it is subjected to this maximum load?

Units used:

$\text{Mg} = 10^6 \text{ g}$

Given:

$r = 100 \text{ mm}$

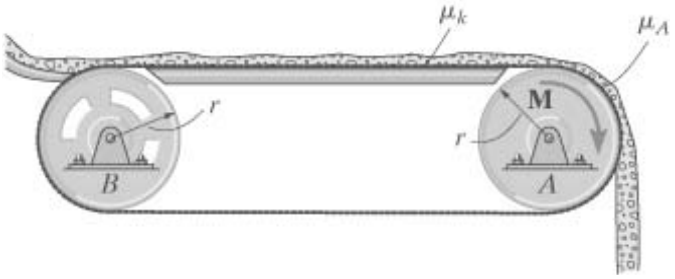
$\mu_A = 0.4$

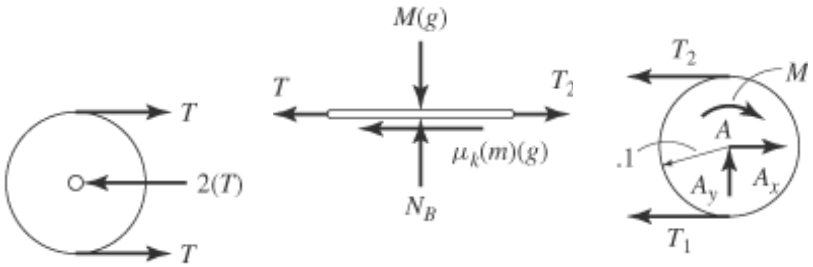
$\mu_k = 0.3$

$\rho = 1500 \frac{\text{kg}}{\text{m}^3}$

$T = 300 \text{ N}$

$g = 9.81 \frac{\text{m}}{\text{s}^2}$





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Solution:

$T_2 = T e^{(\mu_A)\pi}$ $T_2 = 1053.9$ N

Wheel A :

$\Sigma M_A = 0;$ $-M - T r + T_2 r = 0$ $M = -T r + T_2 r$ $M = 75.4\,\text{N}\cdot\text{m}$

Belt

$\Sigma F_x = 0;$ $T_2 - \mu_k m_1 g - T = 0$ $m_1 = \dfrac{T_2 - T}{\mu_k g}$ $m_1 = 256.2$ kg

$V = \dfrac{m_1}{\rho}$ $V = 0.17\,\text{m}^3$

Problem 8-103

Blocks A and B have a mass M_A and M_B, respectively. If the coefficient of static friction between A and B and between B and C is μ_s and between the ropes and the pegs D and E μ'_s, determine the smallest force **F** needed to cause motion of block B .

Units Used:

kN $= 10^3$ N

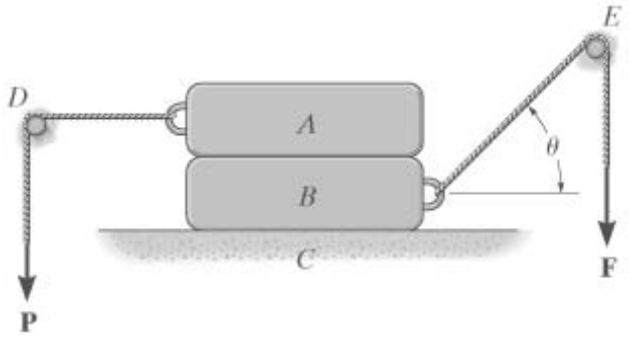
Given:

$\theta = 45$ deg $\mu_s = 0.25$

$M_A = 100$ kg $\mu'_s = 0.5$

$M_B = 150$ kg $P = 30$ N

$g = 9.81 \dfrac{\text{m}}{\text{s}^2}$



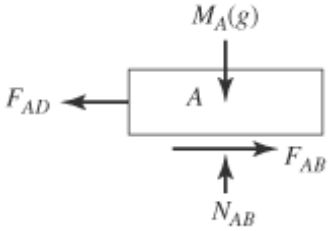
Solution:

Assume no slipping between A & B.

Guesses

$F = 1$ N $N_{AB} = 1$ N $F_{AB} = 1$ N

$N_{BC} = 1$ N $F_{BE} = 1$ N $F_{AD} = 1$ N



Given

$F = F_{BE} e^{\mu'_s\left(\frac{\pi}{2}+\theta\right)}$ $F_{AD} = P e^{\mu'_s \frac{\pi}{2}}$

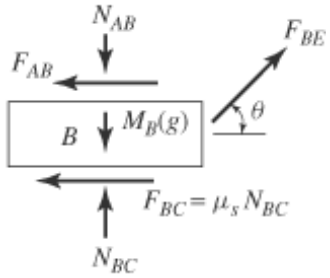
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$$F_{BE}\cos(\theta) - \mu_s N_{BC} - F_{AB} = 0$$

$$F_{BE}\sin(\theta) - N_{AB} - M_B g + N_{BC} = 0$$

$$-F_{AD} + F_{AB} = 0$$

$$N_{AB} - M_A g = 0$$



$$\begin{pmatrix} F \\ N_{AB} \\ F_{AB} \\ N_{BC} \\ F_{BE} \\ F_{AD} \end{pmatrix} = \text{Find}\left(F, N_{AB}, F_{AB}, N_{BC}, F_{BE}, F_{AD}\right) \qquad \begin{pmatrix} F \\ N_{AB} \\ F_{AB} \\ N_{BC} \\ F_{BE} \\ F_{AD} \end{pmatrix} = \begin{pmatrix} 2.49 \\ 0.98 \\ 0.07 \\ 1.91 \\ 0.77 \\ 0.07 \end{pmatrix} \text{kN}$$

Now check assumption $F_{ABmax} = \mu_s N_{AB}$ $F_{ABmax} = 245.25$ N

Since $F_{AB} = 65.8$ N $< F_{ABmax} = 245.3$ N then our assumption is correct

$F = 2.49$ kN

Problem 8-104

Blocks A and B weigh W_1 and W_2, respectively. Using the coefficients of static friction indicated, determine the greatest weight W of block E without causing motion.

Given:

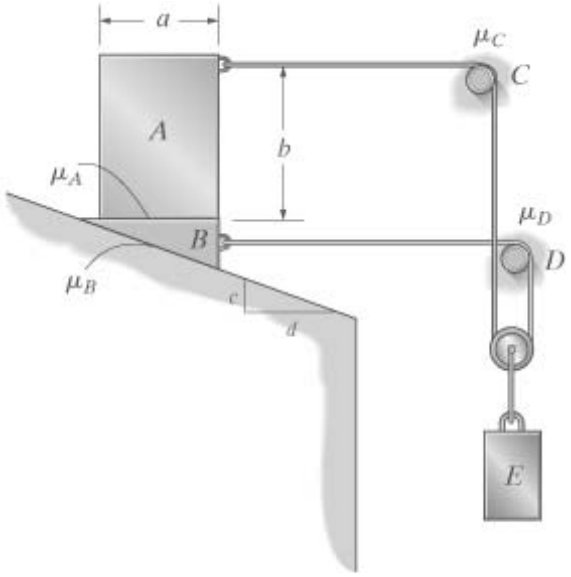
$W_1 = 50$ lb $d = 12$

$W_2 = 30$ lb $\mu_A = 0.3$

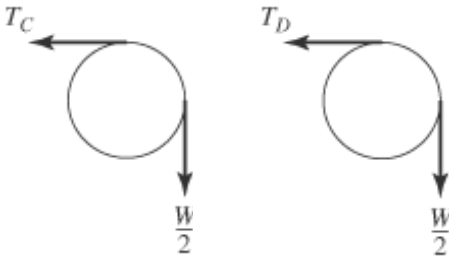
$a = 1.5$ ft $\mu_B = 0.5$

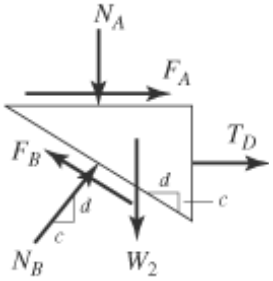
$b = 2$ ft $\mu_C = 0.2$

$c = 5$ $\mu_D = 0.3$



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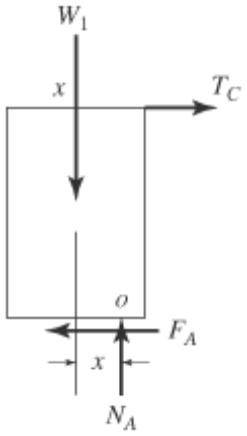


Solution: Assume that the wedge slips on the ground, but the block does not slip on the wedge and the block does not tip.

Guesses $W = 1\ \text{lb}$ $T_D = 1\ \text{lb}$ $T_C = 1\ \text{lb}$

$N_A = 1\ \text{lb}$ $F_A = 1\ \text{lb}$ $N_B = 1\ \text{lb}$

$x = 1\ \text{ft}$ $F_{Amax} = 1\ \text{lb}$



Given

$$\frac{W}{2} = T_C e^{\mu_C \frac{\pi}{2}} \qquad \frac{W}{2} = T_D e^{\mu_D \frac{\pi}{2}}$$

$$F_A - \mu_B N_B \left(\frac{d}{\sqrt{c^2 + d^2}} \right) + N_B \left(\frac{c}{\sqrt{c^2 + d^2}} \right) + T_D = 0$$

$$N_B \left(\frac{d}{\sqrt{c^2 + d^2}} \right) + \mu_B N_B \left(\frac{c}{\sqrt{c^2 + d^2}} \right) - W_2 - N_A = 0$$

$$T_C - F_A = 0 \qquad N_A - W_1 = 0 \qquad -T_C b + N_A x = 0 \qquad F_{Amax} = \mu_A N_A$$

$$\begin{pmatrix} W \\ T_D \\ T_C \\ N_A \\ F_A \\ N_B \\ x \\ F_{Amax} \end{pmatrix} = \text{Find}\left(W, T_D, T_C, N_A, F_A, N_B, x, F_{Amax}\right) \qquad W = 8.15\,\text{lb}$$

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Check assumptions

Since $F_A = 2.97\,\text{lb} < F_{Amax} = 15.00\,\text{lb}$ then the block does not slip relative to the wedge.

Since $x = 0.12\,\text{ft} < \frac{a}{2} = 0.75\,\text{ft}$ then the block does not tip.

Problem 8-105

Block A has mass m_A and rests on surface B for which the coefficient of static friction is μ_{sAB}. If the coefficient of static friction between the cord and the fixed peg at C is μ_{sC}, determine the greatest mass m_D of the suspended cylinder D without causing motion.

Given:

$m_A = 50$ kg

$\mu_{sAB} = 0.25$

$\mu_{sC} = 0.3$

$a = 0.3$ m

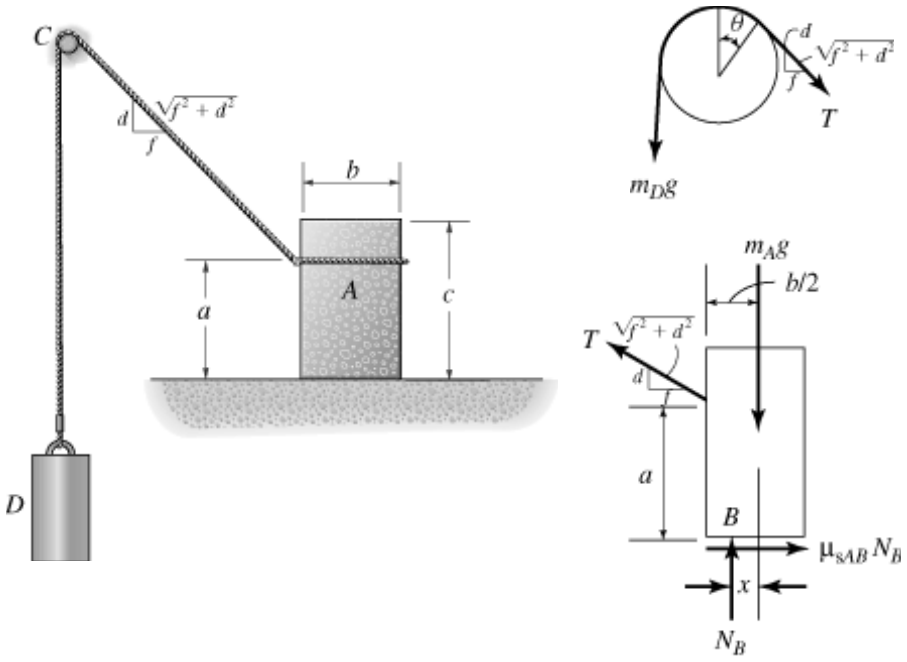
$b = 0.25$ m

$c = 0.4$ m

$d = 3$

$f = 4$

$g = 9.81\ \frac{\text{m}}{\text{s}^2}$



Solution: Assume block A slips but does not tip. $\beta = \pi - \text{atan}\left(\frac{f}{d}\right)$

The initial guesses: $N_B = 100$ N $T = 50$ N $m_D = 1$ kg $x = 10$ mm

Given $m_D g = T e^{\mu_{sC}\beta}$ $\left(\frac{d}{\sqrt{f^2 + d^2}}\right) T - m_A g + N_B = 0$

$\left(\frac{-f}{\sqrt{f^2 + d^2}}\right) T + \mu_{sAB} N_B = 0$ $\left(\frac{f}{\sqrt{f^2 + d^2}}\right) T a - \left(\frac{d}{\sqrt{f^2 + d^2}}\right) T\left(\frac{b}{2}\right) - N_B x = 0$

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